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CLSI M07™

Methods for Dilution Antimicrobial Susceptibility Tests for Bacteria That Grow Aerobically

CLSI M07 covers reference methods for determining minimal inhibitory concentrations of aerobic bacteria by broth macrodilution, broth microdilution, and agar dilution.

A standard for global application developed through the Clinical and Laboratory Standards Institute consensus process.

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Methods for Dilution Antimicrobial Susceptibility Tests for Bacteria That Grow Aerobically

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Abstract

Clinical and Laboratory Standards Institute M07—*Methods for Dilution Antimicrobial Susceptibility Tests for Bacteria That Grow Aerobically* offers guidance for antimicrobial susceptibility testing that is indicated for any organism that contributes to an infectious process warranting antimicrobial chemotherapy, if its susceptibility cannot be reliably predicted from knowledge of the organism's identity. Susceptibility tests are most often indicated when the causative organism is thought to belong to a species capable of exhibiting resistance to commonly used antimicrobial agents.

Various laboratory methods can be used to measure the *in vitro* susceptibility of bacteria to antimicrobial agents. Clinical and Laboratory Standards Institute M07—*Methods for Dilution Antimicrobial Susceptibility Tests for Bacteria That Grow Aerobically* describes standard broth dilution (macrodilution and microdilution [the microdilution method described in CLSI M07 is the same methodology outlined in ISO 20776-1])[1] and agar dilution techniques, and it includes a series of procedures to standardize the way the tests are performed. The performance, applications, and limitations of the current CLSI-recommended methods are also described.

The supplemental information (CLSI M100[2] tables) used with this standard represents the most current information for drug selection, interpretation, and QC using the procedures standardized in CLSI M07.

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Foreword

The most current edition of CLSI M100,[2] an annually published volume of tables, is made available with this standard to ensure users are aware of the latest recommendations related to the methods described in CLSI M02,[3] M07, and M11.[4]

Many other editorial and procedural changes in this edition of CLSI M07 resulted from Subcommittee on Antimicrobial Susceptibility Testing meetings held since 2018. Specific changes to the tables are summarized at the beginning of CLSI M100.[2] The most important changes in CLSI M07 are summarized below.

Overview of Changes

This standard replaces CLSI M07-Ed11, published in 2018. Several changes were made in this edition, including:

- **General:**
  - Revised information for testing and reporting to clarify relevant stakeholders
  - Revised nomenclature from "coagulase-negative staphylococci" (CoNS) to "staphylococci other than *Staphylococcus aureus*" (SOSA)
  - Revised nomenclature from "groups" to "tiers"
  - Revised nomenclature from "microdilution trays" to "microdilution panels"
- **Subchapter 1.4.1, Definitions:**
  - Added definitions for antimicrobial susceptibility testing, EDTA-modified carbapenem inactivation method, heteroresistance, inducible clindamycin resistance, and reference method
  - Deleted definition for D-zone test
- **Subchapter 1.4.2, Abbreviations and Acronyms:**
  - Added abbreviations for Mueller-Hinton fastidious and staphylococci other than *S. aureus*
  - Deleted abbreviations for coagulase-negative staphylococci and Mueller-Hinton broth
- **Subchapter 2.1, Selecting Antimicrobial Agents for Routine Testing and Reporting:**
  - Clarified and updated information on selecting antimicrobial agents for routine reporting
- **Subchapter 2.2, Equivalent Agents:**
  - Replaced former Subchapter 2.2, Routine Reports, with Subchapter 2.2, Equivalent Agents, with information on antimicrobial agents listed in CLSI M100[2] Tables 1
- **Subchapter 2.3, Suggested Guidelines for Routine and Selective Testing and Reporting:**
  - Revised suggested guidelines for routine and selective testing and reporting
  - Added new Table 2, Antimicrobial Agent Test and Report Tiers and Additional Considerations for Agents Listed in CLSI M100[2] Tables 1
  - Added new Table 3, Antimicrobial Agent Test and Report Designations and Additional Considerations for Agents Not Listed in CLSI M100[2] Tables 1
- **Subchapter 2.4, Antimicrobial Classes:**
  - Added new Table 4, Antimicrobial Agent Class Characteristics

- **Subchapter 3.1.3, Preparing Stock Solutions:**
  - Added information stating that if the addition of a surfactant does not result in enhanced or reduced activity of the antimicrobial agent, a small amount of surfactant may be added to the stock solution
- **Subchapter 3.1.4, Number of Concentrations Tested:**
  - Added information about having QC organisms that are on scale and cover most minimal inhibitory concentration ranges included in the concentration being tested
- **Subchapter 3.2.2, Colony Suspension Methods for Inoculum Preparation:**
  - Revised colony suspension method for inoculum preparation
- **Subchapter 3.3, Agar Dilution Procedure:**
  - Added information explaining that data have not been reviewed for all recent antimicrobial agents
  - Added information stating that agar dilution is the only approved method for testing fosfomycin
- **Subchapter 3.5.1, Cation-Adjusted Mueller-Hinton Broth:**
  - Added iron-depleted cation-adjusted Mueller-Hinton broth as the recommended medium for antimicrobial susceptibility testing of commonly isolated, rapidly growing, aerobic or facultative organisms when cefiderocol is tested
- **Subchapter 3.5.2, Broth Media for Testing Fastidious Organisms:**
  - Added Mueller-Hinton fastidious broth to list of media that can be used for testing fastidious organisms
- **Subchapter 3.6.1, Preparing and Storing Diluted Antimicrobial Agents:**
  - Deleted sentence stating that tubes from broth macrodilution can be saved for later use only if they have been frozen
- **Subchapter 3.6.2, Inoculum Preparation, Inoculation, and Incubation:**
  - Added minimum final volume for broth macrodilution
- **Subchapter 3.7.1, Preparing and Storing Diluted Antimicrobial Agents:**
  - Added information stating that an automated instrument or liquid handling device can be used
  - Added information stating that the dilution scheme and volumes should be delivered according to the dispensing system
- **Subchapter 3.7.2, Inoculum Preparation, Inoculation, and Incubation:**
  - Added examples for determining appropriate colony-forming units for each microdilution well when the microdilution method is used
- **Subchapter 3.8.1, Performing Colony Counts From a 0.5 McFarland Suspension:**
  - Added a step-action table for performing colony counts from a 0.5 McFarland suspension
- **Subchapter 3.8.2, Performing Colony Counts From a Microdilution Well or Macrodilution Tube:**
  - Added new subchapter
- **Subchapter 3.9, Determining Broth Macro- or Microdilution End Points:**
  - Added information on interpreting results
  - Revised information about trailing growth and reading minimum inhibitory concentrations

- **Subchapter 3.11, Special Considerations for Fastidious Organisms:**
  - Added information on Mueller-Hinton fastidious broth for testing *Haemophilus influenzae*
- **Subchapter 3.12, Special Considerations for Detecting Resistance:**
  - Added information about *mecC* isolates to Subchapter 3.12.1.3.2, Reporting Oxacillin Resistance
  - Revised information on vancomycin-resistant *S. aureus* and vancomycin-intermediate *S. aureus* in Subchapter 3.12.1.4, Vancomycin Resistance in *S. aureus*
  - Revised information on vancomycin agar screen for *S. aureus* in Subchapter 3.12.1.4.2, Vancomycin Agar Screen for *S. aureus*
  - Revised information on AmpC enzymes in Subchapter 3.12.4.2, AmpC Enzymes
  - Revised information on carbapenemases in Subchapter 3.12.4.3, Carbapenemases (Carbapenem-Resistant Gram-Negative Bacilli)
  - Deleted former Subchapter 3.12.1.6, Linezolid Resistance
  - Renumbered former Table 2, Methods for Detecting Oxacillin Resistance in Staphylococci, as Table 6, Methods or Targets for Detection of Methicillin (Oxacillin)-Resistant *Staphylococcus* spp., and revised incubation times and methods
  - Renumbered former Table 3, Procedural Recommendations for Detecting Oxacillin Resistance in Staphylococci, as Table 7, Procedural Recommendations for Detection of Methicillin (Oxacillin)-Resistant *Staphylococcus* spp.
- **Subchapter 3.14.3, Development of Resistance and Testing Repeat Isolates:**
  - Revised information on development of resistance
- **Subchapter 4.3, Selecting Strains for Quality Control:**
  - Added recommendation for using additional or supplemental QC strains with end points covering as many drug concentrations on the panel as possible
- **Appendix A, Preparation of Supplements, Media, and Reagents:**
  - Added new section A3.3, Iron-Depleted Cation-Adjusted Mueller-Hinton Broth
  - Added new section A3.5, Mueller-Hinton Fastidious Broth
- **Appendix B, Conditions for Dilution Antimicrobial Susceptibility Tests:**
  - Table B1, Conditions for Dilution Antimicrobial Susceptibility Tests for Nonfastidious Organisms
    - Added iron-depleted cation-adjusted Mueller-Hinton broth (cefiderocol) as an acceptable medium for testing some nonfastidious organisms
    - Added water as an acceptable medium for preparing 0.5 McFarland inoculum for testing nonfastidious organisms
  - Table B2, Conditions for Dilution Antimicrobial Susceptibility Tests for Fastidious Organisms
    - Added Mueller-Hinton fastidious agar as an acceptable medium for testing for *H. influenzae* and *Streptococcus pneumoniae*

Summary of CLSI Processes for Establishing Breakpoints and Quality Control Ranges

The Clinical and Laboratory Standards Institute (CLSI) is an international, voluntary, not-for-profit, interdisciplinary, standards-developing, and educational organization accredited by the American National Standards Institute that develops and promotes the use of consensus-developed standards and guidelines within the health care community. These consensus standards and guidelines are developed in an open and consensus-seeking forum to cover critical areas of diagnostic testing and patient health care. CLSI is open to anyone or any organization that has an interest in diagnostic testing and patient care. Information about CLSI can be found at www.clsi.org.

The CLSI Subcommittee on Antimicrobial Susceptibility Testing reviews data from a variety of sources and studies (eg, *in vitro*, pharmacokinetics/pharmacodynamics, and clinical studies) to establish antimicrobial susceptibility test methods, breakpoints, and QC parameters. The details of the data necessary to establish breakpoints, QC parameters, and how the data are presented for evaluation are described in CLSI M23.[5]

Over time, a microorganism's susceptibility to an antimicrobial agent may decrease, resulting in a lack of clinical efficacy and/or safety. In addition, microbiological methods and QC parameters may be refined to ensure more accurate and better performance of susceptibility test methods. Because of these types of changes, CLSI continually monitors and updates information in its documents. Although CLSI standards and guidelines are developed using the most current information available at the time, the field of science and medicine is always changing; therefore, standards and guidelines should be used in conjunction with clinical judgment, current knowledge, and clinically relevant laboratory test results to guide patient treatment.

Additional information, updates, and changes in this standard are found in the meeting summary minutes of the CLSI Subcommittee on Antimicrobial Susceptibility Testing at https://clsi.org/meetings/ast-file-resources/.

CLSI Subcommittee on Antimicrobial Susceptibility Testing Mission Statement

The CLSI Subcommittee on Antimicrobial Susceptibility Testing is composed of representatives from the professions, government, and industry, including microbiology laboratories, government agencies, health care providers and educators, and pharmaceutical and diagnostic microbiology industries. Using the CLSI voluntary consensus process, the subcommittee develops standards that promote accurate antimicrobial susceptibility testing and appropriate reporting. The mission of the CLSI Subcommittee on Antimicrobial Susceptibility Testing is to:

- Develop standard reference methods for antimicrobial susceptibility tests.
- Provide quality control parameters for standard test methods.
- Establish breakpoints and interpretive categories for the results of standard antimicrobial susceptibility tests and provide epidemiological cutoff values when breakpoints are not available.
- Provide suggestions for testing and reporting strategies that are clinically relevant and cost-effective.
- Continually refine standards and optimize detection of emerging resistance mechanisms through development of new or revised methods, breakpoints, and QC parameters.
- Educate users through multimedia communication of standards and guidelines.
- Foster a dialogue with users of these methods and those who apply them.

The ultimate purpose of the subcommittee's mission is to provide useful information to enable laboratories to assist the clinician in the selection of appropriate antimicrobial therapy for patient care. The standards and guidelines are meant to be comprehensive and to include all antimicrobial agents for which the data meet established CLSI guidelines. The values that guide this mission are quality, accuracy, fairness, timeliness, teamwork, consensus, and trust.

NOTE: The content of this standard is supported by the CLSI consensus process and does not necessarily reflect the views of any single individual or organization.

KEY WORDS

agar dilution

antimicrobial susceptibility

broth dilution

broth macrodilution

broth microdilution

minimal inhibitory concentration

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Chapter 1

Introduction

Methods for Dilution Antimicrobial Susceptibility Tests for Bacteria That Grow Aerobically

1 Introduction

1.1 Scope

This standard describes standard broth (macrodilution and microdilution) and agar dilution methods for determining *in vitro* susceptibility to antimicrobial agents for bacteria that grow aerobically and includes:

- Broth and agar dilution test preparation
- Testing conditions, including inoculum preparation and standardization, incubation time, and incubation temperature
- Results interpretation
- QC procedures
- Dilution test method limitations

To assist the medical laboratory, recommendations are provided for selecting antimicrobial agents for routine testing and reporting.

Standards for testing the *in vitro* antimicrobial susceptibility of bacteria that grow aerobically using the antimicrobial disk testing method are found in CLSI M02.[3] Standards for testing the *in vitro* antimicrobial susceptibility of bacteria that grow anaerobically are found in CLSI M11.[4] Guidelines for standardized antimicrobial susceptibility testing (AST) of infrequently isolated or fastidious bacteria that are not included in CLSI M02,[3] M07, or M11[4] are available in CLSI M45.[6] The AST methods provided in this standard can be used in laboratories around the world, including but not limited to:

- Medical laboratories
- Public health laboratories
- Research laboratories
- Food laboratories
- Environmental laboratories

1.2 Background

Either broth or agar dilution methods may be used to quantitatively measure the *in vitro* activity of an antimicrobial agent against a given bacterial isolate. To perform the tests, a series of tubes or microtiter wells with a broth (for broth dilution) or agar medium (for agar dilution) are prepared to which various concentrations of the antimicrobial agents are added. The tubes or microtiter wells (for broth dilution) or plates (for agar dilution) are then inoculated with a standardized suspension of the test organism. After incubating for the appropriate time interval, the tests are read, the minimal inhibitory concentration (MIC) is determined, and the results are analyzed using approved breakpoints. The final result is significantly influenced by methodology, which must be carefully controlled if reproducible results (intra- and interlaboratory) are to be achieved.

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This standard describes reference broth dilution (macrodilution and microdilution) and agar dilution methods. The basic components of these methods are largely derived from information contained in published recommendations.[7] Although these methods are standard reference methods, some are sufficiently practical for routine use in medical or public health laboratories.

Commercial systems based primarily or in part on some of these methods are available and may provide results essentially equivalent to the CLSI methods described. CLSI does not approve or endorse commercial products or devices.

The methods described in this standard are intended primarily for testing commonly isolated aerobic or facultative bacteria that grow well after overnight incubation in unsupplemented Mueller-Hinton agar (MHA) or cation-adjusted Mueller-Hinton broth (CAMHB). Alternative media and methods for some fastidious or uncommon organisms are described in Subchapter 3.11 and CLSI M100[2] Tables 2. Methods for testing anaerobic bacteria are provided in CLSI M11[4] and in CLSI M100[2] Tables 2. Methods for testing infrequently isolated or fastidious bacteria not included in CLSI M02[3] and CLSI M07 are found in CLSI M45.[6]

This standard, along with CLSI M100,[2] describes methods, QC, breakpoints, and interpretive categories currently recommended for dilution susceptibility tests. When new problems are recognized or improvements in these criteria are developed, changes will be incorporated into future editions of this standard and CLSI M100.[2]

1.3 Standard Precautions

Because it is often impossible to know what isolates or specimens might be infectious, all patient and laboratory specimens are treated as infectious and handled according to "standard precautions." Standard precautions are guidelines that combine the major features of "universal precautions and body substance isolation" practices. Standard precautions cover the transmission of all known infectious agents and thus are more comprehensive than universal precautions, which are intended to apply only to transmission of bloodborne pathogens. Published guidelines are available that discuss the daily operations of diagnostic medicine in humans and animals while encouraging a culture of safety in the laboratory.[8] For specific precautions for preventing the laboratory transmission of all known infectious agents from laboratory instruments and materials and for recommendations for the management of exposure to all known infectious diseases, refer to CLSI M29.[9]

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1.4 Terminology

CLSI, as a global leader in standardization, is firmly committed to achieving global harmonization whenever possible. Harmonization is a process of recognizing, understanding, and explaining differences while taking steps to achieve worldwide uniformity. CLSI recognizes that medical conventions in the global metrological community have evolved differently in different countries and regions and that legally required use of terms, regional usage, and different consensus timelines are all important considerations in the harmonization process. CLSI recognizes its important role in these efforts, and its consensus process focuses on harmonization of terms to facilitate the global application of standards and guidelines. Table 1 is provided to clarify the intended interpretations of common terms.

Table 1. Common Terms or Phrases With Intended Interpretations

Term or Phrase	Intended Interpretation
"Needs to" or "must"	Explains an action directly related to fulfilling a regulatory and/or accreditation requirement or is indicative of a necessary step to ensure patient safety or proper fulfillment of a procedure
"Require"	Represents a statement that directly reflects a regulatory, accreditation, performance, product, or organizational requirement or a requirement or specification identified in an approved documentary standard
"Should"	Describes a recommendation provided in laboratory literature, a statement of good laboratory practice, or a suggestion for how to meet a requirement

1.4.1 Definitions

For purposes of CLSI M07, the terms and definitions listed below apply. Consult CLSI's Harmonized Terminology Database at https://htd.clsi.org for related terms and definitions.

antimicrobial susceptibility testing (AST) – procedures used to determine the susceptibility of bacteria to a specific antimicrobial agent(s); **NOTE:** There are two major testing types, disk diffusion and dilution (broth and agar).

breakpoint – minimal inhibitory concentration (MIC) or zone diameter value used to categorize an organism as susceptible, susceptible-dose dependent, intermediate, resistant, or nonsusceptible; **NOTE 1:** MIC or zone diameter values generated by a susceptibility test can be interpreted based on established breakpoints; **NOTE 2:** Because breakpoints are largely based on pharmacologically and clinically rich datasets using *in vitro* and *in vivo* data, they are considered robust predictors of likely clinical outcome; **NOTE 3:** Also known as "clinical breakpoint"; **NOTE 4:** See **interpretive category.**

Carba NP test – colorimetric microtube assay for detecting carbapenemase production.[10,11]

EDTA-modified carbapenem inactivation method (eCIM) – assay used to differentiate metallo-β-lactamases from serine carbapenemases in Enterobacterales isolates that are positive for the modified carbapenem inactivation method (mCIM).

epidemiological cutoff value (ECV) – the minimal inhibitory concentration (MIC) or zone diameter value that separates microbial populations into those with and without phenotypically detectable resistance (non-wild-type or wild-type, respectively). The ECV defines the highest MIC or smallest zone diameter for the wild-type population of isolates.

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EXAMPLE:

Category	Example of Categories for an ECV of 4 µg/mL and 19 mm	
	MIC, µg/mL	Zone Diameter, mm
Wild-type	≤ 4	≥ 20
Non-wild-type	≥ 8	≤ 19

- **wild-type** – an interpretive category defined by an ECV that describes the microbial population with no phenotypically detectable mechanisms of resistance or reduced susceptibility for the antimicrobial agent being evaluated.
- **non-wild-type** – an interpretive category defined by an ECV that describes the microbial population with phenotypically detectable mechanisms of resistance and reduced susceptibility for the antimicrobial agent being evaluated.

heteroresistance – presence of preexisting subpopulations of microbial cells exhibiting higher levels of resistance toward an antimicrobial agent while the majority of the cell population within the same culture is phenotypically susceptible to the same antimicrobial agent.

inducible clindamycin resistance – detection of the presence of a clindamycin resistance mechanism that is inducible (referred to as D-zone test when performed via disk diffusion).

interpretive category – category derived from microbiological characteristics, pharmacokinetic/pharmacodynamic parameters, and clinical outcome data, when available; **NOTE 1:** Minimal inhibitory concentration (MIC) or zone diameter values generated by a susceptibility test can be interpreted based on established breakpoints; **NOTE 2:** See **breakpoint.**

EXAMPLE:

Interpretive Category	Breakpoints	
	MIC, µg/mL	Zone Diameter, mm
Susceptible	≤ 4	≥ 20
Susceptible-dose dependent	8–16	15–19
Intermediate	8–16	15–19
Resistant	≥ 32	≤ 14
Nonsusceptible	> 1	< 17

MIC or zone diameter value breakpoints and interpretive categories are established per CLSI M23[5] for categories of susceptible, intermediate, and resistant (and susceptible-dose dependent and nonsusceptible, when appropriate). CLSI susceptible (S) or susceptible-dose dependent (SDD) breakpoints added or revised since 2010 have been based on specific dosage regimen(s); these dosage regimens are listed in CLSI M100[2] Table 2, Dosages. Antimicrobial Agent Dosage Regimens Used to Establish Susceptible or Susceptible-Dose Dependent Breakpoints (referred to as "CLSI M100[2] Table 2 Dosages").

- **susceptible (S)** – a category defined by a breakpoint that implies that isolates with an MIC at or below or a zone diameter at or above the susceptible breakpoint are inhibited by the usually achievable concentrations of antimicrobial agent when the dosage recommended to treat the site of infection is used, resulting in likely clinical efficacy.

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- **susceptible-dose dependent (SDD)** – a category defined by a breakpoint that implies that susceptibility of an isolate depends on the dosage regimen that is used in the patient. To achieve levels that are likely to be clinically effective against isolates for which the susceptibility testing results (either MICs or zone diameters) are in the SDD category, it is necessary to use a dosage regimen (ie, higher doses, more frequent doses, or both or extended infusion) that results in higher drug exposure than that achieved with the dose that was used to establish the susceptible breakpoint. Consideration should be given to the maximum, literature-supported dosage regimen, because higher exposure gives the highest probability of adequate coverage of an SDD isolate. CLSI M100[2] Table 2 Dosages lists the doses used when establishing SDD categories. The drug label should be consulted for recommended doses and adjustment for organ function; **NOTE:** The SDD category may be assigned when doses well above those used to calculate the susceptible breakpoint are supported by the literature, widely used clinically, and/or approved and for which sufficient data to justify the designation exist and have been reviewed. This category also includes a buffer zone for inherent variability in test methods, which should prevent small, uncontrolled, technical factors from causing major discrepancies in interpretations, especially for drugs with narrow pharmacotoxicity margins. See CLSI M100[2] appendixes for additional information.
- **intermediate (I)** – a category defined by a breakpoint that includes isolates with MICs or zone diameters within the intermediate range that approach usually attainable blood and tissue levels and/or for which response rates may be lower than for susceptible isolates; **NOTE:** An I with a ^ in CLSI M100[2] Tables 2 indicates agents that have the potential to concentrate in the urine. The I^ is for informational use only. The decision to report I^ is best made by each laboratory based on institution-specific guidelines and in consultation with appropriate medical personnel. The I category also includes a buffer zone for inherent variability in test methods, which should prevent small, uncontrolled, technical factors from causing major discrepancies in interpretations, especially for drugs with narrow pharmacotoxicity margins.
- **resistant (R)** – a category defined by a breakpoint that implies that isolates with an MIC at or above or a zone diameter at or below the resistant breakpoint are not inhibited by the usually achievable concentrations of the agent with normal dosage schedules and/or that demonstrate MICs or zone diameters that fall in the range in which specific microbial resistance mechanisms are likely, and clinical efficacy of the agent against the isolate has not been reliably shown in treatment studies.
- **nonsusceptible (NS)** – a category used for isolates for which only a susceptible breakpoint is designated because of the absence or rare occurrence of resistant strains. Isolates for which the antimicrobial agent MICs are above or the zone diameters are below the value indicated for the susceptible breakpoint should be reported as nonsusceptible; **NOTE 1:** An isolate that is interpreted as nonsusceptible does not necessarily mean that the isolate has a resistance mechanism. It is possible that isolates with MICs above the susceptible breakpoint that lack resistance mechanisms may be encountered within the wild-type distribution after the time the susceptible-only breakpoint was set; **NOTE 2:** The term "nonsusceptible" should not be used when the text is describing an organism/drug category with intermediate and resistant interpretive categories. Isolates that are in the categories of "intermediate" or "resistant" could be called "not susceptible" rather than "nonsusceptible."

minimal inhibitory concentration (MIC) – the lowest concentration of an antimicrobial agent that prevents visible growth of a microorganism in an agar or broth dilution susceptibility test.

modified carbapenem inactivation method (mCIM) – phenotypic assay for detecting carbapenemase production in which the ability of an isolate to degrade meropenem in a 10-µg disk is assessed.

on scale – indicates that minimal inhibitory concentrations to be tested include at least one dilution above and one dilution below the minimal inhibitory concentration end point.

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quality assurance (QA) – **1)** part of quality management focused on providing confidence that quality requirements will be fulfilled[12]; **2)** a comprehensive set of policies, procedures, and practices used to monitor the laboratory's entire testing process and ensure that the testing site's results are reliable; **NOTE:** The practice that encompasses all procedures and activities directed toward ensuring that a specified quality of product or service is achieved and maintained. In the testing environment, this process includes monitoring all the raw materials, supplies, instruments, and procedures; sample collection, transport, storage, and processing; recordkeeping; calibrating and maintaining equipment; quality control; proficiency testing; training of personnel; and all else involved in the production of the data reported.

quality control (QC) – part of quality management focused on fulfilling quality requirements[12]; **NOTE:** A system for ensuring maintenance of proper standards by periodic inspection of the results and the operational techniques that are used to ensure accuracy and reproducibility.

reference method – a thoroughly investigated method in which exact and clear descriptions of the necessary conditions and procedures are given for the accurate determination of one or more property values, and in which the documented trueness and precision of the method are commensurate with the method's use for assessing the trueness of other methods for measuring the same property values or for assigning reference method values to reference materials.

routine test – disk diffusion or broth or agar dilution minimal inhibitory concentration tests for routine clinical testing.

saline – a solution of 0.85% to 0.9% NaCl (w/v).

screening test – a test that provides presumptive results; **NOTE:** Additional testing is typically needed only for a specific result (eg, only when screening test result is positive).

supplemental (not routine) test – test that detects susceptibility or resistance to an antimicrobial agent or drug class by a method other than routine disk diffusion or broth or agar dilution and does not need additional tests to confirm susceptibility or resistance; **NOTE 1:** Some supplemental tests identify a specific resistance mechanism; **NOTE 2:** A supplemental test may be required or optional for reporting specific clinical results.

surrogate agent test – test performed with an agent that replaces a test performed with the antimicrobial agent of interest and is used when the agent of interest cannot be tested due to availability or performance issues (eg, surrogate agent performs better than the agent of interest).

test method – the method (ie, either the routine laboratory method or automated method) that is compared with the reference method.

test system – system that includes instructions and all the instrumentation, equipment, reagents, and/or supplies needed to perform an assay or examination and generate test results.

1.4.2 Abbreviations and Acronyms

AST	antimicrobial susceptibility testing
ATCC®[a]	American Type Culture Collection
BHI	brain heart infusion

[a] ATCC® is a registered trademark of the American Type Culture Collection.

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CAMHB cation-adjusted Mueller-Hinton broth
CFU colony-forming unit(s)
CSF cerebrospinal fluid
dMHB dehydrated Mueller-Hinton broth
eCIM EDTA-modified carbapenem inactivation method
EDTA ethylenediaminetetraacetic acid
ESBL extended-spectrum β-lactamase
FDA US Food and Drug Administration
HPLC high-performance liquid chromatography
HTM *Haemophilus* test medium
hVISA heteroresistant vancomycin-intermediate *Staphylococcus aureus*
LHB lysed horse blood
mCIM modified carbapenem inactivation method
MHA Mueller-Hinton agar
MH-F Mueller-Hinton fastidious
MIC minimal inhibitory concentration
MRS methicillin (oxacillin)-resistant staphylococci
MRSA methicillin (oxacillin)-resistant *Staphylococcus aureus*
PBP penicillin-binding protein
PBP2a penicillin-binding protein 2a
PBP2c penicillin-binding protein 2c
pH negative logarithm of hydrogen ion concentration
QA quality assurance
QC quality control
RNA ribonucleic acid
SDD susceptible-dose dependent
SOSA staphylococci other than *Staphylococcus aureus*
VRE vancomycin-resistant enterococci

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Chapter 2

Indications for Performing Antimicrobial Susceptibility Tests

2 Indications for Performing Antimicrobial Susceptibility Tests

AST is indicated for any organism that contributes to an infectious process warranting antimicrobial chemotherapy if its susceptibility cannot be reliably predicted from knowledge of the organism's identity (eg, intrinsic resistance). Antimicrobial susceptibility tests are most often indicated when the causative organism is thought to belong to a species capable of exhibiting resistance to commonly used antimicrobial agents. Resistance mechanisms may include:

- Production of drug-inactivating enzymes
- Alteration of drug targets
- Altered drug uptake or efflux

Some organisms have predictable susceptibility to antimicrobial agents, and empiric therapy for these organisms is widely accepted. Antimicrobial susceptibility tests are seldom necessary when the infection is caused by a microorganism recognized as susceptible to a highly effective drug (eg, the continued susceptibility of *Streptococcus pyogenes* to penicillin). Antimicrobial susceptibility tests are also important in epidemiological studies of resistance, in studies of new antimicrobial agents, and in antimicrobial resistance surveillance.

When the nature of the infection is not clear and the specimen contains mixed growth or normal flora (in which the organisms probably bear little relationship to the infectious process being treated), antimicrobial susceptibility tests are often unnecessary, and the results may be misleading.

Isolated colonies of each organism type that may be pathogenic should be selected from primary agar plates and tested individually for susceptibility. Identification procedures are often performed at the same time. Mixtures of different microorganism types should not be tested on the same susceptibility test.

An organism's MIC is often within a 3-dilution range (or a ± 1 doubling dilution) because of assay variation of the reference broth microdilution test. Even under the best controlled conditions, the isolate's inherent biological variability and other factors may lead to a range of MIC values similar to what is observed with replicate testing with QC strains. This variation can only be convincingly demonstrated with repeat testing. To avoid even greater variability, the dilution test must be standardized and strictly controlled as described herein.

Whenever MIC results are reported to clinicians to direct therapy, an interpretive category (ie, susceptible, susceptible-dose dependent [SDD], intermediate, resistant, or nonsusceptible) should accompany the MIC result based on the criteria outlined in CLSI M100[2] Tables 2. When tests in which four or fewer consecutive doubling dilutions are tested or when nonconsecutive doubling dilutions are tested, an interpretive category result, if available for the organism, should be reported. The MIC result may also be reported, if desired.

2.1 Selecting Antimicrobial Agents for Routine Testing and Reporting

To make routine AST relevant and practical, the antimicrobial agents tested should be limited. CLSI M100[2] Tables 1 list antimicrobial agents that fulfill the basic needs for routine use in most medical laboratories. The tables are based on specific organisms or organism groups, with the various antimicrobial agents indicated in tiers for testing to assist laboratories in selecting their routine testing batteries.

Selecting the most appropriate antimicrobial agents to test and report is a decision best made by each laboratory in consultation with the antimicrobial stewardship team and other relevant institutional stakeholders, as appropriate. Care should be taken to include the specific antimicrobial agents in the formulary as well as those that physicians prescribe daily.

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The suggestions for each organism group in CLSI M100[2] Tables 1 include agents of proven efficacy that show acceptable *in vitro* test performance. Considerations in the assignment of agents to specific tiers include:

- Clinical efficacy
- Prevalence of resistance
- Minimizing emergence of resistance
- US Food and Drug Administration (FDA) clinical indications for use
- Current consensus recommendations for first-choice and alternative drugs
- Cost

Tests on selected agents may be useful for infection-prevention purposes.

2.2 Equivalent Agents

Antimicrobial agents listed together in a single box in CLSI M100[2] Tables 1 are agents for which interpretive categories (susceptible, intermediate, SDD, nonsusceptible, or resistant) and clinical efficacy are similar. A laboratory will often test only one agent from a box routinely, typically the agent that is on its formulary. In some cases, a laboratory may not test any agents from a box, depending on institutional needs.

In some boxes, agents will be listed with an "or" between them. The "or" identifies agents for which cross-resistance and cross-susceptibility are nearly complete. Results from one agent connected by an "or" can be used to predict results for the other agent (ie, equivalent agents). For example, Enterobacterales susceptible to cefotaxime can be considered susceptible to ceftriaxone. The results obtained from testing cefotaxime could be reported along with a comment that the isolate is also susceptible to ceftriaxone. For drugs connected with an "or," combined major and very major errors are fewer than 3%, and minor errors are fewer than 10%, based on a large population of bacteria tested (see CLSI M23[5] for description of error types). In addition, to qualify for an "or," at least 100 strains with resistance to the agents in question must be tested, and a result of "resistant" must be obtained with all agents for at least 95% of the strains. "Or" is also used for comparable agents when tested against organisms for which "susceptible-only" breakpoints are provided (eg, cefotaxime or ceftriaxone with *Haemophilus influenzae*). When no "or" connects agents within a box, testing of one agent cannot be used to predict results for another, owing either to discrepancies or insufficient data (see CLSI M100[2] Instructions for Use of Tables).

2.3 Suggested Guidelines for Routine and Selective Testing and Reporting

The recommended antimicrobial agents are considered appropriate for testing and reporting following the recommendations in CLSI M100[2] Tables 1. The recommendations in CLSI M100[2] Tables 1 for each organism group include antimicrobial agents of proven efficacy that show acceptable *in vitro* test performance. Test and report Tiers 1, 2, 3, 4 and "urine only" are noted in CLSI M100[2] Tables 1. CLSI M100[2] Tables 2 contain two additional test and report tier designations, "other" and "investigational."

Antimicrobial agents other than those appropriate for use in therapy may also be tested to provide taxonomic data (eg, colistin for identification purposes) and epidemiological information (eg, testing streptomycin vs *Salmonella* by public health laboratories for surveillance purposes), but they should not be included on patient reports. However, such results should be available to the infection control practitioner and/or hospital epidemiologist.

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Each laboratory should decide which antimicrobial agents in CLSI M100[2] Tables 1 to routinely report (Tier 1) and which might be reported only selectively (Tiers 2 to 4). Each laboratory should consider developing these selective and/or cascade reporting rules in consultation with the antimicrobial stewardship team and other relevant institutional stakeholders, as appropriate. Selective and cascade reporting is done to encourage appropriate antimicrobial agent use. The positioning of drugs in tiers in CLSI M100[2] Tables 1 can be used to guide development of routine, selective, and/or cascade reporting rules. In addition, each laboratory should develop a protocol to test additional antimicrobial agents on isolates that are confirmed as resistant to all antimicrobial agents on its routine test panel. This protocol should include options for testing additional agents in-house or sending the isolate to a referral laboratory.

Selective reporting involves reporting results for specific antimicrobial agents based on defined criteria unrelated to results obtained from AST (eg, organism identification, body site, clinical setting, or patient demographics) (see CLSI M100[2] for examples).

Cascade reporting involves reporting results for specific agents based on the overall antimicrobial susceptibility profile of an isolate. Results for secondary or broader-spectrum agents (eg, Tier 2 or 3) are reported only if the isolate is resistant to primary or narrower-spectrum agents (eg, Tier 1) (see CLSI M100[2] for examples). A "resistant" result for a broader-spectrum agent (eg, Tier 2) should always be reported even if the organism tests "susceptible" to the narrower-spectrum agent (eg, Tier 1). Such unexpected resistant results should be confirmed (see CLSI M100[2] Tables 1 and CLSI M100[2] appendixes).

Cascade rules can be created for agents within the same tier or between tiers. Agents listed in the same row between tiers in CLSI M100[2] Tables 1 can be used as a guide for creating cascade reporting rules.

Table 2 describes the antimicrobial agent test and report tiers and additional considerations for agents listed in CLSI M100[2] Tables 1. Table 3 includes the test and report designations and additional considerations for agents not listed in CLSI M100[2] Tables 1.

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Table 2. Antimicrobial Agent Test and Report Tiers and Additional Considerations for Agents Listed in CLSI M100[2] Tables 1

Tier	Definition	Test	Report[a]	Additional Testing and Reporting Considerations
1	Antimicrobial agents that are appropriate for routine primary testing and reporting	Routine	Routine	
2	Antimicrobial agents that are appropriate for routine primary testing but may be reported following cascade reporting rules established at each institution	Routine	Cascade	• Report following cascade reporting rules because of resistance to agent(s) in Tier 1. • May be reported routinely based on institution-specific guidelines.
3	Antimicrobial agents that are appropriate for routine primary testing in institutions that serve patients at high risk for MDROs but should only be reported following cascade or selective reporting rules[b]	Routine or by request	Cascade	Test routinely based on institution-specific guidelines or by clinician request and report following cascade reporting rules because of resistance to agent(s) in Tiers 1 and 2.
4	Antimicrobial agents that may warrant testing and reporting by clinician request if antimicrobial agents in other tiers are not optimal because of various factors	By request	By request	• Test and report by clinician request because of: – Unavailability of preferred drug for clinical use – Patient underlying condition, including allergies – Unusual susceptibility profile of the organism, including resistance to agents in Tiers 1, 2, and 3 – Polymicrobial infection • May also warrant testing as an epidemiological aid
Urine only	Antimicrobial agents designated by a "(U)" in CLSI M100[2] Tables 2 should be reported only on organisms isolated from the urinary tract.	Routine	Report as appropriate	Agents in Tiers 1, 2, and 3 may also be reported on urine isolates, as appropriate, following the testing and reporting guidance for the respective tiers.

Abbreviations: MDRO, multidrug-resistant organism; UTI, urinary tract infection.
[a] Antimicrobial agents should be reported selectively, as appropriate (eg, because it is effective in treating uncomplicated UTIs only, nitrofurantoin would be reported only on isolates from urine).
[b] Identification of patients at high risk for MDROs will likely be communicated by infection preventionists. For examples of criteria used to identify patients at high risk for MDROs, see https://www.cdc.gov/hai/organisms/ESBL.html and https://www.cdc.gov/mrsa/community/index.html.

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Table 3. Antimicrobial Agent Test and Report Designations and Additional Considerations for Agents Not Listed in CLSI M100[2] Tables 1

Designation	Definition	Test	Report[a]	Additional Testing and Reporting Considerations
Other	Antimicrobial agents with established clinical breakpoints designated by * in CLSI M100[2] Tables 2 that are generally not candidates for testing and reporting in the United States	By request	By request	• Test and report only by clinician request and only following consultation with the antimicrobial stewardship team and other relevant institutional stakeholders to ensure appropriateness of the request. • Agents with an "other" designation may not reflect current consensus recommendations for first-choice and alternative drugs for the specific organism or organism group.
Inv.	Antimicrobial agents that are investigational for the organism group designated by "Inv." in CLSI M100[2] Tables 2 have not yet been approved by the FDA for use in the United States.	By request	By request	Test and report only by clinician request and only following consultation with the antimicrobial stewardship team and other relevant institutional stakeholders to ensure appropriateness of the request. These agents would likely be clinically available for compassionate use only.

Abbreviations: FDA, US Food and Drug Administration; inv, investigational.
[a] Antimicrobial agents should be reported selectively, as appropriate (eg, daptomycin should not be reported on isolates recovered from the respiratory tract because the agent interacts with pulmonary surfactant, resulting in inhibition of antibacterial activity).

2.4 Antimicrobial Classes

To minimize confusion, all antimicrobial agents should be reported using official nonproprietary (generic) names. The many currently available antimicrobial agents may be grouped together by drug classes, emphasizing their relatedness.

Table 4 provides brief descriptions of common antimicrobial agent classes, separated into β-lactams and non–β-lactams. For a complete list, see CLSI M100[2] glossaries.

Table 4. Antimicrobial Agent Class Characteristics

β-Lactams[a]
Description: • Share a common, central, 4-member β-lactam ring. • Majority bind to at least 2 PBPs. • Inhibit the transpeptidation step in bacterial cell wall synthesis. • Classification as a penicillin, β-lactam combination agent, cephem, monobactam, or penem determined by ring structures or substituent groups added to the β-lactam ring.

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Table 4. (Continued)

β-Lactams[a] (Continued)		
Class	**Characteristics**	**Subclasses**
Penicillins	• Primarily active against non–β-lactamase–producing gram-positive aerobes, some fastidious gram-negative aerobes, and some anaerobic bacteria. • Penicillinase-stable penicillins are not hydrolyzed by penicillinase-producing staphylococci.	• Penicillinase-labile penicillins, which are divided into subclasses: penicillin, aminopenicillins, carboxypenicillins, amdinocillin, and ureidopenicillins • Penicillinase-stable penicillins
β-Lactam combination agents	• β-Lactam class antimicrobial agent plus second agent that functions as an inhibitor of some β-lactamases, potentiating the activity of the β-lactam antimicrobial agent • Broad-spectrum activity against gram-positive and gram-negative bacteria • Organisms that test susceptible to the β-lactam agent alone are also considered susceptible to the β-lactam combination agent. However, organisms that test susceptible to the β-lactam combination agent cannot be assumed to be susceptible to the β-lactam agent alone. Similarly, organisms that test SDD, intermediate, or resistant to the β-lactam agent alone may be susceptible to the β-lactam combination agent.	
Cephems	• Spectrum of activity depends on subclass (gram-positive, gram-negative, anaerobes). • Generally bind with the greatest affinity to PBP3 in enteric bacteria in addition to PBP1 and/or PBP1b. • Not all agents within a specific group or generation have the same spectrum of activity. • Because of differing activity, representatives of each group may be selected for routine testing. • Cephems are also separated based on oral or parenteral delivery.	• Often referred to as first-, second-, third-, or fourth-generation cephalosporins – Third- and fourth-generation cephalosporins are oxyimino-cephalosporins and sometimes referred to as extended-spectrum cephems. • Cephalosporins with anti-MRSA activity: cephamycins, oxycephem, and siderophore cephalosporins
Monobactams	• Monocyclic β-lactams • High affinity to PBP3 in gram-negative aerobic bacteria • Not hydrolyzed by metallo-β-lactamases	
Penems	• Differ slightly in structure from penicillins. • More stable than penicillins and cephems against β-lactamase hydrolysis. • Broad-spectrum activity against many gram-positive and gram-negative bacteria. • Hydrolyzed by class A, B, and D carpenemases.	Carbapenems

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Table 4. (Continued)

Non–β-Lactams[a]		
Class	**Characteristics**	**Subclasses**
Aminoglycosides	• Interfere with protein synthesis by irreversibly binding to 16S ribosomal RNA within the 30S ribosomal subunit. • Variously affected by AMEs, resulting in some differences in the spectrum of activity. • Plazomicin is more stable against common AMEs; however, it is affected by 16 RNA methyl transferases. • Active only in aerobic conditions.	
Ansamycins	• Lipophilic macrocyclic antibiotic that easily diffuses through membranes • Aromatic moiety bridged by an aliphatic chain • Antibacterial action results from high-affinity binding to the DNA-dependent RNA polymerase of prokaryotes (RNAP) and inhibition of RNA synthesis. • Broad-spectrum activity against many gram-positive bacteria, gram-negative bacteria, and mycobacteria.	Rifamycins
DNA-gyrase and topoisomerase inhibitors	• Inhibit DNA-gyrase or topoisomerase activity. • Active against many gram-positive and gram-negative bacteria and some anaerobes and fastidious microorganisms. • Spectrum differences necessitate separate testing of the individual antimicrobial agents.	• Quinolones, fluoroquinolones, benzoquinolizine, spiropyrimidinetrione, and triazaacenaphthylenes • Spiropyrimidinetriones (eg, zoliflodacin) are active against *Neisseria gonorrhoeae*. Triazaacenaphthylenes (eg, gepotidacin) are active against pathogens causing uncomplicated UTI against *N. gonorrhoeae*. These drugs bind different targets and are less affected by common mechanisms of resistance.
Folate pathway antagonists	• Antimetabolic chemotherapeutic antimicrobial agents • Inhibit sequential steps in the folate pathway. • Active against some gram-positive and gram-negative bacteria. • Examples: – Trimethoprim: inhibits dihydrofolate reductase, which catalyzes the formation of tetrahydrofolate from dihydrofolate – Sulfamethoxazole: inhibits dihydropteroate synthase, which catalyzes the formation of dihydrofolate from *para*-aminobenzoic acid • Usually tested in combination: trimethoprim-sulfamethoxazole	Dihydrofolate reductase inhibitors and sulfonamides

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Table 4. (Continued)

Non–β-Lactams[a] (Continued)		
Class	**Characteristics**	**Subclasses**
Glycopeptides	• Linear, cross-linked heptapeptides with a variable number of sugar moieties; may have lipid substituents. • Block subunit cross-linking during cell wall formation by binding to the D-alanyl-D-alanine termini in the stem peptide of nascent peptidoglycan. • Used for treating infections caused by aerobic gram-positive bacteria, especially β-lactam–resistant, gram-positive bacteria (eg, MRSA and some enterococci). • Used for treating gram-positive infections in penicillin-allergic patients. • Used as an oral treatment for *Clostridioides* (formerly *Clostridium*) *difficile* infection (ie, vancomycin).	Glycopeptides, lipoglycopeptides, and lipoglycodepsipeptides
Lipopeptides	• Target the cell membrane. • May be linear or cyclic. • Strongly influenced by the presence of divalent cations in the medium used in testing. • Polymyxins: – Active against gram-negative organisms only. – Bind to the outer cell membrane, leading to membrane-permeability changes and cell death. – Bind to and neutralize lipopolysaccharide. – Excess calcium cations inhibit activity. • Lipopeptides: – Active against gram-positive organisms only. – Form a calcium complex and insert into the cell membrane. – Disrupt cell membrane function with potassium efflux and membrane depolarization. – Presence of physiological calcium ion levels (50 mg/L) essential for optimal activity.	Polymyxins

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Table 4. (Continued)

Non–β-Lactams[a] (Continued)		
Class	**Characteristics**	**Subclasses**
Macrocyclic lactone	Inhibit RNA synthesis by bacterial RNA polymerase at a site distinct from that of rifamycins, eg, fidaxomicin.	
Macrolides	• Inhibit bacterial protein synthesis by binding to the 23S RNA of the 50S ribosomal subunit. • Active against gram-positive and some gram-negative bacteria (eg, *Salmonella* and *Shigella* spp.). • Active against some fastidious bacteria (eg, *Haemophilus* spp.).	Ketolides and fluoroketolides
Nitroheterocyclics	• Nitrofurans: – Reduced intracellularly to a reactive metabolite that binds to macromolecules. – Used extensively to treat uncomplicated UTIs. • Nitroimidazoles: – Converted intracellularly in susceptible organisms to metabolites that bind to DNA and other cellular macromolecules. – Metronidazole is active only against strictly anaerobic bacteria and some protozoan parasites.	Nitrofuran, nitroimidazoles, and thiazolides
Oxazolidinones	• Inhibit protein synthesis • Binds to the 50S ribosome subunit, preventing peptide chain initiation • Active against gram-positive organisms and mycobacteria	
Phosphonates	• Inhibit bacterial cell wall by inactivating the enzyme UDP-N-acetylglucosamine-3-3-enolpryuvyltransferase (MurA). • Broad-spectrum activity against both gram-positive and gram-negative bacteria. • CLSI breakpoints for fosfomycin are established for *Escherichia coli* and *Enterococcus faecalis*. • Fosfomycin-trometamol is mainly used to treat uncomplicated UTIs.	Fosfomycin
Tetracyclines	• Inhibit protein synthesis by binding to the 30S ribosomal subunit and blocking binding of aminoacyl transfer RNA. • Active in certain gram-positive and gram-negative bacteria. – Glycylcyclines (tigecycline) and fluorocyclines (eravacycline) are also active against anaerobic bacteria. • Closely related group, and with few exceptions, only tetracycline might need to be tested routinely.	Fluorocycline, glycylcycline, and aminomethylcycline

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Table 4. (Continued)

Other Non–β-Lactam Drug Classes With One or Two Agents[a]		
Class	**Characteristics**	**Subclasses**
Protein synthesis inhibitors	Protein synthesis inhibition by several mechanisms like binding to the 50S ribosomal subunit and blocking the path by which nascent peptides exit the ribosome (eg, lincosamides) or binding to the 16S rRNA helix 34 of the 30S subunit of the bacterial ribosome, blocking the translocation step of protein synthesis (eg, spectinomycin)	• Aminocyclitols (ie, spectinomycin) • Lincosamides (ie, clindamycin) • Phenicols (ie, chloramphenicol) • Pleuromutilins (ie, lefamulin) • Pseudomonic acid (ie, mupirocin) • Quinolonyl oxazolidinone (ie, cadazolid) • Steroid (ie, fusidic acid) • Streptogramins (ie, quinupristin-dalfopristin)
Lysins	• Hydrolytic enzyme that cleaves the bacterial cell wall • Often bacterial species specific, eg, exebacase	
Peptide	Peptide antibiotics interfere with cell wall development by blocking the attachment of new cell wall subunits (eg, muramyl pentapeptides).	Magainin

Abbreviations: AME, aminoglycoside-inactivating enzyme; DNA, deoxyribonucleic acid; MRSA, methicillin (oxacillin)-resistant *Staphylococcus aureus*; PBP, penicillin-binding protein; RNA, ribonucleic acid; rRNA, ribosomal ribonucleic acid; SDD, susceptible-dose dependent; UTI, urinary tract infection.

[a] Refer to CLSI M100[2] glossaries.

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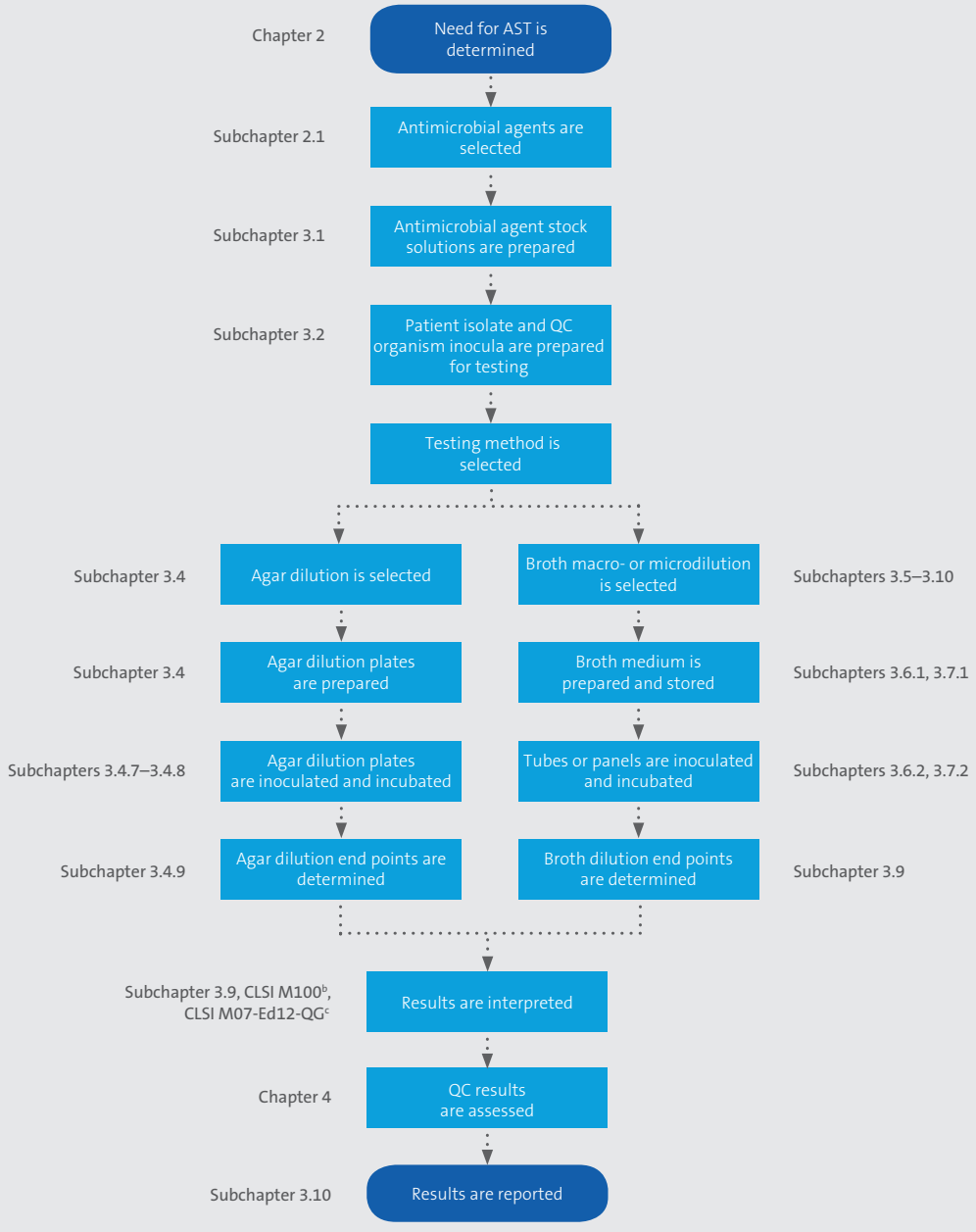
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Chapter 3

Broth and Agar Dilution Antimicrobial Susceptibility Testing Process

3 Broth and Agar Dilution Antimicrobial Susceptibility Testing Process

Figure 1 provides an overview of the disk diffusion AST process. Detailed information for each step is provided in each designated chapter or subchapter.



Abbreviations: AST, antimicrobial susceptibility testing; QC, quality control.
[a] Three basic symbols are used in this process flow chart: oval (signifies the beginning or end of a process), arrow (connects process activities), box (designates process activities).
[b] See CLSI M100.[2]
[c] See CLSI *M07 Minimal Inhibitory Concentration Reading Guide*.[13]

Figure 1. Broth and Agar Dilution AST Process[a]

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3.1 Antimicrobial Agents

3.1.1 Source

Antimicrobial standards or reference powders should be obtained directly from the drug manufacturer, the US Pharmacopeial Convention (http://www.usp.org), or appropriate commercial sources. Parenteral preparations should not be used for AST. Acceptable powders bear a label that states the drug's generic name, lot number, potency (usually expressed in µg or IU per mg of powder), and expiration date. Powders should be stored as recommended by the manufacturer or at ≤ –20°C in a desiccator (preferably in a vacuum). When the desiccator is removed from the refrigerator or freezer, it should be warmed to room temperature before opening to avoid water condensation.

3.1.2 Weighing Antimicrobial Powders

All antimicrobial agents are assayed for standard units of activity. The assay units may differ widely from the powder's actual weight and often may differ between drug production lots. Thus, a laboratory must standardize its antimicrobial solutions based on assays of antimicrobial powder lots that are used to make stock solutions.

The value for potency supplied by the manufacturer should include consideration for:

- Purity measures (usually by high-performance liquid chromatography [HPLC] assay)
- Water content (eg, by Karl Fischer titration or by weight loss on drying)
- The salt/counter-ion fraction (if the compound is supplied as a salt instead of free acid or base)

The potency may be expressed as a percentage or in units of µg/mg (w/w).

A certificate of analysis with values for each of these components may be provided with some antimicrobial agent powders. In this case, an overall potency value may not be provided but can be calculated from HPLC purity, water content, and when applicable, the active fraction for drugs supplied as a salt (eg, hydrochloride, mesylate). However, when these calculations are applied, if any value is unknown or not clearly determined from the certificate of analysis, it is advisable that the factors used in this calculation be confirmed with the supplier or the manufacturer. The following demonstrates an example calculation.

Example: meropenem trihydrate

Certificate of analysis data:

- Assay purity (by HPLC): 99.8%
- Measured water content (by Karl Fischer titration): 12.1% (w/w)
- Active fraction: 100% (supplied as the free acid and not a salt)

Potency calculation using data above:

- Potency = (assay purity) • (active fraction) • (1 – water content)
- Potency = (998) • (1.0) • (1 – 0.121)
- Potency = 877 µg/mg or 87.7%

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Either of the following formulas can be used to determine the amount of powder (1) or diluent (2) needed for a standard solution:

$$\text{Weight}\ (\text{mg}) = \frac{\text{Volume}\ (\text{mL}) \cdot \text{Concentration}(\mu\text{g/mL})}{\text{Assay potency}\ (\mu\text{g/mg})} \quad (1)$$

or

$$\text{Volume}\ (\text{mL}) = \frac{\text{Weight}\ (\text{mg}) \cdot \text{Assay potency}\ (\mu\text{g/mg})}{\text{Concentration}\ (\mu\text{g/mL})} \quad (2)$$

The antimicrobial agent powder should be weighed on an analytical balance that has been calibrated by approved reference weights from a national metrology organization. If possible, more than 10 mg of powder should be weighed. It is advisable to accurately weigh a portion of the antimicrobial agent in excess of that needed and to calculate the diluent volume needed to obtain the final concentration desired, as in formula (2) above.

Example: To prepare 100 mL of a stock solution containing 1280 µg/mL of antimicrobial agent with antimicrobial powder that has a potency of 750 µg/mg, 170 to 200 mg of the antimicrobial powder should be accurately weighed. If the actual weight is 182.6 mg, the diluent volume needed is:

$$\text{Volume}\ (\text{mL}) = \frac{(\text{Actual powder weight}) \cdot (\text{Assay potency})}{(\text{Desired concentration})}$$

$$\text{Volume}\ (\text{mL}) = \frac{182.6\ \text{mg} \cdot 750\ \mu\text{g/mg}}{1280\ \mu\text{g/mL}} = 107.0\ \text{mL} \quad (3)$$

Therefore, 182.6 mg of antimicrobial powder should be dissolved in 107.0 mL of diluent.

3.1.3 Preparing Stock Solutions

Antimicrobial agent stock solutions should be prepared at concentrations of at least 1000 µg/mL (eg, 1280 µg/mL) or 10 times the highest concentration to be tested, whichever is greater. However, some antimicrobial agents have limited solubility, and lower stock concentrations may need to be prepared. In all cases, directions provided by the drug's manufacturer as part of determining solubility should be considered. Some drugs must be dissolved in solvents other than water. In such cases:

- A minimum amount of solvent should be used to solubilize the antimicrobial powder.
- The final stock concentration dilution should be completed with water or the appropriate diluent as indicated in CLSI M100[2] Tables 6.
- For potentially toxic solvents, the safety data sheets available from the manufacturer should be consulted (see CLSI M100[2] Tables 6).

Additionally, if there are data to support that the addition of a surfactant does not result in enhanced or reduced activity of the antimicrobial agent, a small amount of surfactant may be added to the stock solution. Surfactant reduces surface tension and aids in even filling of the stock solution into secondary receptacles.

Because microbial contamination is extremely rare, solutions that have been prepared aseptically but not filter sterilized are generally acceptable. However, if desired, solutions may be sterilized by membrane filtration,

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typically with 0.45 µm pore size cellulose nitrate membrane filters. Paper, asbestos, or sintered glass filters, which may adsorb appreciable amounts of certain antimicrobial agents, must not be used. Whenever filtration is used, it is important to document the absence of adsorption by appropriate assay procedures.

Small volumes of the sterile stock solutions should be dispensed into sterile glass, polypropylene, polystyrene, or polyethylene vials, carefully sealed, and stored (preferably at –60°C or below but never at a temperature warmer than –20°C and never in a self-defrosting freezer). Vials may be thawed as needed and used the same day. Any unused stock solution should be discarded at the end of the day. Most antimicrobial agent stock solutions can be stored at –60°C or below for six months or more without significant loss of activity. In all cases, directions provided by the drug's manufacturer must be considered in addition to these general recommendations. Any significant antimicrobial agent deterioration should be reflected in the AST results from testing QC strains.

3.1.4 Number of Concentrations Tested

The concentrations tested for a particular antimicrobial agent should adequately cover the breakpoints provided in CLSI M100[2] Tables 2 and/or on the FDA Susceptibility Test Interpretive Criteria (STIC) website.[14] The number of concentrations tested is the laboratory's decision; however, it is advisable to choose a range that allows at least one QC organism to have on-scale values. It is ideal to include QC organisms that are on scale and cover most of the ranges of MICs included in the concentration being tested, although this is difficult to achieve for some drugs, particularly those with a susceptible-only breakpoint. Unusual concentrations may be tested for special purposes (eg, high concentrations of gentamicin and streptomycin may be tested to indicate a synergistic effect with a penicillin or glycopeptide against enterococci).

3.2 Preparing Inoculum for Dilution Susceptibility Tests

3.2.1 Turbidity Standard for Inoculum Preparation

To standardize the inoculum density for an antimicrobial susceptibility test, a $BaSO_4$ turbidity standard equivalent to a 0.5 McFarland standard or its optical equivalent (eg, latex particle suspension) should be used. A $BaSO_4$ 0.5 McFarland standard should be prepared as described in Appendix A4.1. Alternatively, a photometric device or nephelometer along with the manufacturers recommended standards can be used.

3.2.2 Colony Suspension Methods for Inoculum Preparation

The colony suspension method (formerly the direct colony suspension method) is the most convenient method for inoculum preparation. This method can be used with most organisms and is recommended for testing the fastidious organisms *H. influenzae, Haemophilus parainfluenzae, Neisseria gonorrhoeae, Neisseria meningitidis,* and streptococci (see Subchapter 3.11) and for testing staphylococci to detect methicillin (oxacillin) resistance. The steps for preparing the inoculum by the colony suspension method are listed in the following table.

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Step	Action	Comment
1	Make a direct suspension of well-isolated colonies of the same morphological type from an 18- to 24-hour agar plate. The suspension should be made in saline, sterile distilled water, tryptic soy broth, or other broth that is not antagonistic to the antimicrobial agents tested. For *N. gonorrhoeae*, prepare the suspension in MHB or 0.9% phosphate-buffered saline, pH 7.	Use a nonselective medium, such as blood agar. For *Haemophilus* spp., *N. gonorrhoeae*, and *N. meningitidis*, colonies should be picked from chocolate agar. For *Streptococcus* spp., use sheep blood agar.
2	Adjust the suspension to achieve a turbidity equivalent to a 0.5 McFarland standard. This step results in a suspension containing approximately 1–2 × 10^8 CFU/mL for *E. coli* ATCC® 25922. The inoculum should be used within 15 minutes.	Use a calibrated photometric device or nephelometer. If the suspension adjustment is performed visually, use adequate light to compare the inoculum tube and the 0.5 McFarland standard against a card with a white background and contrasting black lines (eg, Wickerham card[15]).

Abbreviations: ATCC®, American Type Culture Collection; CFU, colony-forming unit(s); MHB, Mueller-Hinton broth; pH, negative logarithm of hydrogen ion concentration.

3.2.3 Broth Culture Method for Inoculum Preparation

Alternatively, the broth culture method (formerly the growth method) can be used and is sometimes preferable when colony growth is difficult to suspend directly and a smooth suspension cannot be made. It can also be used for nonfastidious organisms (except staphylococci) when fresh (18- to 24-hour) colonies are not available. The steps for preparing the inoculum using the broth culture method are listed in the following table.

Step	Action	Comment
1	Select at least 3–5 well-isolated colonies of the same morphological type from an agar plate culture.	
2	Touch the top of each colony with a loop or sterile swab and transfer the growth into a tube containing 4–5 mL of a broth medium, such as tryptic soy broth or other broth that is not antagonistic to the antimicrobial agents tested.	
3	Incubate the broth culture at 35°C ± 2°C until it achieves or exceeds the turbidity of the 0.5 McFarland standard (usually 2–6 hours).	
4	Adjust the turbidity of the actively growing broth culture with sterile saline or broth to achieve a turbidity equivalent to that of a 0.5 McFarland standard. This step results in a suspension containing approximately 1–2 × 10^8 CFU/mL for *E. coli* ATCC® 25922. The inoculum should be used within 15 minutes.	• Use a calibrated photometric device or nephelometer. If the suspension adjustment is performed visually, use adequate light to compare the inoculum tube and the 0.5 McFarland standard against a card with a white background and contrasting black lines (eg, Wickerham card[15]). • **NOTE:** Avoid extremes in inoculum density that might arise from prolonged growth because this can affect organism viability. Never use undiluted overnight broth cultures or other unstandardized inocula for performing a dilution susceptibility test.

Abbreviations: ATCC®, American Type Culture Collection; CFU, colony-forming unit(s).

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3.3 Agar Dilution Procedure

The agar dilution method for determining antimicrobial susceptibility is well established.[16] However, for some recently introduced antimicrobial agents, the committee has not reviewed performance data for agar dilution. Agar dilution is the only approved method for testing fosfomycin. The antimicrobial agent is incorporated into the agar medium, with each plate containing a different concentration of the agent. The inocula can be applied rapidly and simultaneously to the agar surfaces using an inoculum-replicating apparatus[17] capable of transferring 32 to 36 inocula to each plate. The general procedure for performing agar dilution is provided in this subchapter. Variations for testing fastidious organisms are provided in Subchapter 3.11.

3.3.1 Test Media

3.3.1.1 Mueller-Hinton Agar

MHA is considered the best medium for routine AST of nonfastidious bacteria because:

- It shows acceptable batch-to-batch reproducibility for AST.
- It is low in inhibitors that affect sulfonamide, trimethoprim, and tetracycline susceptibility test results.
- It supports satisfactory growth of most nonfastidious pathogens.
- A substantial amount of data and experience has been collected about antimicrobial susceptibility tests performed with this medium.

Although MHA is generally reliable for AST, results obtained with some brands and batches may, on occasion, vary significantly. MHA should be prepared as described in Appendix A. Only MHA formulations that have been tested according to and that meet the acceptance limits as defined by the International Organization for Standardization[18] should be used. To ensure the reliability of the prepared media:

- Performance of new lots or new manufacture of medium must be tested before use as outlined in Subchapter 4.5.
- The pH of each MHA batch must be checked as described in Appendix A.
- Additions to MHA that must be made for testing specific antimicrobial agents are:
  - 2% NaCl when staphylococci and oxacillin are tested
  - 25 µg/mL glucose-6-phosphate when fosfomycin is tested
- Calcium or magnesium cations must not be added to MHA.
- MHA must be made fresh on the day that drug dilutions are added to agar plates when tigecycline and omadacycline are tested.

3.3.1.2 Agar Media for Testing Fastidious Organisms

Media for testing fastidious organisms using methods described in this standard are:

- MHA with 5% sheep blood
- GC agar base + 1% defined growth supplement

See Appendix A for instructions for preparing these media.

3.3.2 Materials: Inoculum Replicators

Most currently available inoculum replicators transfer 32 to 36 inocula to each plate.[1] Replicators with pins 3 mm in diameter deliver approximately 2 µL (range, 1 to 3 µL) onto the agar surface. Those with smaller 1-mm pins deliver 10-fold less, approximately 0.1 to 0.2 µL.[19]

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3.4 Preparing Agar Dilution Plates

3.4.1 Dilution Scheme for the Reference Method

Intermediate (10×) antimicrobial agent solutions are prepared by making successive 1:2, 1:4, and 1:8 dilutions using the dilution format described in CLSI M100[2] Table 7 or by making serial twofold dilutions. Next, one part of the 10× antimicrobial solution is added to nine parts of molten agar. At least two plates with no antimicrobial solution should be prepared as growth control and purity plates.

3.4.2 Procedure

The steps for preparing the agar medium are listed in the following table.

Step	Action	Comment
1	Add appropriate dilutions (see Subchapter 3.2) of the antimicrobial solution to molten test agars that have been equilibrated in a water bath to 45 to 50°C. Add sterile water to tubes for growth-control plates.	
2	Mix the tubes thoroughly and pour the mixture into Petri plates on a level surface to produce an agar depth of 3–4 mm.	• Pour the plates quickly after mixing to prevent cooling and partial solidification in the mixing container. • Avoid generating bubbles by mixing gently.
3	Let the agar solidify at room temperature, and either use the plates immediately or store them in sealed plastic bags at 2 to 8°C for up to 5 days or longer based on the drug's stability and QC results.	• In one study, agar plates containing cefaclor had to be prepared within 48 hours of use because of drug degradation, whereas cefamandole remained stable for longer than the recommended 5 days.[20] In addition to cefaclor, other antimicrobial agents that are particularly labile are ampicillin, clavulanate, and imipenem. • **NOTE:** Do not assume all antimicrobial agents will maintain their potency under these storage conditions. The user should evaluate plate stability from results obtained with QC strains and should develop applicable shelf-life criteria. This information is sometimes available from pharmaceutical manufacturers.
4	Let plates stored at 2 to 8°C equilibrate to room temperature before use. Ensure that the agar surface is dry before inoculating the plates.	If necessary, place the plates in an incubator or laminar flow hood for approximately 30 minutes with the lids ajar to hasten drying of the agar surface. Do not overdry the plates.

Abbreviation: QC, quality control.

3.4.3 Quality Control Frequency

Although weekly QC may be acceptable for most drug and agar combinations, some may need more frequent testing (eg, the labile drugs described in Subchapter 3.4.2 [step 3]). See Chapter 4 for additional details.

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3.4.4 Control Plates

Drug-free plates should be prepared from the base medium (with or without supplements as indicated in Subchapter 3.3.1) and used as growth controls.

3.4.5 Preparing the Inoculum

A standardized inoculum for the agar dilution method can be prepared by either growing microorganisms to the turbidity of the 0.5 McFarland standard or by directly suspending colonies to achieve the same density as described in Subchapter 3.2. Preparing the initial inoculum and final dilution may vary for certain fastidious organisms, such as *N. meningitidis* (see Subchapter 3.11 and Appendix B).

3.4.6 Diluting the Inoculum Suspension

Cultures adjusted to the 0.5 McFarland standard contain approximately 1 to 2 × 10^8 colony-forming unit(s) (CFU)/mL with most species, and the final inoculum needed is 10^4 CFU per 5- to 8-mm diameter spot. When using replicators with 3-mm pins that deliver 2 µL (range, 1 to 3 µL), the 0.5 McFarland suspension should be diluted 1:10 in sterile broth or saline to obtain a concentration of 10^7 CFU/mL. The final inoculum on the agar will be approximately 10^4 CFU per spot. When using replicators with 1-mm pins that deliver 0.1 to 0.2 µL, the initial suspension must not be diluted. Optimally, the adjusted suspensions for final inoculation should be used within 15 minutes of preparation.

3.4.7 Inoculating Agar Dilution Plates

The steps for inoculating agar dilution plates are listed in the following table.

Step	Action	Comment
1	Arrange the tubes containing the adjusted bacterial suspensions in order in a rack.	
2	Place an aliquot of each well-mixed suspension into the corresponding well in the replicator inoculum block.	An appropriate dilution of the inoculum suspension should be made depending on the volume of inoculum delivery so as to obtain a final concentration of 10^4 CFU per spot (see Subchapter 3.4.6).
3	Mark the agar plates for orientation of the inoculum spots.	
4	Apply an aliquot of each inoculum to the agar surface, either with an inocula-replicating device or with standardized loops or pipettes.	
5	Inoculate a growth-control plate (no antimicrobial agent) first and then, starting with the lowest concentration, inoculate the plates containing the different antimicrobial concentrations. Inoculate a second growth-control plate last to ensure there was no contamination or significant antimicrobial carryover during the inoculation.	
6	Streak a sample of each inoculum on a suitable nonselective agar plate and incubate overnight.	This is done to detect mixed cultures and provide freshly isolated colonies in case retesting is needed.

Abbreviation: CFU, colony-forming unit(s).

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3.4.8 Incubating Agar Dilution Plates

The steps for incubating the plates are listed in the following table.

Step	Action	Comment
1	Let the inoculated plates stand at room temperature until the moisture in the inoculum spots has been absorbed into the agar (ie, until the spots are dry).	Let the plates stand for no more than 30 minutes.
2	Invert the plates and incubate at 35°C ± 2°C for 16–20 hours.	• Do not incubate the plates in an atmosphere with increased CO_2 when nonfastidious organisms are tested, because the surface pH may be altered. However, incubate *N. gonorrhoeae* and *N. meningitidis* in an atmosphere containing 5% CO_2. Streptococci may also be incubated in 5% CO_2, if necessary for growth. • See Subchapter 3.11 and Appendix B for exceptions. • Also refer to CLSI M100[2] Tables 2.

Abbreviation: pH, negative logarithm of hydrogen ion concentration.

3.4.9 Determining Agar Dilution End Points

The steps for determining agar dilution end points are listed in the following table.

Step	Action	Comment
1	Read the MICs by placing the plates on a dark, nonreflecting surface to determine the end points.	For the test to be considered valid, there must be growth on the growth-control plate. This is evidenced by confluent growth in the area where the inoculum was spotted on the agar.
2	Record the MIC as the lowest concentration of antimicrobial agent that completely inhibits growth.	• With trimethoprim and the sulfonamides, antagonists in the medium may allow some slight growth; therefore, read the end point at the concentration in which there is 80% or greater reduction in growth compared with the control. • If two or more colonies persist in concentrations of the agent beyond an obvious end point or if there is no growth at lower concentrations but growth at higher concentrations, check the culture purity and repeat the test if needed. • Disregard single colonies or a faint haze caused by the inoculum.
3	Interpret the results.	Refer to the appropriate portion of CLSI M100[2] Tables 2 for breakpoints.

Abbreviation: MIC, minimal inhibitory concentration.

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3.5 Broth Dilution Procedures (Macrodilution and Microdilution)

The general procedures for performing broth dilution are provided in this subchapter. Variations for testing fastidious organisms are provided in Subchapter 3.11.

3.5.1 Cation-Adjusted Mueller-Hinton Broth

CAMHB is the recommended medium for AST of commonly isolated, rapidly growing, aerobic or facultative organisms because:

- It shows acceptable batch-to-batch reproducibility for AST.
- It is low in inhibitors that affect sulfonamide, trimethoprim, and tetracycline susceptibility test results.
- It supports satisfactory growth of most pathogens.
- A substantial amount of data and experience has been collected about antimicrobial susceptibility tests performed with this medium.

Instructions for preparing the media are provided in Appendix A. To ensure the reliability of the prepared media:

- The pH of each CAMHB batch must be checked as described in Appendix A.
- The MIC performance characteristics of each broth batch must be evaluated using a standard set of QC organisms (see Subchapter 4.3). If a new CAMHB lot does not yield the expected MICs, the cation content should be investigated along with other variables and components of the test.
- To determine the suitability of the medium for sulfonamide and trimethoprim tests, each batch of broth should be tested using *Enterococcus faecalis* ATCC® 29212. The end points should be easy to read (as 80% or greater reduction in growth compared with the control). If the MIC for trimethoprim-sulfamethoxazole is ≤ 0.5/9.5 µg/mL, the medium may be considered adequate.
- Additions or changes to CAMHB that must be made for testing specific antimicrobial agents are:
  - 2% NaCl when oxacillin and staphylococci are tested
  - 0.002% polysorbate-80 when dalbavancin, oritavancin, and telavancin are tested
  - Additional calcium to equal 50 mg/L when daptomycin is tested
  - Iron-depleted CAMHB supplemented with calcium (20-25 mg/L), magnesium (10-12.5 mg/L), and zinc (0.5-1.0 mg/L) when cefiderocol is tested. Refer to CLSI M100.[2]
- CAMHB must be made fresh on the day that drug dilutions are added to broth when tigecycline and omadacycline are tested.

3.5.2 Broth Media for Testing Fastidious Organisms

Media that can be used for testing fastidious organisms using methods described in this standard are:

- CAMHB + 2.5% to 5% lysed horse blood (LHB)
- *Haemophilus* test medium (HTM) broth
- Mueller-Hinton fastidious (MH-F) broth

Instructions for preparing these media are provided in Appendix A.

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3.6 Broth Macrodilution (Tube) Method

3.6.1 Preparing and Storing Diluted Antimicrobial Agents

Sterile 13 × 100-mm test tubes with screw caps, plastic or metal closures, or cotton plugs should be used to conduct the test. A growth-control tube containing broth without antimicrobial agent should be used for each organism tested.

The steps for preparing and storing diluted antimicrobial agents for the broth macrodilution method are listed in the following table.

Step	Action	Comment
1	Prepare the final twofold (or other) dilutions of antimicrobial agent volumetrically in the broth using one pipette for measuring all diluents and then for adding the stock antimicrobial solution to the first tube. For each subsequent dilution step, use of a new pipette tip is recommended.	• A minimum final volume of 1 mL of each dilution is needed for the test. • The scheme in CLSI M100[2] Tables 8 provides a convenient and reliable procedure for preparing dilutions. • Because there is a 1:2 dilution of the drugs when an equal volume of inoculum is added, the antimicrobial dilutions are often prepared at double the desired final concentration.
2	Use the tubes on the day of preparation or immediately place them in a freezer at ≤ –20°C (preferably at ≤ –60°C) until needed.	• Although the antimicrobial agents in frozen tubes usually remain stable for several months, certain agents (eg, clavulanate, imipenem) are more labile than others and should be stored at ≤ –60°C. • Do not store the tubes in a self-defrosting freezer. Thawed antimicrobial solutions must not be refrozen, because repeated freeze-thaw cycles accelerate the degradation of some antimicrobial agents, particularly β-lactams.

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3.6.2 Inoculum Preparation, Inoculation, and Incubation

The steps for preparing the inoculum and inoculating and incubating the tubes are listed in the following table.

Step	Action	Comment
1	Prepare a standardized inoculum. A minimum final volume of 1 mL per dilution tested is needed.	Use either the colony suspension or broth culture method as described in Subchapter 3.2.3.
2	Within 15 minutes of preparation, dilute the adjusted inoculum suspension in broth.	This step can be accomplished by diluting the 0.5 McFarland suspension 1:150, resulting in a tube containing approximately 1×10^6 CFU/mL. The subsequent 1:2 dilution in step 3 brings the final inoculum to 5×10^5 CFU/mL (range, 2–8 $\times 10^5$ CFU/mL).
3	Within 15 minutes after standardizing the inoculum as described in step 2, add 1 mL adjusted inoculum to each tube containing 1 mL antimicrobial agent in the dilution series (and a positive control tube containing only broth) and mix.	• This step results in a 1:2 dilution of each antimicrobial concentration and a 1:2 dilution of the inoculum. • It is advisable to perform a purity check on the inoculum suspension by subculturing an aliquot onto a nonselective agar plate for simultaneous incubation.
4	Cap the tubes.	
5	Incubate the inoculated macrodilution tubes at 35°C ± 2°C for 16–20 hours in an ambient air incubator (for exceptions, see Subchapter 3.11 and Appendix B) within 15 minutes of adding the inoculum.	Incubation times may differ for fastidious organisms or some problem organisms with difficult-to-detect resistance. Follow specific instructions in Subchapter 3.11, Appendix B, or in the appropriate CLSI M100[2] table.

Abbreviation: CFU, colony-forming unit(s).

3.7 Broth Microdilution Method

This method is called “microdilution” because it involves using small volumes of broth dispensed in sterile, plastic microdilution panels that have round- or conical-bottom wells. Each well should contain a final volume of 0.1 mL of broth. The broth microdilution method described in CLSI M07 is the same methodology outlined in published international standards.[18]

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3.7.1 Preparing and Storing Diluted Antimicrobial Agents

The steps for preparing and storing antimicrobial agent solutions for the broth microdilution method are listed in the following table.

Step	Action	Comment
1	Prepare the microdilution panels by making intermediate twofold (or other) dilutions of antimicrobial agent volumetrically in broth or sterile water. Use 1 pipette for measuring all diluents and for adding the stock antimicrobial solution to the first tube. For each subsequent dilution, use of a new pipette tip is recommended.	• For the intermediate (10×) antimicrobial solutions, dilute the concentrated antimicrobial stock solution (see Subchapter 3.1.3) as described in CLSI M100[2] Tables 8 or by making serial twofold dilutions.
2	Dispense the antimicrobial agent/broth solutions into the plastic microdilution panels using a dispensing device, pipette, or automated liquid handling instrument.	• The most convenient method for preparing microdilution panels is by using a dispensing device and antimicrobial dilutions made in at least 10 mL of broth. The dispensing device then delivers 0.1 (± 0.02) mL into each of the 96 wells of a standard MIC panel. • If the inoculum is to be added by pipette (as described in Subchapter 3.7.2), prepare the antimicrobial agent solutions at twice the desired final concentration and fill the wells with 0.05 mL instead of 0.1 mL. • Alternatively, an automated instrument or liquid handling device can be used to prepare the antimicrobial dilutions and dispense the antimicrobial agent solutions into the 96-well microdilution panels. Prepare the dilution scheme as determined by the volumes delivered according to the dispensing system. • Each MIC panel should include a growth-control well for each growth medium tested and a sterility (uninoculated) well.
3	Seal the filled panels in plastic bags and immediately place in a freezer at ≤ –20°C (preferably at ≤ –60°C) until needed.	• Although the antimicrobial agents in frozen panels usually remain stable for several months, certain agents (eg, clavulanate, imipenem) are more labile than others and should be stored at ≤ –60°C. • Do not store the MIC panels in a self-defrosting freezer. Thawed antimicrobial solutions must not be refrozen because repeated freeze-thaw cycles accelerate the degradation of some antimicrobial agents, particularly β-lactams.

Abbreviation: MIC, minimal inhibitory concentration.

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3.7.2 Inoculum Preparation, Inoculation, and Incubation

The steps for preparing the inoculum and inoculating and incubating the plates are listed in the following table.

Step	Action	Comment
1	Let the microdilution panels thaw completely at ambient room temperature.	• Thawed MIC panels must be inoculated and incubated within 4 hours from the time they are removed from the freezer. • Do not refreeze the panels.
2	Prepare a standardized inoculum using either the colony suspension or broth culture method.	See Subchapter 3.2.
3	Within 15 minutes of preparation, dilute the adjusted inoculum suspension in water, saline, or broth.	• Each well should contain approximately 5×10^5 CFU/mL (range, 2–8 × 10^5 CFU/mL) when inoculated. • The dilution procedure to obtain this final inoculum varies according to the method for delivering the inoculum to the individual wells and must be calculated for each situation. • For microdilution tests, the exact inoculum volume delivered to the wells must be known to make this calculation. • Example 1: When the volume of broth in the well is 0.1 mL and the inoculum volume is 0.01 mL, the 0.5 McFarland suspension (1–2 × 10^8 CFU/mL) should be diluted 1:20 to yield approximately 5×10^6 CFU/mL. When 0.01 mL of this suspension is inoculated into the broth microdilution panel, the final test bacterial concentration is approximately 5×10^5 CFU/mL. A colony count should be performed directly from a microdilution well to ensure accuracy of the organism concentration. See Subchapter 3.8.2. • Example 2: When the volume of broth in the well is 0.05 mL and the inoculum volume is 0.05 mL, the 0.5 McFarland suspension (1–2 × 10^8 CFU/mL) should be diluted 1:100 to yield approximately 1×10^6 CFU/mL. When 0.05 mL of this suspension is inoculated into the broth microdilution panel, the final test bacterial concentration is approximately 5×10^5 CFU/mL. A colony count should be performed directly from a microdilution well to ensure accuracy of the organism concentration. See Subchapter 3.8.2.

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Step	Action	Comment
4	Within 15 minutes after the inoculum has been diluted (see step 3), inoculate each well of a microdilution panel, using either an inoculator device (ie, an inoculator), pipette, or automated liquid handling instrument.	• If an inoculator is used, it should deliver a volume that does not exceed 10% of the volume in the well (eg, ≤ 10 µL of inoculum in 0.1 mL antimicrobial agent solution). • If a 0.05-mL pipette is used, the result will be a 1:2 dilution of the contents of each well (containing 0.05 mL), as with the macrodilution procedure. • It is advisable to perform a purity check of the inoculum suspension by subculturing an aliquot onto a nonselective agar plate for simultaneous incubation.
5	To prevent drying, seal each microdilution panel or stack of 4 MIC panels in a plastic bag, with plastic tape, or with a tight-fitting plastic cover before incubating.	To maintain the same incubation temperature for all cultures, do not stack microdilution panels more than 4 high (if using panel covers, place 1 on the top of the stack only).
6	Incubate the inoculated microdilution panels at 35°C ± 2°C for the appropriate length of time (generally 16–20 hours) in an ambient air incubator within 15 minutes of adding the inoculum.	Incubation times may differ for fastidious organisms, some non-fermenters, and some problem organisms with difficult-to-detect resistance (eg, MRSA). For exceptions, follow specific instructions in Subchapters 3.11 and 3.12, Appendix B, or in the appropriate CLSI M100[2] table.

Abbreviations: CFU, colony-forming unit(s); MIC, minimal inhibitory concentration; MRSA, methicillin (oxacillin)-resistant *Staphylococcus aureus*.

3.8 Inoculum Suspension Colony Counts

Laboratories are encouraged, although not required, to perform colony counts on inoculum suspensions at least quarterly. Laboratories are also encouraged to sample the growth control for each isolate tested to ensure purity and sufficient inoculum and provide fresh colonies for repeat or additional testing if needed, without delay.

3.8.1 Performing Colony Counts From a 0.5 McFarland Suspension

The concentration of organisms when taken directly from a 0.5 McFarland suspension should closely approximate 1.5×10^8 CFU/mL for *E. coli* ATCC® 25922.

The steps for performing colony counts are listed in the following table.

Step	Action	Comment
1	Remove a 0.01-mL aliquot from the 0.5 McFarland suspension and dilute it in 10 mL saline.	This 1:1000 dilution results in a 10^5 CFU/mL suspension.
2	Mix thoroughly.	
3	Remove 0.1 mL of the suspension created in step 1 from the tube and dilute it in 10 mL saline.	This 1:100 dilution results in a 10^3 CFU/mL suspension.
4	Mix thoroughly.	
5	Spread a 0.1-mL aliquot over the surface of a suitable agar medium.	This 1:10 dilution results in a 10^2 CFU/mL suspension or approximately 150 colonies per plate.
6	Incubate the media.	After incubation, the presence of approximately 150 colonies indicates an inoculum density of 1.5×10^8 CFU/mL or $1–2 \times 10^8$ CFU/mL.

Abbreviation: CFU, colony-forming unit(s).

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3.8.2 Performing Colony Counts From a Microdilution Well or Macrodilution Tube

The concentration of organism when taken directly from a microdilution well or macrodilution tube should closely approximate 5×10^5 CFU/mL for *E. coli* ATCC® 25922.

The steps for performing colony counts are shown in the following table.

Step	Action	Comment
1	Remove a 0.01-mL aliquot from the growth-control well or tube immediately after inoculating and dilute it in 10 mL saline.	This 1:1000 dilution results in a 10^2 CFU/mL suspension.
2	Mix thoroughly.	
3	Spread 0.1 mL aliquot over the surface of a suitable agar medium.	This 1:10 dilution results in an approximately 5×10^1 CFU/mL suspension or approximately 50 colonies per plate.
4	Incubate the media.	After incubation, the presence of approximately 50 colonies indicates an inoculum density of 5×10^5 CFU/mL.[a]

Abbreviation: CFU, colony-forming unit(s).
[a] Based on the dilution scheme, the colony count range is 20–100; however, colony counts of 30–40 are most frequently observed.

3.9 Determining Broth Macro- or Microdilution End Points

The steps for reading and interpreting broth macro- or microdilution end points are listed in the following table. For additional figures, refer to the CLSI M07 *Minimal Inhibitory Concentration Reading Guide*.[13]

Step	Action	Comment
1	Read the MIC as the lowest concentration of antimicrobial agent that completely inhibits organism growth in the tubes or microdilution wells as detected by the unaided eye.	• See step 2 for exceptions. • Viewing devices intended to facilitate reading microdilution tests and recording results may be used as long as there is no compromise in the ability to discern growth in the wells.

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Step	Action	Comment
2	Compare the amount of growth in the wells or tubes containing the antimicrobial agent with the amount of growth in the growth-control wells or tubes (no antimicrobial agent) used in each set of tests when the growth end points are determined.	For a test to be considered valid, acceptable growth (definite turbidity and/or button of growth) must occur in the growth-control well (see Figure 2). Generally, microdilution MICs for gram-negative bacilli are the same or one twofold dilution lower than the comparable macrodilution MICs. Exceptions to reading complete growth inhibition include: • Trailing growth can occur and make end-point determination difficult for all bacteria with certain classes of antimicrobial agents (eg, lincosamides, tetracyclines, macrolides, phenicols, oxazolidinones, streptogramins, fusidic acid). Read the end point at the first well where trailing in growth occurs. Tiny buttons of growth should be ignored (see Figures 3 and 4). • Trailing growth can be seen with gram-positive cocci when chloramphenicol, clindamycin, macrolides (eg, erythromycin, azithromycin, clarithromycin), oxazolidinones (eg, linezolid, tedizolid), tetracyclines (eg, tetracycline, doxycycline, minocycline), and streptogramins (eg, quinupristin/dalfopristin) are tested. • Trailing growth can be seen with gram-negative rods when tetracyclines (eg, tetracycline, tigecycline, minocycline), chloramphenicol, macrolides (eg, azithromycin), and cefiderocol are tested. • With trimethoprim and the sulfonamides, antagonists in the medium may allow some slight growth; therefore, read the end point at the concentration in which there is 80% or greater reduction in growth compared with the control (see Figure 5). • When a single skipped well occurs in a microdilution test, read the MIC as described in step 1. If skipped wells are observed across multiple concentrations of a single drug, do not report the results of that drug and repeat the test (see Figure 6). • If MIC end points are difficult to determine, consult other laboratorians. If a consensus result cannot be determined, repeat the test and, if possible, perform AST by an alternative method particularly if critical to patient care (eg, MIC is near the breakpoint).

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Step	Action	Comment
3	Interpret the results.	The MIC is read as the lowest concentration of antimicrobial agent that inhibits organism growth as described above. If there is no growth in all the concentrations of antimicrobial agent tested, including the lowest concentration, the MIC is reported as less than or equal to the lowest concentration tested. If there is growth in all the concentrations of antimicrobial agent tested, including the highest concentration, the MIC is reported as greater than the highest concentration tested. Refer to the appropriate portion of CLSI M100[2] Tables 2 for breakpoints.

Abbreviations: AST, antimicrobial susceptibility testing; MIC, minimal inhibitory concentration.

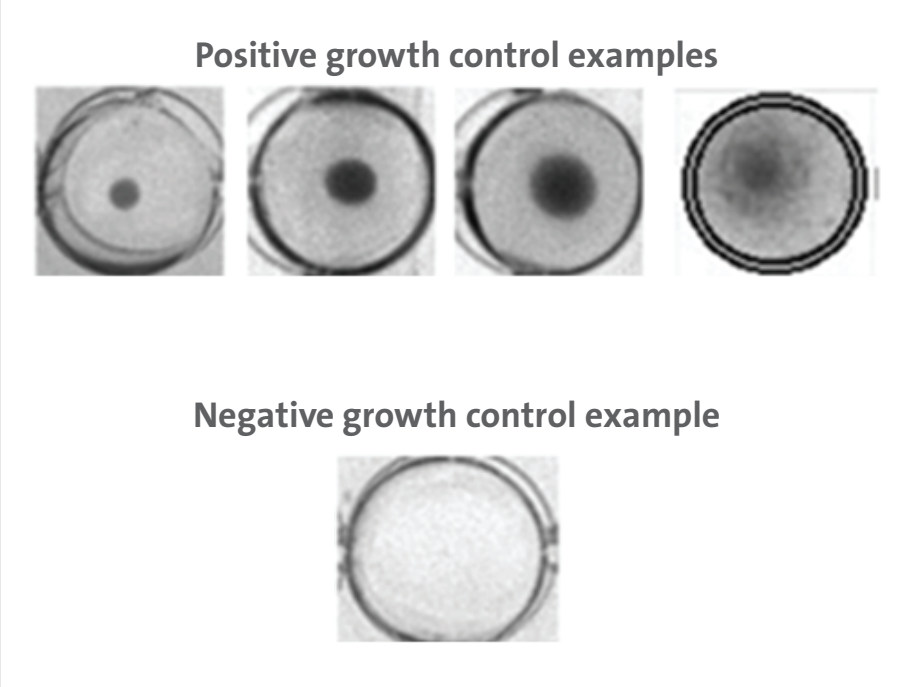
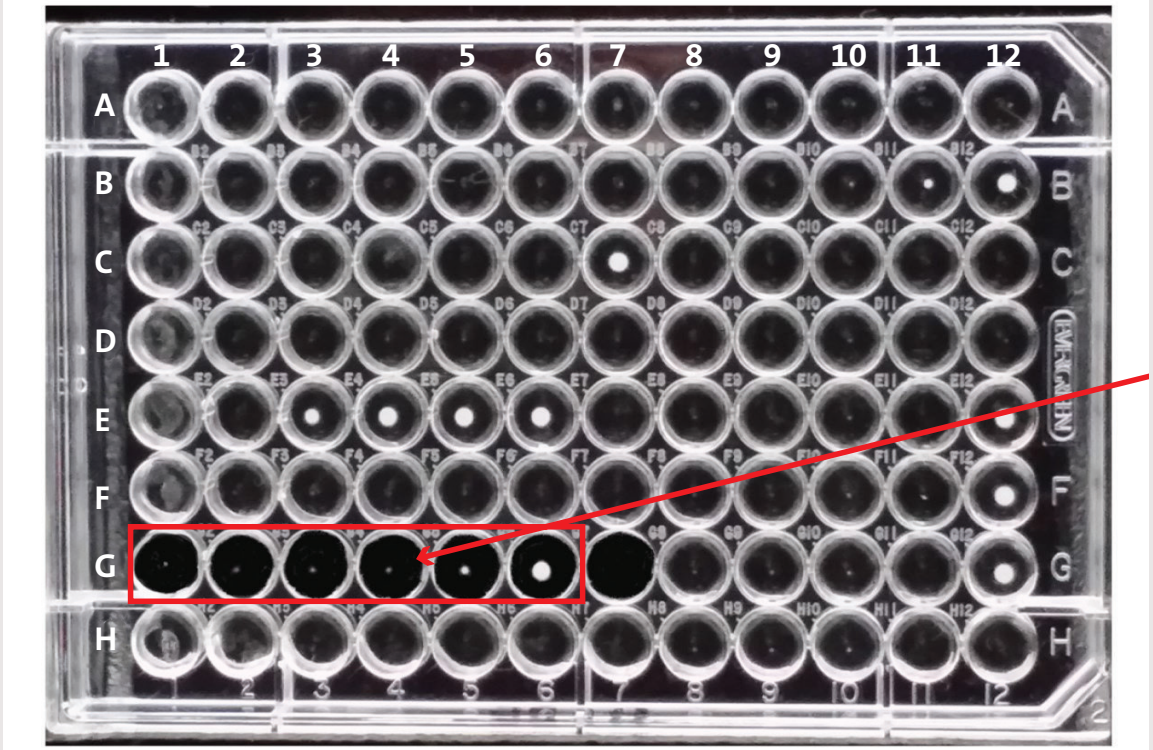


Figure 2. Growth Control Examples. (Courtesy of Thermo Fisher Scientific. Used with permission.)

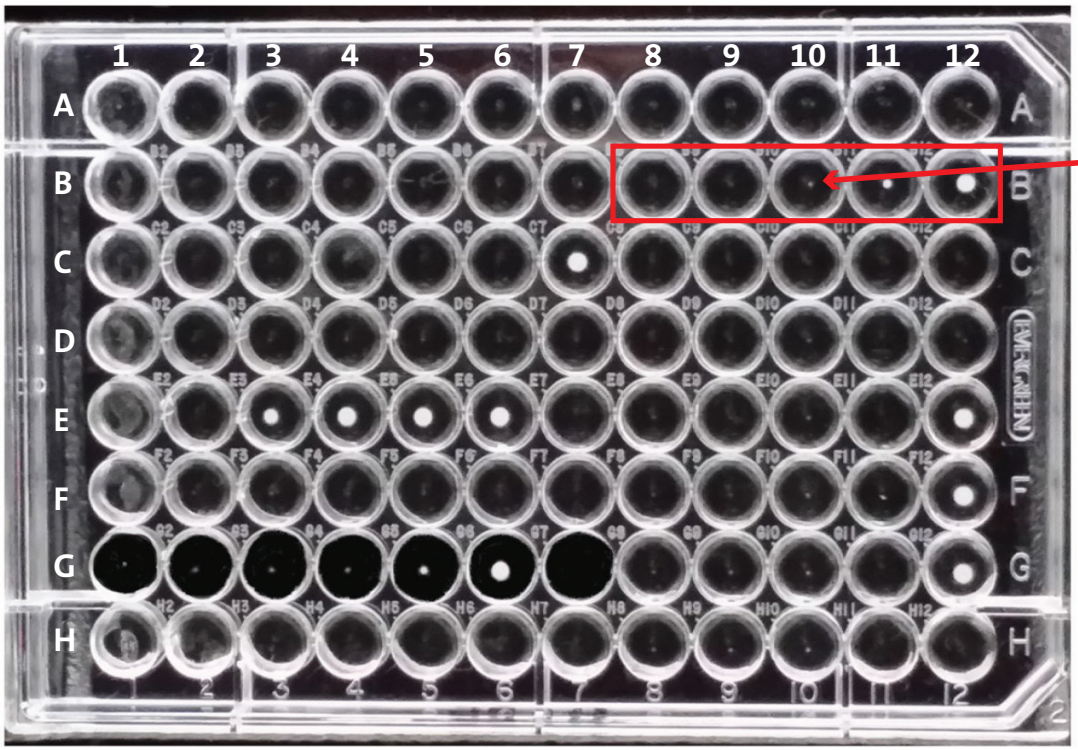
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Abbreviation: MIC, minimal inhibitory concentration.

Figure 3. Erythromycin: Trailing End Points. From left to right, wells G1 to G6 are 8 to 0.25 µg/mL. The MIC is well G4 (designated by the arrow). Well G12 is the growth control with no antimicrobial agent, and well H12 is the negative growth control.

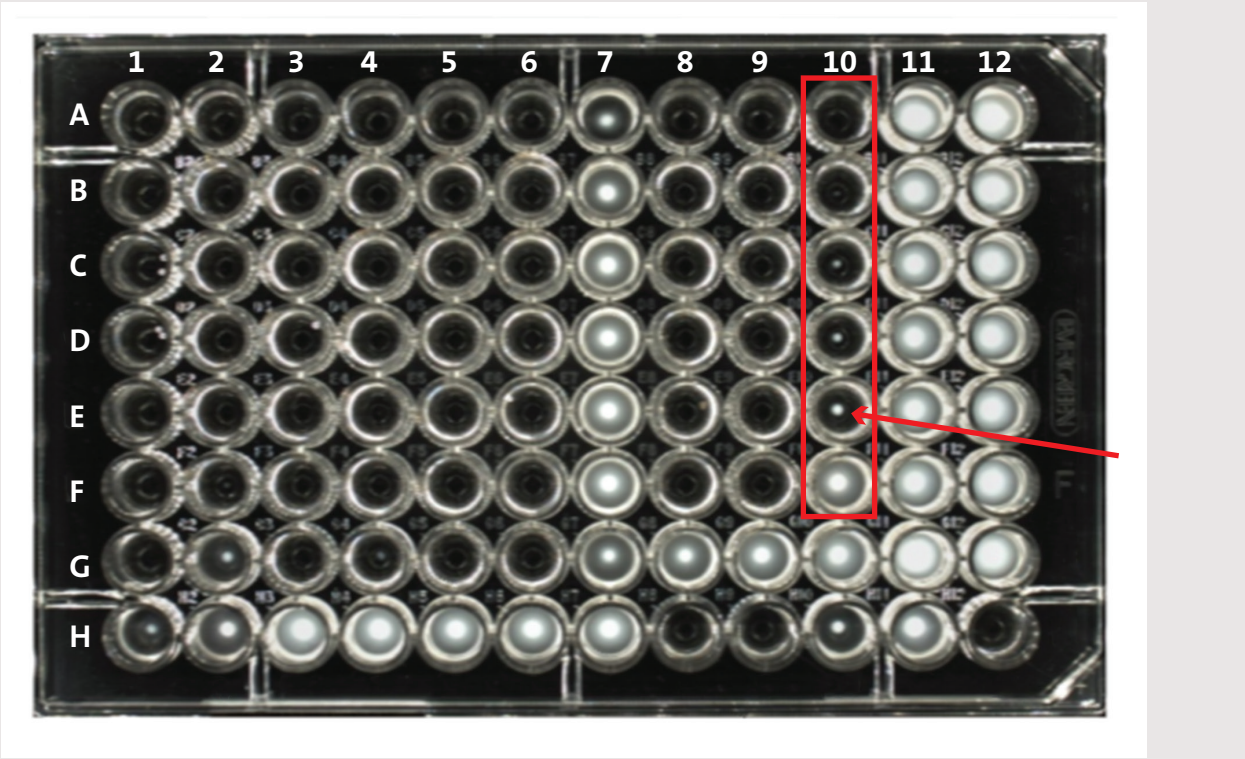
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Abbreviation: MIC, minimal inhibitory concentration.

Figure 4. Linezolid: Trailing End Points. From left to right, wells B8 to B12 are 16 to 1 µg/mL. The MIC is well B10 (designated by the arrow). Well G12 is the growth control with no antimicrobial agent, and well H12 is the negative growth control.

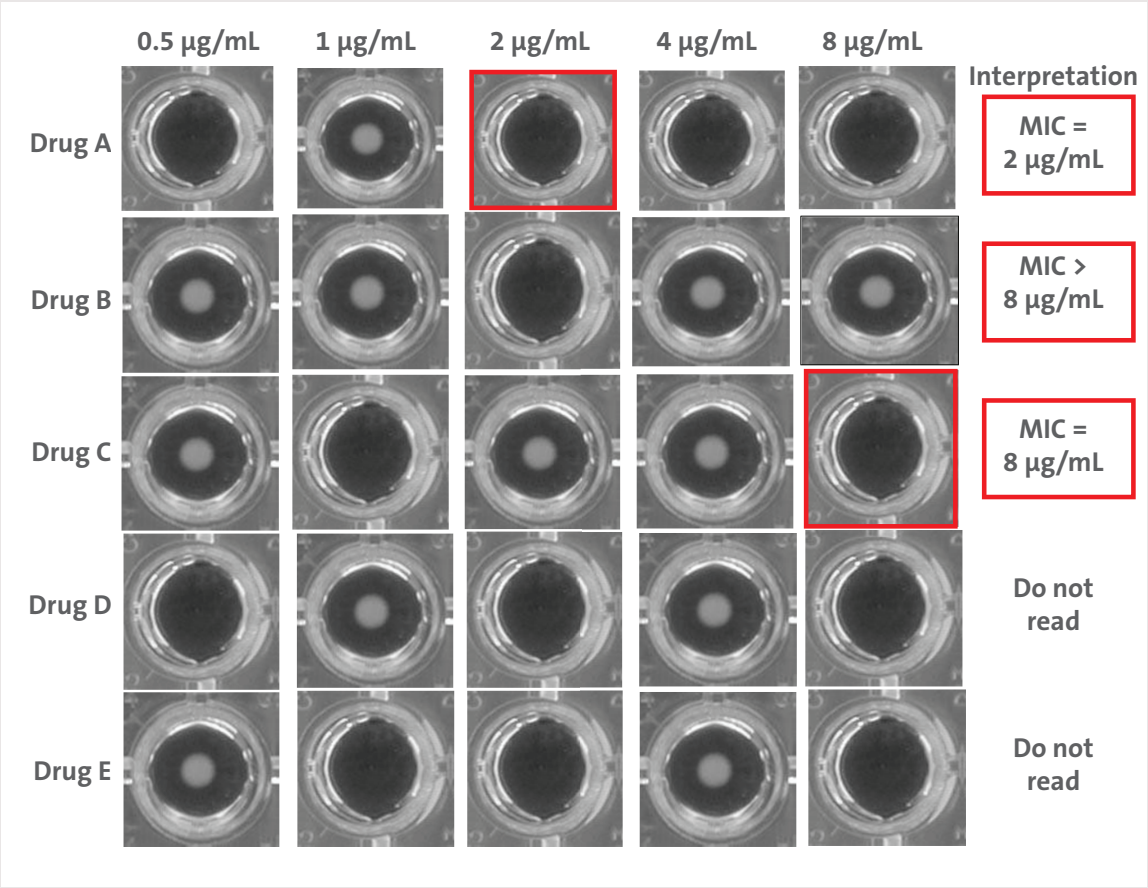
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Abbreviation: MIC, minimal inhibitory concentration.

Figure 5. Trimethoprim-Sulfamethoxazole: 80% Inhibition End Point. From top to bottom, wells A10 to F10 are 152/8–9.5/0.5 µg/mL. The MIC is well E10 (designated by the arrow). Well G12 is the growth control with no antimicrobial agent, and well H12 is the negative growth control.

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Abbreviation: MIC, minimal inhibitory concentration.

Figure 6. Example of Skipped Wells

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3.10 Reporting Minimal Inhibitory Concentration Results

The MIC results determined as described in this standard may be reported directly to clinicians for patient care purposes. However, for all clinicians to understand the data, it is essential that an interpretive category result is also routinely provided. Recommended interpretive categories for various MIC values are included in the CLSI M100[2] tables for each organism group and are based on evaluation of data as described in CLSI M23.[5] See Subchapter 1.4.1 for definitions of the interpretive categories susceptible, SDD, intermediate, resistant, and nonsusceptible.

3.11 Special Considerations for Fastidious Organisms

Mueller-Hinton medium, previously described for the rapidly growing aerobic pathogens, is not adequate for AST of fastidious organisms (see Table 5). If MIC tests are performed with fastidious organisms, the medium, QC procedures, and breakpoints must be modified to fit each organism (see Table 5). Dilution tests are described in CLSI M100[2] and CLSI M45[6] for the following organisms:

- *H. influenzae* and *H. parainfluenzae*, using HTM (broth dilution only); MH-F broth can be used for testing *H. influenzae* with selected antimicrobial agents.
- *N. gonorrhoeae*, using GC agar base medium + 1% defined growth supplement (agar dilution only).
- *N. meningitidis*, using 2.5% to 5% LHB-supplemented CAMHB or MHA with 5% sheep blood.
- Streptococci, using 2.5% to 5% LHB-supplemented CAMHB or MHA with 5% sheep blood.

Details of the conditions needed for testing these organisms are included in Table 5 and summarized in Appendix B. Other fastidious bacteria may be tested by a dilution or disk diffusion method as described in CLSI M45.[6] Methods for testing anaerobic bacteria are provided in CLSI M11.[4]

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Table 5. Testing Considerations for Fastidious Organisms

Step	Action	Organisms (see NOTES for specific organisms below table)			
		H. influenzae and *H. parainfluenzae*	*N. gonorrhoeae*	*N. meningitidis* (refer to NOTE 2 before testing)	*Streptococcus pneumoniae* and Other *Streptococcus* spp.
1	Using the colony suspension procedure (see Subchapter 3.2.2), prepare a suspension in broth, sterile distilled water, or saline from organisms grown and incubated on the source plates indicated in the organism-specific columns in this step. Adjust the suspension with broth, sterile distilled water, or saline to achieve a turbidity equivalent to a 0.5 McFarland standard.	• Incubate chocolate agar plates for 20–24 hours in 5% CO_2. • Adjust the suspension using a calibrated photometric device, using care in preparation (the suspension will contain approximately 1–4 × 10^8 CFU/mL), because higher inoculum concentrations may lead to false-resistant results with some β-lactam antimicrobial agents, particularly when β-lactamase–producing strains of *H. influenzae* are tested.	• Incubate the chocolate agar plate for 20–24 hours in 5% CO_2. • Prepare a 0.5 McFarland suspension in MHB or 0.9% phosphate-buffered saline, pH 7.	Incubate the chocolate agar plate for 20–24 hours in 5% CO_2.	Incubate the sheep blood agar plate for 18–20 hours (no longer) in 5% CO_2 as necessary to obtain adequate growth.

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Table 5. (Continued)

Step	Action	Organisms (see NOTES for specific organisms below table)			
		H. influenzae and *H. parainfluenzae*	*N. gonorrhoeae*	*N. meningitidis* (refer to NOTE 2 before testing)	*Streptococcus pneumoniae* and Other *Streptococcus* spp.
2	Inoculate the panels prepared with the recommended medium within 15 minutes after adjusting the turbidity of the inoculum suspension (see Appendix A for medium preparation or obtain commercially).	• **Broth dilution:** Use HTM or MH-F broth as described below. • The HTM method has been validated for *H. influenzae* and *H. parainfluenzae* only. MH-F broth has been validated for *H. influenzae* only. • If sulfonamides or trimethoprim are tested, aseptically add 0.2 IU thymidine phosphorylase to the medium. • The agar dilution method using HTM or MH-F broth has not been validated.	• **Agar dilution:** Use GC agar to which a 1% defined growth supplement is added after autoclaving. • Cysteine-free growth supplement is needed for agar dilution tests with carbapenems and clavulanate. Cysteine-containing defined growth supplement does not significantly alter dilution test results with other drugs. • Broth microdilution cannot be used for testing.	• **Broth dilution:** Use CAMHB with 2.5% to 5% LHB. • **Agar dilution:** Use MHA supplemented with 5% sheep blood.	• **Broth dilution:** Use CAMHB with 2.5% to 5% LHB. • CAMHB should be supplemented with calcium for daptomycin when β-hemolytic group or viridans group are tested. • Recent studies using the agar dilution method have not been performed and reviewed. If agar dilution is performed, follow the general test procedure in Subchapter 3.3 with the exceptions noted in this table and incubate in 5% CO_2 if necessary for growth.
3	Incubate panels using conditions and time as shown.	Incubate at 35°C ± 2°C for 20–24 hours.	Incubate at 36°C ± 1°C (do not exceed 37°C) in 5% CO_2 for 20–24 hours.	Incubate at 35°C ± 2°C in 5% CO_2 for 20–24 hours.	Incubate at 35°C ± 2°C in ambient air (CO_2 if necessary for agar dilution) for 20–24 hours.
4	Read the MICs (see Subchapters 3.4.9 and 3.9) and interpret results using specific MIC breakpoints in CLSI M100.[2]	See CLSI M100[2] Tables 2.			

Abbreviations: AST, antimicrobial susceptibility testing; BSC, biological safety cabinet; BSL-2, biosafety level 2; BSL-3, biosafety level 3; CAMHB, cation-adjusted Muller-Hinton broth; CFU, colony-forming unit(s); HTM, *Haemophilus* test medium; LHB, lysed horse blood; MHA, Mueller-Hinton agar; MHB, Mueller-Hinton broth; MH-F, Mueller-Hinton fastidious; MIC, minimal inhibitory concentration; pH, negative logarithm of hydrogen ion concentration.

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Table 5 NOTEs (for specific organisms):

NOTE 1 for *N. gonorrhoeae*:

- *N. gonorrhoeae* with 10-unit penicillin disk zone diameters of ≤ 19 mm are likely to be β-lactamase–producing strains. However, the β-lactamase test remains preferable to other susceptibility methods for rapid, accurate recognition of this plasmid-mediated penicillin resistance (see Subchapter 3.13.2).
- Culture and AST (agar dilution and disk diffusion methods) of *N. gonorrhoeae* should be considered in cases of suspected or documented treatment failure. Antimicrobial agents recommended for routine testing include, at minimum, azithromycin, ceftriaxone, cefixime, ciprofloxacin, and tetracycline.
- For azithromycin, currently there is only a susceptible breakpoint. This breakpoint presumes that azithromycin is used in an approved regimen that includes an additional antimicrobial agent.

NOTE 2 for *N. meningitidis*:

- **Recommended precautions:** Perform all AST of *N. meningitidis* in a BSC.[21-23] Manipulating *N. meningitidis* outside a BSC is associated with increased risk for contracting meningococcal disease. Laboratory-acquired meningococcal disease is associated with a case fatality rate of 50%.[24] Exposure to droplets or aerosols of *N. meningitidis* is the most likely risk for laboratory-acquired infection. Rigorous protection from droplets or aerosols is mandated when microbiological procedures (including AST) are performed on all *N. meningitidis* isolates.[22]
- If a BSC is unavailable, manipulation of these isolates should be minimized, limited to Gram staining or serogroup identification using phenolized saline solution, while wearing a laboratory coat and gloves and working behind a full-face splash shield. BSL-3 practices, procedures, and containment equipment for activities must be used with a high potential for droplet or aerosol production and for activities involving production quantities or high concentrations of infectious materials. If BSL-2 or BSL-3 facilities are not available, isolates should be forwarded to a referral or public health laboratory with a minimum of BSL-2 facilities.
- Laboratorians who are exposed routinely to potential aerosols of *N. meningitidis* should consider vaccination according to the current recommendations of the Centers for Disease Control and Prevention Advisory Committee on Immunization Practices (http://www.cdc.gov/vaccines/acip/index.html).
- Susceptibility testing with broth microdilution and agar dilution of *N. meningitidis* have been validated for detecting possible emerging resistance with some antimicrobial agents. To date, resistance has mostly been found in older agents used for therapy (penicillin or ampicillin) or agents used for prophylaxis of case contacts (rifampin, ciprofloxacin). Because resistance to antimicrobial agents such as ceftriaxone or cefotaxime that are often used for therapy of invasive diseases has not been confirmed, routine testing of isolates by medical laboratories is not necessary. Prophylaxis of close contacts should not be delayed while awaiting susceptibility testing results. Ceftriaxone and cefotaxime nonsusceptible isolates are rare. If testing is performed, any isolate found to be nonsusceptible should be confirmed and submitted to a public health laboratory.

NOTE 3 for *S. pneumoniae* and Other *Streptococcus* spp.:

- Inducible clindamycin resistance can be identified in *S. pneumoniae* and β-hemolytic streptococci using the method described in Subchapter 3.13.1.

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3.12 Special Considerations for Detecting Resistance

3.12.1 Staphylococci

3.12.1.1 Penicillin Resistance and β-Lactamase

Most staphylococci are resistant to penicillin, which is rarely an option for treating staphylococcal infections. Penicillin-resistant staphylococcal strains produce β-lactamase, and testing penicillin instead of ampicillin is preferred. Penicillin should be used to test the susceptibility of all staphylococci to all penicillinase-labile penicillins (see CLSI M100[2] glossaries).

Some β-lactamase–producing staphylococcal isolates test susceptible to penicillin. Because staphylococcal β-lactamase is readily inducible, there is a risk of this occurring if penicillin were used to treat such strains. For this reason, it is recommended that *Staphylococcus* spp. isolates with penicillin MICs ≤ 0.12 µg/mL or zone diameters ≥ 29 mm be tested for β-lactamase production before the isolate is reported as penicillin susceptible. Several tests for β-lactamase production have been described. These include nitrocefin-based tests or evaluating the zone edge of a penicillin disk diffusion zone-edge test (ie, a fuzzy zone edge indicates no β-lactamase production, whereas a sharp edge indicates β-lactamase production for *Staphylococcus aureus* that have penicillin zones ≥ 29 mm).[25]

Recommended methods for detecting penicillin resistance and β-lactamase by organism group include:

- *S. aureus:*
  - The penicillin disk diffusion zone-edge test is the most sensitive test for detecting β-lactamase production in *S. aureus* and is the recommended phenotypic test to confirm the absence of β-lactamase production and penicillin susceptibility in this species. If disk diffusion testing is done routinely for *S. aureus* and a penicillin disk is tested, no additional steps are needed. A nitrocefin-based test can be performed on *S. aureus* and, if positive, the isolate can be reported as penicillin resistant. However, if the nitrocefin test is negative, a penicillin disk diffusion zone-edge test must be performed before reporting the isolates as penicillin susceptible.
- Staphylococci other than *Staphylococcus aureus* (SOSA):
  - For SOSA, including *Staphylococcus lugdunensis*, only nitrocefin-based tests are recommended.
  - For the most current recommendations to detect β-lactamases in staphylococcal species, see CLSI M100[2] Tables 3.

3.12.1.2 Methicillin (Oxacillin) Resistance

Historically, resistance to the antistaphylococcal, penicillinase-stable penicillins has been referred to as "methicillin resistance," and the abbreviations "MRSA" (methicillin [oxacillin]-resistant *Staphylococcus aureus*) or "MRS" (methicillin [oxacillin]-resistant staphylococci) are still commonly used, even though methicillin is no longer available for treatment. In this standard, resistance to these agents may be referred to using several terms (eg, "MRS," "methicillin resistance," or "oxacillin resistance"). Most resistance to oxacillin in staphylococci is mediated by the *mecA* or *mecC* gene, which directs the production of supplemental penicillin-binding proteins (PBPs), called penicillin-binding protein 2a (PBP2a) or penicillin-binding protein 2c (PBP2c), respectively, during bacterial cell replication. Resistance is expressed either homogeneously or heterogeneously. Homogeneous expression of resistance is easily detected with standard testing methods, because nearly all bacterial cell progeny express the resistance phenotype. Heterogeneous expression may be more difficult to detect, because only a fraction of the progeny population (eg, 1 in 100 000 cells) expresses resistance. MICs for strains of *S. aureus* with *mecC* are typically in the resistant range for cefoxitin and/or oxacillin. *mecC* resistance cannot be detected by tests directed only at *mecA* or PBP2a.

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3.12.1.3 Methods for Detecting Oxacillin Resistance

Methods and testing conditions recommended for detecting oxacillin resistance in staphylococci are provided in Tables 6 and 7. Oxacillin disk diffusion methods should not be used except in the case of *Staphylococcus epidermidis, Staphylococcus pseudintermedius,* and *Staphylococcus schleiferi.* Cefoxitin disk diffusion is tested as a surrogate for oxacillin for all staphylococci except for *S. pseudintermedius* and *S. schleiferi*, for which oxacillin disk diffusion should be used. For the most current recommendations regarding testing and reporting, refer to CLSI M100[2] Tables 2. Recommended methods for detecting oxacillin resistance by organism group include:

- ***S. aureus* and *S. lugdunensis*:**
  - For *S. lugdunensis*, oxacillin-susceptible, *mecA*-negative strains exhibit oxacillin MICs in the range of 0.25 to 1 µg/mL, whereas *mecA*-positive strains usually exhibit MICs ≥ 4 µg/mL, characteristics more like *S. aureus* than other SOSA. Therefore, *mecA*-mediated resistance in *S. lugdunensis* is detected more accurately using *S. aureus* rather than SOSA breakpoints.
  - Cefoxitin disk diffusion and MIC tests are equivalent to oxacillin MIC tests in sensitivity and specificity for *S. aureus* and *S. lugdunensis* and are better predictors of *mecA*-mediated resistance than oxacillin agar-based methods, including the oxacillin salt agar screening plate. Therefore, cefoxitin can be used as a surrogate for oxacillin MIC tests.
- ***S. pseudintermedius* and *S. schleiferi*:**
  - For *S. pseudintermedius* and *S. schleiferi*, methicillin (oxacillin) resistance is best determined with oxacillin MIC or oxacillin disk diffusion tests. Neither cefoxitin MIC nor cefoxitin disk tests are reliable for detecting *mecA*-mediated resistance in *S. pseudintermedius* and *S. schleiferi* (see Table 6).
- **SOSA (excluding the species listed above):**
  - For SOSA, the cefoxitin disk diffusion test has sensitivity equivalent to oxacillin MIC tests but greater specificity (ie, the cefoxitin disk test more accurately identifies oxacillin-susceptible strains than the oxacillin MIC test) (see Table 6).

Table 6. Methods or Targets for Detection of Methicillin (Oxacillin)-Resistant *Staphylococcus* spp.

Organism	Disk Diffusion		MIC		*mecA*	PBP2a	Oxacillin Salt Agar
	Cefoxitin	Oxacillin	Cefoxitin	Oxacillin			
S. aureus	Yes (16–18 h)	No	Yes (16–20 h)	Yes (24 h)	Yes	Yes	Yes (24 h)
S. lugdunensis	Yes (16–18 h)	No	Yes (16–20 h)	Yes (24 h)	Yes	Yes	No
S. epidermidis	Yes (24 h)	Yes (16–18 h)	No	Yes (24 h)	Yes	Yes	No
S. pseudintermedius	No	Yes (16–18 h)	No	Yes (24 h)	Yes	Yes	No
S. schleiferi	No	Yes (16–18 h)	No	Yes (24 h)	Yes	Yes	No
Staphylococcus spp. (not listed above or not identified to the species level)	Yes, with exceptions[a] (24 h)	No	No	Yes (24 h)	Yes	Yes	No

Abbreviations: h, hour(s); MIC, minimal inhibitory concentration; PBP2a, penicillin-binding protein 2a.
[a] The cefoxitin disk diffusion test may not perform reliably for all species (eg, *S. haemolyticus*) that fall into the category of *Staphylococcus* spp. (not listed above or not identified to the species level).[26]

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The test conditions shown in Table 7 improve *mecA*-mediated oxacillin resistance detection in staphylococci, especially heteroresistant MRSA.

Table 7. Procedural Recommendations for Detection of Methicillin (Oxacillin)-Resistant *Staphylococcus* spp.

Recommendation Type	Comments
General	• Prepare the inoculum using the colony suspension method (see Subchapter 3.2.2). • Incubate at temperatures between 33 to 35°C. Testing at temperatures above 35°C may not detect resistance when oxacillin-based tests are used.
Oxacillin-based tests	• Add NaCl (2% w/v; 0.34 mol/L) to the medium for both agar and broth dilution oxacillin testing but not for agar medium used for disk diffusion testing. • Incubate oxacillin MIC tests for a full 24 hours before reporting as susceptible. A resistant result can be reported earlier.
Cefoxitin-based tests	Incubate cefoxitin MIC tests for 16–18 hours for *S. aureus* and *S. lugdunensis*.

Abbreviation: MIC, minimal inhibitory concentration.

3.12.1.3.1 Molecular Detection Methods

Tests for detecting the *mecA* and/or *mecC* genes or the protein produced by *mecA*, PBP2a (also called PBP2'), are also available and are generally accurate methods for predicting resistance to oxacillin. *mecC* resistance cannot be detected by tests directed at *mecA* or PBP2a only.

3.12.1.3.2 Reporting Oxacillin Resistance

When oxacillin test results are reported, considerations include:

- Isolates that test resistant by any of the recommended methods should be reported as oxacillin resistant.
- Resistance may be reported any time growth is observed after incubating for a minimum of 16 hours.
- If cefoxitin is used as a surrogate, the isolate should be reported as oxacillin susceptible or oxacillin resistant based on the cefoxitin result.
- Oxacillin results can be applied to the other penicillinase-stable penicillins.
- In most staphylococcal isolates, oxacillin resistance is mediated by *mecA* or *mecC*, encoding PBP2a and PBP2c, respectively. Isolates that test positive for *mecA* or *mecC* or PBP2a or PBP2c should be reported as oxacillin resistant.
- *S. aureus* isolates that are positive for *mecA* or *mecC* genes with molecular methods and test susceptible to cefoxitin should be investigated for mixed specimen or contamination using a purity plate. AST and molecular methods should be repeated. If the discrepancy is not resolved, the isolates should be reported as methicillin (oxacillin) resistant. See CLSI M100[2] appendixes.
- *S. aureus* isolates that are negative for *mecA* or *mecC* genes, or that do not produce PBP2a for which the phenotypic cefoxitin test is resistant, should be investigated for mixed specimen. The organism identification, AST, and molecular methods should be repeated. If the discrepancy is not resolved, the isolates should be reported as methicillin (oxacillin) resistant. See CLSI M100[2] appendixes.
- Oxacillin-resistant staphylococci are considered resistant to all penicillins, cephems (with exception of ceftaroline), β-lactam combination agents, and carbapenems (refer to CLSI M100[2] glossaries), because most cases of documented MRS infections have responded poorly to β-lactam therapy or because clinical data that demonstrate efficacy for these agents in MRS infections are not available.

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Methicillin (oxacillin)-susceptible staphylococci can be considered susceptible to:

- β-Lactam combination agents (eg, amoxicillin-clavulanate, ampicillin-sulbactam, piperacillin-tazobactam)
- Oral cephems (eg, cefaclor, cefdinir, cefpodoxime, cefprozil, cefuroxime, cephalexin, loracarbef)
- Parenteral cephems including first-generation cephalosporins (eg, cefazolin), second-generation cephalosporins (eg, cefamandole, cefonicid, cefuroxime), third-generation cephalosporins (eg, cefoperazone, cefotaxime, ceftizoxime, ceftriaxone), and fourth-generation cephalosporins (eg, cefepime), cephamycins (eg, ceftmetazole, cefotetan), ceftaroline, and moxalactam
- Carbapenems (eg, doripenem, ertapenem, imipenem, meropenem)

3.12.1.4 Vancomycin Resistance in *S. aureus*

Considerations for detecting vancomycin resistance include:

- **Vancomycin-resistant *S. aureus:***
  - The exact resistance mechanisms that result in elevated MICs are unknown, although they likely involve alterations in the cell wall and changes in several metabolic pathways. To date, most vancomycin-resistant *S. aureus* strains appear to have developed from MRSA. All these strains contained a *vanA* gene similar to that found in enterococci.[27,28]
  - These strains are reliably detected by the broth microdilution reference method and the vancomycin agar screen test (see Subchapter 3.12.2.2) when the tests are incubated for a full 24 hours at 35°C ± 2°C before being reported as susceptible.
  - Breakpoints for vancomycin and *S. aureus* are ≤ 2 µg/mL for susceptible, 4 to 8 µg/mL for intermediate, and ≥ 16 µg/mL for resistant. For SOSA, breakpoints are ≤ 4 µg/mL for susceptible, 8 to 16 µg/mL for intermediate, and ≥ 32 µg/mL for resistant.
- **Vancomycin-intermediate *S. aureus:***
  - When first described, heteroresistant vancomycin-intermediate *Staphylococcus aureus* (hVISA) isolates were noted to contain subpopulations of cells (typically 1 in every 100 000 to 1 000 000 cells) for which the vancomycin MICs were 8 to 16 µg/mL (ie, in the intermediate range). Because a standard broth microdilution test uses an inoculum of 5×10^5 CFU/mL, these resistant subpopulations may go undetected and the vancomycin MICs determined for such isolates would fall in the susceptible range (formerly between 1 and 4 µg/mL). Many physicians and microbiologists initially were skeptical that heteroresistance would result in clinical treatment failures with vancomycin because such strains were susceptible to vancomycin by the standard CLSI broth microdilution reference method. However, CLSI reviewed both clinical and laboratory data and lowered the susceptible breakpoint for vancomycin (for *S. aureus* isolates only, not SOSA) from ≤ 4 to ≤ 2 µg/mL and the resistant breakpoint from ≥ 32 to ≥ 16 µg/mL to avoid designating some heteroresistant strains susceptible and to make the breakpoints more predictive of clinical outcome. These lower breakpoints identify the hVISA strains for which the vancomycin MICs are 4 µg/mL as intermediate. Some susceptible *S. aureus* strains for which the vancomycin MICs are 1 to 2 µg/mL may still be hVISAs.
  - Determining population analysis profiles of *S. aureus* isolates is labor intensive and not suitable for routine medical laboratory application. Unfortunately, there is no standardized technique that is convenient and reliable for detecting hVISA strains. The inability of both automated commercial systems and standard reference susceptibility testing methods to detect the hVISA phenotype makes it impossible to identify all patients whose infections may be caused by hVISAs and may therefore not respond to vancomycin therapy.[29,30]

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3.12.1.4.1 Methods for Detecting *S. aureus* Isolates Not Susceptible to Vancomycin

S. aureus with vancomycin MICs ≥ 8 µg/mL can be detected by either broth dilution vancomycin MIC tests or the vancomycin agar screen test. To recognize staphylococcal strains for which the vancomycin MICs are 4 µg/mL, MIC testing must be performed and the tests incubated for a full 24 hours at 35°C ± 2°C. MIC tests should be performed to determine the susceptibility of all staphylococci isolates to vancomycin. The disk test does not differentiate vancomycin-susceptible isolates of *S. aureus* from vancomycin-intermediate isolates or differentiate among vancomycin-susceptible, -intermediate, and -resistant isolates of SOSA, all of which give similar size zones of inhibition.

3.12.1.4.2 Vancomycin Agar Screen for *S. aureus*

The test should be performed according to the procedure described in CLSI M100[2] Tables 3 by inoculating an isolate of *S. aureus* onto brain heart infusion (BHI) vancomycin screening agar containing 6 µg/mL of vancomycin. The inoculum should be prepared from a colony suspension to obtain a 0.5 McFarland turbidity standard. The incubation temperature is 35°C ± 2°C in ambient air, and the test should be incubated for 24 hours. The agar plates should be examined carefully with transmitted light for more than one colony or light film of growth, which indicate presumptive reduced susceptibility to vancomycin. Many *S. aureus* isolates with vancomycin MICs of 4 µg/mL do not grow on this vancomycin agar screen media. If growth is found on the vancomycin screen agar, a vancomycin MIC test must be performed to establish the MIC value. Also, there are insufficient data to recommend using this agar screen test for SOSA.

3.12.1.4.3 Reporting Vancomycin for Staphylococci

Vancomycin-susceptible staphylococci should be reported following the laboratory's routine reporting protocols. Additional considerations include:

- For strains not susceptible to vancomycin (ie, those with MICs ≥ 4 µg/mL for *S. aureus* or ≥ 8 µg/mL for SOSA), preliminary results should be reported following routine reporting protocols.
- Any *S. aureus* with MICs ≥ 8 µg/mL or SOSA with MICs ≥ 32 µg/mL should be sent to a referral laboratory for confirmation. Public health authorities should be notified according to local recommendations.
- Tables 2 and appendixes in CLSI M100[2] contain the most current recommendations for testing and reporting.

3.12.1.5 Inducible Clindamycin Resistance

Inducible clindamycin resistance can be identified using the method described in CLSI M100[2] Tables 3. See Subchapter 3.13.1 for additional information on this test.

3.12.1.6 Mupirocin Resistance

High-level mupirocin resistance (ie, MICs ≥ 512 µg/mL) occurs in *S. aureus* and is associated with the presence of the plasmid-mediated *mupA* gene.[31-33] Mupirocin use has been reported to increase rates of high-level mupirocin resistance in *S. aureus*.[34] High-level mupirocin resistance can be detected using either disk diffusion or broth microdilution tests (see CLSI M100[2] Tables 3).[35]

3.12.2 Enterococci

3.12.2.1 Penicillin and Ampicillin Resistance

Enterococci may be resistant to penicillin and ampicillin because of production of low-affinity PBPs or, very rarely, β-lactamase. Agar or broth dilution MIC testing and disk diffusion testing accurately detect isolates with altered PBPs but do not reliably detect isolates that produce β-lactamase. β-Lactamase–producing *Enterococcus* strains are best detected by using a direct, nitrocefin-based β-lactamase test (see Subchapter 3.13.2). A β-lactamase test is the only reliable test for detecting β-lactamase–producing *Enterococcus* spp. However, because β-lactamase–

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positive enterococci are rare, this test does not need to be performed routinely but can be used in selected cases if there is suspicion of β-lactamase production, eg, failure of β-lactam therapy for an isolate that tests penicillin susceptible. A positive β-lactamase test result predicts resistance to penicillin and amino- and ureidopenicillins.

Enterococcus strains with ampicillin and penicillin MICs ≥ 16 µg/mL are categorized as resistant. However, enterococci with penicillin MICs ≤ 64 µg/mL or ampicillin MICs ≤ 32 µg/mL may be susceptible to synergistic killing by these penicillins in combination with gentamicin or streptomycin (in the absence of high-level resistance to gentamicin or streptomycin, see Subchapter 3.12.2.3) if high doses of penicillin or ampicillin are used. Enterococci possessing higher penicillin levels (MICs ≥ 128 µg/mL) or ampicillin (MICs ≥ 64 µg/mL) resistance may not be susceptible to the synergistic effect.[36,37] Physicians' requests to determine the actual penicillin or ampicillin MIC for blood and CSF *Enterococcus* isolates should be considered.

3.12.2.2 Vancomycin Resistance

Accurately detecting vancomycin-resistant enterococci (VRE) by agar or broth dilution MIC testing necessitates incubating for a full 24 hours (rather than 16 to 20 hours) and carefully examining the plates, tubes, or wells for evidence of faint growth before reporting as susceptible. With disk diffusion, zones should be examined using transmitted light. The presence of a haze or any growth within the zone of inhibition indicates resistance. Organisms with intermediate zones should be tested by an MIC method. For isolates for which the vancomycin MICs are 8 to 16 µg/mL, biochemical tests or matrix-assisted laser desorption/ionization time-of-flight mass spectrometry (see CLSI M100[2] Tables 3 for information on additional testing and reporting). Other identification methods such as motility and pigment production should be performed to distinguish species with acquired resistance (eg, *vanA* and *vanB*) from those with intrinsic, intermediate-level resistance to vancomycin (eg, *vanC*), such as *Enterococcus gallinarum* and *Enterococcus casseliflavus*, which often grow on the vancomycin screen plate. A vancomycin agar screen test may also be used, as described in CLSI M100[2] Tables 3.

The vancomycin agar screen test can be used in addition to the dilution methods described above for detecting VRE. The test should be performed by inoculating an enterococcal isolate onto BHI-vancomycin screening agar containing 6 µg/mL of vancomycin using the method described in CLSI M100[2] Tables 3. The inoculum should be prepared from a colony suspension to obtain a 0.5 McFarland turbidity standard. The incubation temperature is 35°C ± 2°C in ambient air, and the test should be incubated for 24 hours. The vancomycin screening agar plate should be examined for more than one colony, which indicates presumptive resistance to vancomycin. Identification confirmation should be performed as described above.

3.12.2.3 High-Level Aminoglycoside Resistance

High-level resistance to gentamicin and/or streptomycin indicates that an enterococcal isolate will not be killed by the synergistic action of a penicillin or glycopeptide combined with that aminoglycoside.[36] Agar or broth high-concentration gentamicin (500 µg/mL) and streptomycin (1000 µg/mL with broth microdilution; 2000 µg/mL with agar dilution) tests or disk diffusion tests using high-concentration gentamicin (120-µg) and streptomycin (300-µg) disks can be used to screen for this type of resistance (see CLSI M100[2] Tables 3 for tests and QC recommendations). Other aminoglycosides do not need to be tested, because their activities against enterococci are not superior to gentamicin or streptomycin. Additionally, high-level resistance to both gentamicin and streptomycin implies that resistance will be found for all aminoglycosides, making testing of additional aminoglycosides unnecessary.

3.12.3 *S. pneumoniae*

Penicillin, cefotaxime, ceftriaxone, or meropenem should be tested by an MIC method and reported routinely or as part of selective reporting with CSF isolates of *S. pneumoniae*. Such isolates can also be tested against vancomycin using the MIC or disk diffusion method. Because specific breakpoints must be used depending on

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the infection site and the penicillin formulation used for therapy, CLSI M100[2] Tables 2 should be consulted when penicillins and third-generation cephalosporins are reported. In CLSI M100[2] Tables 2, breakpoints are listed for intravenous penicillin therapy for meningitis and nonmeningitis infections. Separate breakpoints are included for treating less severe infections with oral penicillin.

Amoxicillin, ampicillin, cefepime, cefotaxime, ceftriaxone, cefuroxime, ertapenem, imipenem, and meropenem may be used to treat pneumococcal infections. However, reliable disk diffusion susceptibility tests with these agents do not yet exist. The *in vitro* activity of these antimicrobial agents is best determined using an MIC method.

3.12.4 Gram-Negative Bacilli

The major mechanism of resistance to β-lactam antimicrobial agents in gram-negative bacilli is β-lactamase enzyme production. Many different enzyme types have been reported. β-lactamases may be named after:

- Primary substrates that they hydrolyze
- Biochemical properties of the β-lactamases
- Strains of bacteria from which the β-lactamases were detected
- Patient from whom a β-lactamase–producing strain was isolated[38]

β-Lactamases may be classified as molecular class A, B, C, or D enzymes, as shown in Table 8.[38]

Table 8. Enzyme Classifications for β-Lactamases[39]

Class	Active Site	Characteristics	Examples
A	Serine	Broad-spectrum enzymes, including most ESBLs and serine carbapenemases	• ESBLs: derivatives of TEM, SHV, CTX-M, VEB, and PER • Carbapenemases: KPC, SME, GES, IMI/NMC
B	Zinc	Metallo-β-lactamases hydrolyzing most β-lactams, including carbapenems but not monobactams	Carbapenemases: VIM, IMP, SPM, NDM, SIM
C	Serine	Broad-spectrum enzymes hydrolyzing cephalosporin subclasses III and IV[a] and cephamycins	Cephalosporinases: ACT, CMY, DHA, PDC
D	Serine	Narrow- and broad-spectrum enzymes, oxacillin-hydrolyzing enzymes	• ESBLs: OXA-11, OXA-15 • Carbapenemases: OXA-48 like, OXA-23

Abbreviations: ESBL, extended-spectrum β-lactamase; KPC, *Klebsiella pneumoniae* carbapenemase; NDM, New Delhi metallo-β-lactamase; SME, *Serratia marcescens* enzyme.
[a] Cephalosporin subclasses III and IV include cephalosporins that are sometimes referred to as third-generation or fourth-generation, respectively.

β-Lactamase enzymes in all four classes inactivate β-lactam antimicrobial agents at different rates. The genes encoding β-lactamases may be located on chromosomes and expressed with or without induction or carried on plasmids in single or multiple copies. An isolate may produce β-lactamases and possess other resistance mechanisms, such as porin mutations or efflux pumps, that restrict antimicrobial access to their active binding sites in the bacterial cell. The various β-lactam resistance mechanisms encountered in gram-negative bacteria generate a variety of antimicrobial activities expressed as a range of MIC values. The breakpoint is expected to be the MIC or zone diameter value that differentiates β-lactamase- and/or other resistance mechanism–negative strains (susceptible) from β-lactamase and/or other resistance mechanism–positive strains (resistant). However, weak β-lactamase activity or low-level β-lactamase expression may not necessarily mean that the isolate will be refractory to β-lactam therapy. In practice, some isolates that are interpreted as susceptible produce

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β-lactamases that have clinically inconsequential enzyme activity. These may be extended-spectrum β-lactamase (ESBL), AmpC, or carbapenemase-type enzymes as described in Subchapters 3.12.4.1, 3.12.4.2, and 3.12.4.3, respectively.

Identifying a specific β-lactamase resistance mechanism (eg, ESBL, carbapenemases) is not necessary for determining a susceptible or resistant interpretation. However, in consultation with the antimicrobial stewardship team and other relevant institutional stakeholders, laboratories may decide to perform phenotypic or genotypic testing for β-lactamases, and the results may be used to guide therapeutic management and for infection-prevention purposes. CLSI M100[2] Tables 3 describes tests that can be used to screen for and confirm the presence of ESBLs in *E. coli, Klebsiella pneumoniae, Klebsiella oxytoca,* and *Proteus mirabilis*. CLSI M100[2] Tables 3 describe tests that can be used to screen for and confirm carbapenemase production in Enterobacterales and *Pseudomonas aeruginosa.*

3.12.4.1 Extended-Spectrum β-Lactamases

ESBLs are class A and occasional class D enzymes that hydrolyze extended-spectrum cephalosporins and monobactams and are commonly inhibited by clavulanate[40] and avibactam. These enzymes arise from mutations in genes for common plasmid-encoded β-lactamases (eg, TEM-1 and SHV-1), or they may be only distantly related to a native enzyme, as in the case of the CTX-M β-lactamases. ESBLs may confer resistance to penicillins, cephalosporins, and aztreonam in isolates of *K. pneumoniae, K. oxytoca,*[41] *E. coli, P. mirabilis,*[42] and other genera of the order Enterobacterales.[38] β-Lactam breakpoints are set at MIC and zone diameter values that recognize ESBL activity, in combination with other resistance mechanisms, likely to predict clinical success or failure. Most ESBL-negative gram-negative bacilli will test susceptible; however, some strains that test susceptible may contain ESBL genes producing low enzyme amounts or enzymes that have poor hydrolytic activity. These strains are correctly categorized as susceptible when the current CLSI breakpoints published in CLSI M100[2] are used.

A similar chromosomal enzyme, OXY (formerly KOXY or K1), in *K. oxytoca* acts as an extended-spectrum penicillinase, inactivating amino- and carboxypenicillins. Overproduction of OXY enzymes caused by promoter mutations results in resistance to ceftriaxone and aztreonam (but not ceftazidime), as well as resistance to sulbactam, clavulanic acid, and tazobactam. Strains producing OXY enzymes may result in a positive ESBL confirmatory test. MIC and zone diameter breakpoints correctly predict susceptibility and resistance.

3.12.4.2 AmpC Enzymes

The AmpC β-lactamases are chromosomal or plasmid-encoded enzymes.[38] Isolates that produce AmpC enzymes may have an antimicrobial susceptibility profile similar to those that produce ESBLs in that they show reduced susceptibility to penicillins, cephalosporins, and aztreonam. However, in contrast to ESBLs, AmpC β-lactamases also inactivate cephamycins (ie, bacteria expressing AmpC enzymes test resistant to cefoxitin and cefotetan). In addition, AmpC-producing strains are resistant to the older β-lactamase inhibitor combination agents; however, newer inhibitors such as avibactam, relebactam, and vaborbactam inhibit AmpC enzymes. Isolates expressing AmpC β-lactamases may test resistant to carbapenems if accompanied by a porin mutation or in combination with overexpression of specific efflux pumps.

Chromosomal AmpC β-lactamases are found in *Enterobacter* spp., *Klebsiella* (formerly *Enterobacter*) *aerogenes*, *Citrobacter* spp., *Serratia* spp., and some other gram-negative species. The chromosomal AmpC β-lactamases are usually expressed in low amounts, but the organisms can be induced to produce higher amounts of enzyme by exposure to penicillins, carbapenems, and some cephems (eg, cefoxitin). Third- and fourth-generation cephalosporins do not induce production of AmpC enzymes but can be hydrolyzed by them. Using cephalosporins also may select for stably derepressed chromosomal variants, which can emerge during therapy.[43] Derepressed variants with elevated AmpC production are resistant to most β-lactams.

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Plasmid-mediated AmpC enzymes are transmissible among bacteria and can be found in isolates of *K. pneumoniae, E. coli, Proteus* spp., *Salmonella* spp., and *Shigella* spp. Plasmid-mediated AmpC β-lactamases are expressed constitutively, usually at high levels.

Citrobacter freundii complex, *Enterobacter cloacae* complex, *Hafnia alvei, K. aerogenes, Morganella morganii, Providencia* spp., *Serratia marcescens,* and *Yersinia enterocolitica* may test susceptible to ceftriaxone, cefotaxime, ceftazidime, and ceftaroline, but these agents may be ineffective against these genera within a few days after initiation of therapy due to derepression of inducible AmpC β-lactamase. The risk of AmpC derepression during therapy is moderate to high with *C. freundii* complex, *E. cloacae* complex, and *K. aerogenes* and appears to be less frequent with *M. morganii, Providencia* spp., and *S. marcescens.*[44] Therefore, isolates that are initially susceptible may become resistant. Testing subsequent isolates may be warranted if clinically indicated.

There is no CLSI-validated phenotypic test to confirm the presence of plasmid-encoded AmpC β-lactamases in clinical isolates. Isolates carrying both ESBLs and plasmid-encoded AmpC β-lactamases are common in some geographical regions. Using the current breakpoints for antimicrobial agents affected by these enzyme combinations is the best approach for guiding treatment of these strains.

3.12.4.3 Carbapenemases (Carbapenem-Resistant Gram-Negative Bacilli)

Carbapenemase activity has been detected in patient isolates of gram-negative bacilli with β-lactamase enzymes in classes A, B, and D. In Enterobacterales, KPC-, GES-, IMI-, and SME-type enzymes within class A, NDM-, VIM-, and IMP-type enzymes within class B, and OXA-type enzymes[45] within class D represent major families of clinical importance (see Table 9).

NDM-type and other metallo-β-lactamase enzymes need zinc for activity and are inhibited by substances that bind zinc (eg, EDTA). *Stenotrophomonas maltophilia*, *Bacillus anthracis*, and some strains of *Bacteroides fragilis* produce a chromosomal metallo-β-lactamase. Other metalloenzymes may be carried on plasmids and can occur in *Acinetobacter* spp., *P. aeruginosa, S. marcescens, K. pneumoniae*, and, increasingly, in other Enterobacterales.

Carbapenemase activity in Enterobacterales and *P. aeruginosa* can be confirmed using the Carba NP colorimetric microtube assay or the modified carbapenem inactivation method (mCIM) test described in CLSI M100[2] Tables 3.[46-50] Both the Carba NP and mCIM tests may detect carbapenemase production, but neither of these tests can identify the specific enzyme present. There is no CLSI-validated phenotypic test to confirm the presence of carbapenemase activity in *Acinetobacter* spp. or other taxa listed in Table 9; therefore, a molecular test is the recommended method.

Metallo-β-lactamase activity can be detected in Enterobacterales by using the EDTA-modified carbapenem inactivation method (eCIM) test in combination with the mCIM test as described in CLSI M100[2] Tables 3. The eCIM test enables differentiation of metallo-β-lactamases from serine carbapenemases in Enterobacterales isolates that are positive for mCIM. Using current breakpoints for antimicrobial agents affected by these carbapenemases is the recommended approach for providing guidance for treating infections by Enterobacterales containing both serine and metallo-β-lactamases, including KPC-, OXA-, and NDM-type enzymes. Refer to CLSI M100[2] Tables 3 for the most current recommendations for testing and reporting.

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Table 9. β-Lactamases With Carbapenemase Activity

β-Lactamase Class[a]	Found in	Examples
A	• Enterobacterales • *P. aeruginosa*	• KPC • GES • IMI/NMC • SME (*S. marcescens* only)
B	• Enterobacterales • *P. aeruginosa* • *Acinetobacter* spp. • *B. anthracis* • *Aeromonas* spp. • *S. maltophilia*	Metallo-β-lactamases inhibited by EDTA (IMP, VIM, SPM, NDM, CphA, L1)
D	• Enterobacterales • *P. aeruginosa* • *Acinetobacter* spp.	OXA-type enzymes

Abbreviations: EDTA, ethylenediaminetetraacetic acid; KPC, *Klebsiella pneumoniae* carbapenemase; NDM, New Delhi metallo-β-lactamase; SME, *Serratia marcescens* enzyme.
[a] Carbapenemases have not yet been found in class C.

3.13 Supplemental (Not Routine) Tests

A supplemental (not routine) test detects susceptibility or resistance to a drug or drug class by a method other than routine disk diffusion or broth or agar dilution MIC and does not need additional tests to confirm susceptibility or resistance. Some supplemental tests identify a specific resistance mechanism and may be required or optional for reporting specific clinical results.

3.13.1 Inducible Clindamycin Resistance

Inducible clindamycin resistance can be identified using the method described in CLSI M100[2] Tables 3.

Macrolide-resistant isolates of *S. aureus*, SOSA, *S. pneumoniae*, and β-hemolytic streptococci may express constitutive or inducible resistance to clindamycin (23S ribosomal RNA methylation encoded by the *erm* gene, also referred to as MLS_B [macrolide, lincosamide, and type B streptogramin] resistance) or may be resistant only to macrolides (efflux mechanism encoded by the *msrA* gene in staphylococci or a *mef* gene in streptococci). Infections caused by both staphylococci and streptococci with inducible clindamycin resistance may fail to respond to clindamycin therapy.[51] Inducible clindamycin resistance can be detected for all staphylococci, *S. pneumoniae,* and β-hemolytic streptococci in a single well test using broth microdilution.

For staphylococci, erythromycin 4 µg/mL and clindamycin 0.5 µg/mL are used together in a single well of a broth microdilution panel. For streptococci, the combination of erythromycin 1 µg/mL and clindamycin 0.5 µg/mL is used. The broth microdilution test applies only to isolates that are erythromycin resistant and clindamycin susceptible or intermediate. For these isolates, growth in the well indicates the presence of inducible clindamycin resistance and should be reported as clindamycin resistant. Following incubation, organisms that do not grow in the broth microdilution well should be reported as tested (ie, susceptible or intermediate to clindamycin).

Recommendations for QC and the most current updates for testing are provided in CLSI M100[2] Tables 3 and Tables 5.

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Inducible clindamycin resistance can also be detected in these organisms with the test using erythromycin and clindamycin disks placed together in close proximity (referred to as the D-zone test) (see CLSI M02,[3] Subchapter 3.13.1, and CLSI M100[2] Tables 3).

3.13.2 β-Lactamase Tests

3.13.2.1 Purpose

A rapid β-lactamase test may yield clinically relevant results earlier than a disk diffusion test with *Haemophilus* spp., *N. gonorrhoeae*, and some anaerobes. A β-lactamase test is the only reliable test for detecting β-lactamase–producing *Enterococcus* spp. A positive β-lactamase test result predicts:

- Resistance to penicillin, ampicillin, and amoxicillin among *Haemophilus* spp., *N. gonorrhoeae*, and some anaerobes
- Resistance to penicillin and amino- and ureidopenicillins among staphylococci and enterococci

A negative β-lactamase test result does not rule out β-lactam resistance because of other mechanisms. β-lactamase tests must not be used for members of the Enterobacterales, *Pseudomonas* spp., and other aerobic gram-negative bacilli, because the results may not be predictive of susceptibility to the β-lactams most often used for therapy.

3.13.2.2 Selecting a β-Lactamase Test

Detecting β-lactamase in *S. aureus* is most accurately accomplished by using the penicillin disk diffusion zone-edge test (see Subchapter 3.13.1 and CLSI M100[2] Tables 3). Nitrocefin-based tests can be used for *S. aureus*, but negative results should be confirmed with the penicillin zone-edge test before reporting penicillin as susceptible. Nitrocefin-based tests are the preferred method for testing SOSA, enterococci, *Haemophilus* spp., *Moraxella catarrhalis*, and *N. gonorrhoeae*[52] but need enzyme induction in SOSA, including *S. lugdunensis*. Induction can be easily accomplished by testing the growth from the zone margin surrounding a cefoxitin disk test. Acidimetric β-lactamase tests have generally produced acceptable results with *Haemophilus* spp., *N. gonorrhoeae*, and staphylococci. Iodometric tests may be used for testing *N. gonorrhoeae*. To ensure accurate results when these assays are performed, known positive and negative control strains should be tested at the same time as patient isolates (see the manufacturer's recommendations for commercial tests).

3.14 Dilution Test Method Limitations

3.14.1 Application to Various Organism Groups

The dilution methods described in this standard are standardized for testing rapidly growing pathogens, including *Staphylococcus* spp., *Enterococcus* spp., Enterobacterales, *P. aeruginosa*, *Acinetobacter* spp., *Burkholderia cepacia* complex, and *S. maltophilia*, and have been modified for testing fastidious organisms such as *H. influenzae*, *H. parainfluenzae, N. gonorrhoeae, N. meningitidis,* and streptococci (see CLSI M100[2] Tables 2). For organisms excluded from CLSI M100[2] Tables 2 and not covered in other CLSI documents (eg, CLSI M45[6]), studies are not yet adequate to develop reproducible, definitive standards to interpret results. These organisms may:

- Need different media.
- Need different incubation conditions.
- Show marked strain-to-strain variation in growth rate.

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For these microorganisms, an infectious diseases specialist should be consulted for guidance in determining the need for AST and for interpreting results. Published reports in the medical literature and current consensus recommendations for therapy of uncommon microorganisms may obviate the need for AST of such organisms. If testing is necessary, a dilution method usually is the most appropriate testing method, and the organism may need to be submitted to a referral laboratory.

3.14.2 Warning

Dangerously misleading results can occur when certain antimicrobial agents are tested and reported as susceptible against specific organisms. These combinations include but may not be limited to:

- Aminoglycosides, first- and second-generation cephalosporins, and cephamycins against *Salmonella* and *Shigella* spp.
- Routine testing of other β-lactam agents, except penicillin, oxacillin, cefoxitin, and ceftaroline, against *Staphylococcus* spp. is not advised.
- Aminoglycosides (except for high-level resistance testing) against *Enterococcus* spp.

In addition, some bacterial species have inherent or innate antimicrobial resistance, which is reflected in wild-type antimicrobial patterns of all or almost all species representatives. Intrinsic resistance is so common that AST is unnecessary. For example, some *Citrobacter* spp. are intrinsically resistant to ampicillin. A small percentage (1% to 3%) may appear susceptible because of method variation, mutation, or low levels of resistance expression. Testing such species is unnecessary; however, if tested, the result for an antimicrobial agent/organism combination listed as having intrinsic resistance should be reported as resistant. Refer to the intrinsic resistance tables in CLSI M100[2] appendixes for a list of such species and the antimicrobial agents to which they are resistant.

3.14.3 Development of Resistance and Testing Repeat Isolates

Isolates that are initially susceptible may become intermediate or resistant after initiating therapy. Therefore, subsequent isolates of the same species from a similar anatomical site should be tested to detect resistance that may have developed. Development of resistance can occur within as little as one to four days and has been noted most frequently in:

- *E. cloacae* complex, *C. freundii* complex, and *K. aerogenes* with third-generation cephalosporins
- *P. aeruginosa* with all antimicrobial agents
- Staphylococci with fluoroquinolones

For *S. aureus*, vancomycin-susceptible isolates may become vancomycin intermediate during the course of prolonged therapy.

In certain circumstances, the decision to perform susceptibility tests on subsequent isolates necessitates knowledge of the specific situation and the severity of the patient's condition (eg, an isolate of *E. cloacae* complex from a blood culture on a premature infant or MRSA from a patient with prolonged bacteremia). Laboratory guidelines on when to perform susceptibility tests on repeat isolates should be determined after consultation with the antimicrobial stewardship team and other relevant institutional stakeholders.

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Chapter 4

Quality Control and Quality Assurance

4 Quality Control and Quality Assurance

4.1 Quality Control Purpose

QC includes the procedures to monitor the test system to ensure accurate and reproducible results. This outcome is achieved by but not limited to testing carefully selected QC strains with known susceptibility to the antimicrobial agents tested. The goals of a QC program are to monitor:

- Precision (reproducibility) and accuracy of susceptibility test procedures
- Performance of reagents and equipment used in the tests
- Performance of persons who carry out the tests and report the results

A comprehensive QA program helps to ensure that testing materials and processes consistently provide quality results. QA includes but is not limited to:

- Monitoring
- Evaluating
- Taking corrective actions (if necessary)
- Recordkeeping
- Calibrating and maintaining equipment
- Proficiency testing
- Training
- Competence assessment
- QC

4.2 Quality Control Responsibilities

This subchapter is intended to apply to the standard reference methods; however, it may be applicable to certain commercially available AST systems that are based primarily or in part on methods described in CLSI M07.

Manufacturers and users of AST systems have a shared responsibility to ensure products meet performance specifications. The primary purpose of QC testing performed by manufacturers (in-house reference methods or commercial methods) is to ensure that testing materials and reagents have been appropriately manufactured and that the QC strains will provide acceptable results in the users' hands. The primary purpose of QC testing performed by laboratories (users) is to ensure that current and new testing materials and reagents are performing and maintained properly, and testing is performed according to established protocols. A logical division of responsibility and accountability may be described as:

- Manufacturers (in-house or commercial products) are responsible for:
  - Antimicrobial labeling
  - Antimicrobial agent potency or concentration
  - Antimicrobial stability
  - Compliance with good manufacturing practices (eg, quality management system standards)
  - Product integrity
  - Accountability and traceability to consignee

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- Laboratories (users) are responsible for:
  - Antimicrobial agent storage under the manufacturer's recommended environmental conditions (to prevent antimicrobial agent deterioration)
  - Proficiency of personnel performing tests
  - Using current CLSI standards (or manufacturers' instructions for use) and adhering to the established procedure (eg, inoculum preparation, incubation conditions, end-point determination, results interpretation)

Manufacturers should design and recommend a QC program that enables users to evaluate variables (eg, inoculum density, storage, and shipping conditions) that are most likely to adversely affect test results and to determine that the test results are accurate and reproducible when the test is performed according to established protocols.

NOTE: Laboratories should familiarize themselves with applicable regulatory and accreditation requirements for QC (see CLSI EP23™[53]).

4.3 Selecting Strains for Quality Control

Each QC strain should be obtained from a recognized source (eg, ATCC®). CLSI-recommended QC strains appropriate for the antimicrobial agent and reference method should produce results within the expected ranges listed in CLSI M100[2] that were established according to the procedures described in CLSI M23.[5] If specialty media is required for testing, additional QC organisms to ensure accuracy of the test results may be warranted. It is the manufacturer's responsibility to ensure their products pass QC. For MIC methods, the ideal QC range is defined by the 95% confidence interval calculated around the mode ± one-doubling dilution for a three-dilution range. In some cases, a four-dilution QC range is required to include at least 95% of the observed MIC results (eg, bimodal distribution). Laboratories should continuously monitor their QC results for trending at the low or high ends of the range, especially for those organism-antimicrobial agent combinations that span a four-dilution range. If trends are observed, troubleshooting should be initiated. Commercial system users should follow the QC recommendations in that system's instructions for use.

Ideal strains for dilution test QC demonstrate MICs that fall near the middle of the concentration range tested for all antimicrobial agents (eg, an ideal QC strain would be inhibited at the fourth dilution of a seven-dilution series, but strains with MICs at either the third or fifth dilution would also be acceptable). With newer, more potent antimicrobial agents, it may be necessary to test additional QC strains to provide on-scale values.

When three or fewer consecutive doubling dilutions of an antimicrobial agent are tested, QC procedures must be modified. For example, both QC strain 1 with a QC range of 0.5 to 2 µg/mL (mode = 1 µg/mL) and QC strain 2 with a QC range 2 to 8 µg/mL (mode = 4 µg/mL) would be needed to provide adequate QC coverage when dilutions of 0.5 to 4 µg/mL are tested. Deterioration of drug concentration could be detected by an MIC result of > 4 µg/mL with QC strain 1, and excess drug concentration could be detected by an MIC result of 1 with QC strain 2. When dilutions of 0.25 to 16 µg/mL are tested, either QC strain would provide adequate coverage, because their QC ranges would be on scale.

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Laboratories should consider using additional or supplemental QC strains with end points covering as many drug concentrations on the panel as possible, especially when production lots of MIC panels are prepared. These strains could be selected from QC strains listed in CLSI M100[2] appendixes or from well-characterized stock or clinical organisms, as appropriate. For isolates without established QC ranges, the acceptable QC range would be determined by repetitive testing of the isolate on multiple production lots of MIC panels.

QC strains and their characteristics are described in CLSI M100[2] appendixes. Refer to CLSI M100[2] Tables 5 for recommendations for routine QC. Routine QC strains are tested regularly (eg, daily or weekly) to ensure the test system is working and produces results that fall within the specified limits in CLSI M100.[2] The QC strains recommended in CLSI M100[2] should be included when a laboratory performs CLSI reference MIC testing as described herein.

Other QC strains are considered supplemental and are used to assess particular test or test system characteristics in select situations, or they may represent alternative QC strains. For example, *H. influenzae* ATCC® 10211 is more fastidious than *H. influenzae* ATCC® 49247 or *H. influenzae* ATCC® 49766 and is used to ensure HTM can adequately support the growth of *H. influenzae* and *H. parainfluenzae* patient isolates. Supplemental QC strains may possess susceptibility or resistance characteristics specific for one or more special tests listed in CLSI M02,[3] M07, and M100.[2] They can be used to assess a new test, for training new personnel, and for competence assessment. It is not necessary to include supplemental QC strains in routine daily or weekly AST QC programs.

Expected QC ranges for routine QC strains are listed in CLSI M100[2] Tables 5. When recommended, acceptable QC ranges for supplemental QC strains are also included in CLSI M100[2] Tables 3, which describe screening and confirmatory tests for specific resistance mechanisms.

4.4 Maintaining and Testing Quality Control Strains

Proper organism storage and maintenance is needed to ensure acceptable QC strain performance (also refer to Appendix C). Considerations include:

- For long-term storage, stock cultures should be maintained at ≤ −20°C (preferably at ≤ −60°C or in liquid nitrogen) in a suitable stabilizer (eg, 50% fetal calf serum in broth, 10% to 15% glycerol in tryptic soy broth, defibrinated sheep blood, or skim milk) or in a freeze-dried state.
- Some QC strains, particularly those with plasmid-mediated resistance (eg, *E. coli* ATCC® 35218), have been shown to lose the plasmid when stored at temperatures above −60°C or repeatedly subcultured. The footnotes in CLSI M100[2] appendixes highlight special considerations for handling certain QC strains. These considerations include testing β-lactamase–producing QC strains with an antimicrobial agent, which is known to be inactivated by the β-lactamase produced.
- Appendix C illustrates workflow for subculturing and testing QC strains.
- If an unexplained QC error occurs that might be because of a change in the organism's inherent susceptibility or resistance, a fresh stock culture of the QC strain should be obtained from an external source (see Subchapter 4.8 for additional guidance).
- It is important to always test the QC strains using the same materials and methods that are used to test patient isolates.

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4.5 Batch or Lot Quality Control

Guidelines for QC of each batch, shipment, or media lot include:

- Each new batch, shipment, or lot of broth macrodilution tubes, broth microdilution panels, or agar dilution plates should be tested with the appropriate QC strains before or concurrent with their first use for testing patient isolates. If MICs obtained with the batch or lot do not fall within the acceptable ranges (see CLSI M100[2] Tables 5), corrective action should be taken. Patient results can be reported for that day if QC results are within the acceptable limits (see Subchapter 4.8 for corrective action and Subchapter 4.9 to determine whether patient test results can be reliably reported). Records to maintain include, at a minimum, the lot numbers, expiration dates, and date of use of all materials and reagents used in performing susceptibility tests. See CLSI M100[2] Tables 5 for suggested QC frequency when modifications are made to antimicrobial susceptibility test systems.
- For MHA or supplemented MHA tubes prepared in-house, at least one uninoculated tube, broth dilution panel or agar plate from each batch or lot should be incubated overnight to confirm the medium's sterility.
- Dehydrated Mueller-Hinton broth (dMHB) used to prepare broth macrodilution tubes or microdilution panels can be obtained from manufacturers with or without added cations (see Appendix A). For dMHB without added cations, information regarding the cation levels initially present in the medium must be obtained from the manufacturer. Then, the medium must be supplemented accordingly. For dMHB provided by the manufacturer as CAMHB, additional supplementation is not necessary. Some antimicrobial agents are more likely to be affected by variations in cation content or pH (eg, aminoglycosides and daptomycin; see CLSI M100[2] Tables 5), and unacceptable cation content or pH should be suspected if QC errors are encountered with these antimicrobial agents. For example, acceptable cation content can be determined by testing gentamicin and *P. aeruginosa* ATCC® 27853, and if the MIC results are out of range on the low end (see CLSI M100[2] Tables 5), the broth may need additional cation supplementation (see Subchapter 3.5.1 and Appendix A).

4.6 Minimal Inhibitory Concentration Control Ranges

Acceptable MIC QC ranges are listed in CLSI M100[2] Tables 5. These ranges were developed following a specific protocol outlined in CLSI M23.[5]

4.7 Quality Control Testing Frequency (also refer to Appendix D and CLSI M100[2] Tables 5)

Appropriate QC strains should be tested on each day the test is performed on patient isolates. Alternatively, one of two plans can be adopted to demonstrate acceptable performance to reduce the frequency of QC tests from daily to weekly. Either plan enables weekly QC testing once satisfactory performance with daily testing of QC strains is documented (see Subchapters 4.7.2.1 and 4.7.2.2). The weekly QC testing options are not applicable when tests are performed less than once a week.

4.7.1 Daily Quality Control Testing

A laboratory can perform QC testing daily. Daily (vs weekly) QC testing must be performed on each day patient isolates are tested if tests are performed less than once a week. "Daily" QC testing or testing on "consecutive" test days means testing QC strains on each day tests are performed on patient isolates. It does not refer to calendar days.

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4.7.2 Performance Criteria for Reducing Quality Control Frequency to Weekly

Two plans are available to demonstrate satisfactory performance with daily QC testing before going to weekly QC testing. These are:

- The 20- or 30-day plan
- The 15-replicate (3- × 5-day) plan

4.7.2.1 The 20- or 30-Day Plan

When the 20- or 30-day plan is implemented:

- All applicable QC strains are tested for 20 or 30 consecutive test days, and results are documented.
- Recommended actions as described in Appendix D should be followed.
- If no more than one of 20 or three of 30 MIC results for each antimicrobial agent and organism combination are outside the acceptable QC range listed in CLSI M100[2] Tables 5, it is acceptable to go to weekly QC testing.
- If completion of the 20- or 30-day plan is unsuccessful, corrective action should be taken as appropriate and daily QC testing continued.
- If a laboratory routinely tests QC strains on each day of use and desires to convert to a weekly QC plan, it is acceptable to retrospectively analyze QC data from consecutive tests available during the previous two years, providing no aspects of the test system have changed.

4.7.2.2 The 15-Replicate (3- × 5-Day) Plan

When the 15-replicate (3- × 5-day) plan is used:

- Three replicates of each applicable QC strain are tested using individual inoculum preparations for five consecutive test days and the results are documented.
- Recommended actions as described in Appendix D and Table 10 should be followed.
- Following successful completion of the 15-replicate (3- × 5-day) plan, it is acceptable to go to weekly QC testing.
- If completion of the 15-replicate (3- × 5-day) plan is unsuccessful, corrective action should be taken, as appropriate, and daily QC testing continued.

Table 10. 15-Replicate (3- × 5-Day) Plan: Acceptance Criteria and Recommended Action[a]

Number Out of Range With Initial Testing (based on 15 replicates)	Conclusion From Initial Testing (based on 15 replicates)	Number Out of Range After Repeat Testing (based on all 30 replicates)	Conclusion After Repeat Testing
0–1	Plan is successful. Convert to weekly QC testing.	N/A	N/A
2–3	Test another 3 replicates for 5 days.	2–3	Plan is successful. Convert to weekly QC testing.
≥ 4	Plan fails. Investigate and take corrective action as appropriate. Continue QC each test day.	≥ 4	Plan fails. Investigate and take corrective action as appropriate. Continue QC each test day.

Abbreviations: N/A, not applicable; QC, quality control.
[a] Each QC strain and antimicrobial agent combination is assessed separately.

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For background information that supports the 3- × 5-day plan, refer to the Statisticians' Summary for Alternative QC Frequency Testing Proposal (available at https://clsi.org/media/1719/statisticians_summary_alternative_qc_frequency_testing_proposal.pdf).

4.7.2.3 Implementing Weekly Quality Control Testing

Weekly QC testing may be implemented once satisfactory performance with daily QC testing has been documented (see Subchapters 4.7.2.1 and 4.7.2.2).

- QC testing must be performed once per week and whenever any test reagent component (eg, a new lot of broth from the same manufacturer) is changed.
- If any of the weekly QC results are out of range, corrective action should be taken.
- Refer to CLSI M100[2] Tables 5 for guidance on QC frequency with new materials or test modifications.

The plans listed here for conversion to weekly QC can also be used for testing systems in which an MIC is determined using three or fewer consecutive doubling dilutions of an antimicrobial agent.

For some antimicrobial agents, QC records may indicate the need for testing to be done more frequently than once a week, because of the relatively rapid drug degradation (see Subchapters 3.4.2 [ie, step 3] and 3.7.1 [ie, step 3]).

4.8 Out-of-Range Results With Quality Control Strains and Corrective Action

Out-of-range QC results can be categorized into random, identifiable, or system related:

- QC ranges are established to include ≥ 95% of results obtained from routine testing of QC strains. A small number of (random) out-of-range QC results may be obtained even when the test method is performed correctly, and materials are maintained according to recommended protocols. Such occurrences can be observed based on normal variability of the test.
- Out-of-range results with QC strains caused by random or identifiable errors can usually be resolved by a single repeat of the QC test. However, out-of-range QC results that are caused by a problem with the test system usually do not correct when the QC test is repeated and may indicate a serious problem that can adversely affect patient results. Every out-of-range QC result must be investigated.

NOTE: See the MIC troubleshooting guide in CLSI M100[2] Tables 5 for troubleshooting and corrective action for out-of-range results with QC strains.

4.8.1 Daily or Weekly Quality Control Testing: Out-of-Range Results Caused by Identifiable Error

If the reason for an out-of-range result can be identified and easily corrected:

1. Correct the problem.
2. Document the reason.
3. Retest the QC strain on the day the error is observed.

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If the repeated result is within range, no additional corrective action is needed. Identifiable reasons for the out-of-control results should be documented and may include but are not limited to:

- QC strain:
  - Changes in the organism (eg, mutation, loss of plasmid)
  - Contamination
  - Improper storage
  - Inadequate maintenance (eg, preparing a QC plate for testing from a plate other than the F2 subculture plate that is less than seven days old)
  - Nonviability
  - Using the wrong QC strain
- Testing supplies:
  - Contamination
  - Inadequate volume of broth in tubes or wells
  - Incorrectly calculated antimicrobial stock concentrations
  - Improper storage or shipping conditions
  - Using damaged (eg, cracked, leaking) panels, plates, cards, tubes
  - Using expired materials
- Testing process:
  - Incorrectly prepared or adjusted inoculum suspensions
  - Contamination
  - Inoculum prepared from a plate incubated for incorrect time frame
  - Inoculum prepared from differential or selective media containing antimicrobial agents or other growth-inhibiting compounds
  - Using the wrong incubation temperature or conditions
  - Using incorrect incubation time
  - Using the wrong reagents and/or ancillary supplies
  - Incorrectly reading or interpreting test results
  - Transcription error
- Equipment:
  - Not functioning properly or out of calibration (eg, pipettes, photometric device)

4.8.2 Daily Quality Control Testing: Out-of-Range Result Not Caused by Identifiable Error

Corrective action needs to be taken if results for a QC strain and antimicrobial agent combination are out of range and the error is not identifiable and occurs on two consecutive testing days. Corrective action also needs to be taken if more than three results for a QC strain and antimicrobial agent combination are out of range during 30 consecutive testing days (see QC ranges listed in CLSI M100[2] Tables 5).

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4.8.3 Weekly Quality Control Testing: Out-of-Range Result Not Caused by Identifiable Error

If the reason for the out-of-range result with the QC strain cannot be identified, corrective action needs to be taken to determine whether the error is random or system related:

- The out-of-range antimicrobial agent and organism combination must be tested on the day the error is observed or as soon as an F2 or F3 subculture of the QC strain is available (see Appendix C).
- If the repeat results are in range, all QC results available for the antimicrobial agent and organism combination when the same lot numbers of materials that were used when the out-of-range QC result was observed are used, they need to be evaluated. If five acceptable QC results are available, no additional days of QC testing are needed.

The following tables illustrate two scenarios that might be encountered, along with suggested actions.

Scenario 1: Ampicillin *E. coli* ATCC® 25922; acceptable range, 2 to 8 µg/mL

Week	Day	Lot Number	MIC Result, µg/mL	Action
1	1	3564	4	
2	1	3564	8	
3	1	3564	8	
4	1	3564	4	
5	1	3564	16	Out of range. Repeat QC same day.
5	2	3564	8	In range. Five acceptable in-range QC tests for *E. coli* ATCC® 25922 and ampicillin with lot 3564. Resume weekly QC testing.

Abbreviations: ATCC®, American Type Culture Collection; MIC, minimal inhibitory concentration; QC, quality control.

Conclusion: Random QC error.

Scenario 2: Ampicillin *E. coli* ATCC® 25922; acceptable range, 2 to 8 µg/mL

Week	Day	Lot Number	MIC Result, µg/mL	Action
1	1	9661	4	
2	1	9661	8	
3	1	9661	16	Out of range. Repeat QC same day.
3	2	9661	8	In range. Three acceptable in-range QC tests for *E. coli* ATCC® 25922 and ampicillin with lot 9661. Repeat QC for 2 more consecutive days.
3	3	9661	8	In range.
3	4	9661	8	In range. Five acceptable in-range QC tests for *E. coli* ATCC® 25922 and ampicillin with lot 9661. Resume weekly QC testing.

Abbreviations: ATCC®, American Type Culture Collection; MIC, minimal inhibitory concentration; QC, quality control.

Conclusion: Random QC error.

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4.8.4 Additional Corrective Action

If repeat results with QC strains are still out of range, additional corrective action is needed.

- It is possible that the problem is caused by a system error rather than a random error (see Subchapter 4.8.1 and CLSI M100[2] Tables 5).
- Daily QC tests must be continued until the problem is resolved.
- If necessary, a new QC strain (either from stock cultures or a reliable source) and new lots of materials (including new turbidity standards) need to be obtained, possibly from different manufacturers. If the problem appears to be related to a manufacturer, the manufacturer should be contacted and provided with the test results and lot numbers of materials used. It may be helpful to exchange QC strains and materials with another laboratory using the same method to determine the root cause of out-of-range QC results when the reason is not identifiable. Until the problem is resolved, it may be necessary to use an alternative test method.

4.9 Reporting Patient Results When Out-of-Range Quality Control Results Are Observed

When an out-of-range result occurs when QC strains are tested or when corrective action is necessary, each patient test result must be carefully examined to determine whether it can be reliably reported. Factors to consider may include but are not limited to:

- Size and direction of QC strain error (eg, slightly or significantly increased or decreased MIC)
- Actual patient result and its proximity to the breakpoint
- Results with other QC organisms
- Results with other antimicrobial agents
- Usefulness of the particular QC strain and antimicrobial agent as an indicator for a procedural or storage issue (eg, inoculum dependent, heat labile) (refer to CLSI M100[2] Tables 5)

Options to consider for patient results include:

- Suppressing the results for an individual antimicrobial agent
- Reviewing individual patient or cumulative data for unusual patterns
- Using an alternative test method or a referral laboratory until the problem is resolved

4.10 Confirming Results When Patient Isolates Are Tested

Multiple test parameters are monitored by following the QC recommendations described in this standard. However, acceptable results derived from testing QC strains do not guarantee accurate results when patient isolates are tested. It is important to review all results obtained from all antimicrobial agents tested on a patient's isolate before reporting the results. This should include ensuring that:

- AST results are consistent with the isolate's identification.
- AST results for individual antimicrobial agents within a specific drug class follow the established hierarchy of activity rules (eg, third-generation cephalosporins are more active than first- or second-generation cephalosporins against Enterobacterales).
- The isolate is susceptible to antimicrobial agents for which resistance has not been documented (eg, vancomycin and *Streptococcus* spp.) and for which only "susceptible" breakpoints exist in CLSI M100[2] Tables 2.

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Unusual or inconsistent results should be confirmed by checking for:

- Previous results on the patient (eg, whether the patient previously had the same isolate with an unusual antibiogram)
- Previous QC performance (eg, whether there is a similar trend or observation with recent QC testing, such as smaller QC zone sizes with a new lot of materials and an increase in resistance with patient isolates)
- Problems with the testing supplies, process, or equipment (see Subchapter 4.8.1 and CLSI M100[2] Tables 5)

If a reason for the unusual or inconsistent result for the isolate cannot be ascertained, a repeat of the antimicrobial susceptibility test or the identification, or both, may be needed. Sometimes, it is helpful to use an alternative test method for the repeat test. A suggested list of results that may need verification is included in CLSI M100[2] appendixes. Each laboratory must develop its own policy for confirming unusual or inconsistent AST results. This policy should emphasize results that may significantly affect patient care.

4.11 End-Point Interpretation Control

End-point interpretation needs to be monitored to minimize variation in MIC end-point interpretation among observers. All laboratory personnel who perform these tests should independently read a selected set of dilution tests. The results should be recorded and compared with the results obtained by an experienced reader or, when QC strains are used, compared with the expected results from CLSI M100[2] Tables 5. All readers should agree within ± 1 twofold concentration dilution for readings of the same dilution scheme on the same panel or plate.[54] This process may be included as a part of annual and semiannual competence assessment.

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Chapter 5

Conclusion

5 Conclusion

This standard presents a process workflow for standard broth dilution (macrodilution and microdilution) and agar dilution techniques, which includes broth and agar dilution test preparation, testing conditions (including inoculum preparation and standardization, incubation time, and incubation temperature), reporting of MIC results, dilution test method limitations, and QC procedures. Using the methods in this standard, along with the corresponding CLSI M100[2] supplement, enables laboratories to assist the clinician in selecting appropriate antimicrobial therapy for patient care.

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Chapter 6

Supplemental Information

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Appendix A. Preparation of Supplements, Media, and Reagents

Abbreviations for Appendix A

CAMHB cation-adjusted Mueller-Hinton broth

dMHB dehydrated Mueller-Hinton broth

HTM *Haemophilus* test medium

ID-CAMHB iron-depleted cation-adjusted Mueller-Hinton broth

LHB lysed horse blood

MHA Mueller-Hinton agar

MHB Mueller-Hinton broth

MH-F Mueller-Hinton fastidious

NAD β-nicotinamide adenine dinucleotide

P-80 polysorbate 80

pH negative logarithm of hydrogen ion concentration

A1 Supplements

A1.1 Cation Stock Solutions

The steps for preparing cation stock solutions are listed in the following table.

Step	Action	Comment
1	Prepare a magnesium stock solution by dissolving 8.36 g of $MgCl_2 \cdot 6H_2O$ in 100 mL deionized water.	This solution contains 10 mg Mg^{++}/mL.
2	Prepare a calcium stock solution by dissolving 3.68 g of $CaCl_2 \cdot 2H_2O$ in 100 mL deionized water.	This solution contains 10 mg Ca^{++}/mL.
3	Sterilize the solutions by membrane filtration.	
4	Store the solutions at 2 to 8°C.	

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Appendix A. (Continued)

A1.2 Lysed Horse Blood (50%)

The steps for preparing lysed horse blood (LHB) are listed in the following table.

Step	Action	Comment
1	Aseptically mix equal volumes of mechanically defibrinated horse blood and sterile, deionized water.	The solution is now 50% LHB. The 50% LHB can be stored indefinitely frozen at ≤ –20°C.
2	Freeze and thaw the 50% LHB until the cells are thoroughly lysed.	This takes approximately 5–7 freeze-thaw cycles.
3	Clear the combined broth and LHB by centrifuging the 50% LHB at 12 000 × g for 20 minutes.	Necessary for use in broth dilution tests.
4	Decant the supernatant and recentrifuge if necessary.	
5	Aseptically add appropriate amounts of the 50% LHB to CAMHB.	• Blood may be added to microdilution panels either when they are first dispensed or after they are thawed just before inoculation. If added after plate preparation, the blood must be added along with the inoculum so additional dilution of the antimicrobial agent in the panel does not occur (see Subchapter 3.7.2 of this standard), and the final concentration of LHB in the well is 2.5% to 5%. • Alternatively, add 5 µL 50% LHB to each 100-µL well before panel inoculation. This is only acceptable if the total volume of liquids added to the well does not exceed 10 µL.
6	After aseptically adding the blood to the autoclaved and cooled broth, check the pH.	

Abbreviations: CAMHB, cation-adjusted Mueller-Hinton broth; LHB, lysed horse blood; pH, negative logarithm of hydrogen ion concentration.

NOTE: LHB prepared for use in broth microdilution is available from commercial sources.

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Appendix A. (Continued)

A2 Agar Media

A2.1 Mueller-Hinton Agar

The steps for preparing the media for use in agar dilution plates are listed in the following table.

Step	Action	Comment
1	Prepare medium according to the manufacturer's instructions.	MHA from a commercially available dehydrated base should be used. Only MHA formulations that have been tested according to and that meet the acceptance limits should be used.[1]
2	Autoclave the medium and let cool to 45 to 50°C in a water bath.	
3	Aseptically add the antimicrobial solutions and heat-labile supplements to the cooled media.	See Subchapter 3.4.2 of this standard.
4	Pour the plates and let the media solidify.	
5	Check the pH of each batch of MHA when prepared.	• The method used depends on the type of equipment available. Check the pH by: – Macerating enough agar to submerge the tip of a pH electrode – Letting a small amount of agar solidify around the tip of a pH electrode in a beaker or cup – Using a surface electrode • The agar medium should have a pH between 7.2 and 7.4 at room temperature and must therefore be checked after solidifying. If the pH is < 7.2, certain drugs will appear to lose potency (eg, aminoglycosides, macrolides), whereas other antimicrobial agents may appear to have excessive activity (eg, tetracyclines). If the pH is > 7.4, the opposite effects can be expected.

Abbreviations: MHA, Mueller-Hinton agar; pH, negative logarithm of hydrogen ion concentration.

NOTE: Supplemental calcium or magnesium cations must not be added to MHA.[1]

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Appendix A. (Continued)

A2.2 Mueller-Hinton Agar With Additives

To prepare Mueller-Hinton agar (MHA) + 2% NaCl, the steps above in section A2.1 should be followed, except that for oxacillin, methicillin, or nafcillin testing of staphylococci, 20 g NaCl is added to 1 L MHA (2% w/v) before autoclaving.

To prepare MHA with 5% sheep blood:

1. Prepare MHA as described above in section A2.1 (steps 1 and 2). When the MHA has cooled to 45 to 50°C, add 50 mL defibrinated sheep blood to 1 L MHA.
2. Check the pH after the blood is aseptically added to the autoclaved and cooled medium. The final pH should be the same as unsupplemented MHA (7.2 to 7.4).

A2.3 GC Agar + 1% Defined Growth Supplement

The steps for preparing GC agar + 1% defined growth supplement are listed in the following table.

Step	Action	Comment
1	Prepare 1 L single-strength GC agar base from a commercially available dehydrated base according to the manufacturer's instructions.	
2	Autoclave the media.	
3	Cool the media to 45 to 50°C in a 45 to 50°C water bath.	
4	Prepare a 1% defined growth supplement containing the specified ingredients per L.	• 1.1 g L-cystine • 0.03 g guanine HCl • 0.003 g thiamine HCl • 0.013 g *para*-aminobenzoic acid • 0.01 g vitamin B_{12} • 0.1 g thiamine pyrophosphate (cocarboxylase) • 0.25 g NAD • 1 g adenine • 10 g L-glutamine • 100 g glucose • 0.02 g ferric nitrate • 25.9 g L-cysteine HCl
5	Add 10 mL 1% defined growth supplement to the cooled GC agar.	
6	Check the pH of the medium after the defined growth supplement is aseptically added to the autoclaved and cooled medium.	The pH should be 7.0–7.4.

Abbreviations: NAD, β-nicotinamide adenine dinucleotide; pH, negative logarithm of hydrogen ion concentration.

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Appendix A. (Continued)

A3 Broth Media

A3.1 Cation-Adjusted Mueller-Hinton Broth

Dehydrated Mueller-Hinton broth (dMHB) base should be obtained from a commercially available source that is accessible in either of the following two forms:

- Containing the desired concentration of cations (20 to 25 mg Ca^{++}/L and 10 to 12.5 mg Mg^{++}/L)
- Without supplemental cation, in which case the cation content is usually inadequate.

If prepared according to the manufacturer's recommendations, the first form does not need additional cation adjustment (except for testing daptomycin, when the broth must be supplemented to 50 mg/L Ca^{++}).

For the second form, the amount of Ca^{++} and Mg^{++} cations present in the medium must be determined before it is prepared. This information is usually available in the certificate of analysis for the lot of medium obtained. If it is not, the manufacturer needs to be contacted for that information. The cation concentrations already present in the broth must be accounted for when the amount of Ca^{++} or Mg^{++} is calculated, if any needs to be added.

The instructions for cation adjustments below should be followed only when the initial Mueller-Hinton broth (MHB) has been certified by the manufacturer or measured by the user to contain no or inadequate amounts of Ca^{++} and Mg^{++} when assayed for total divalent cation content by atomic absorption spectrophotometry. Inadequate cation content may lead to erroneous results. For testing daptomycin, MHB must be supplemented to 50 mg/L Ca^{++}.

The steps for preparing cation-adjusted Mueller-Hinton broth (CAMHB) are listed in the following table.

Step	Action	Comment
1	Prepare MHB according to the manufacturer's recommendations.	
2	Autoclave the media.	
3	Chill overnight at 2 to 8°C.	If the broth is to be used the same day, it can be chilled to 2 to 8°C in an ice bath.
4	While stirring, add 0.1 mL chilled Ca^{++} or Mg^{++} stock solution per L broth for each desired increment of 1 mg/L in the final concentration in the adjusted MHB.	• See the example below. • It is preferable to add the Mg^{++} before adding the Ca^{++}.

Abbreviation: MHB, Mueller-Hinton broth.

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Example for preparing MHB that contains 4.3 mg/L Ca^{++} and 3.8 mg/L Mg^{++}:

Step	Action	Comment
1	Calculate the amount of Ca^{++} needed using this formula: final amount needed – amount in medium = amount to be added	• For Ca^{++}, the final amount needed is 20 to 25 mg/L. Per the calculations below, add 20.7 mg/L to the medium. • 25 mg/L – 4.3 mg/L = 20.7 mg/L
2	Add 2.07 mL Ca^{++} stock per L (0.1 mL for each 1 mg/L needed).	20.7 mg/L • 0.1 mL = 2.07 mL
3	Calculate the amount of Mg^{++} needed using the same formula as in step 1.	• For Mg^{++}, the final amount needed is 10 to 12.5 mg/L. Per the calculations below, add 8.7 mg/L. • 12.5 mg/L – 3.8 mg/L = 8.7 mg/L
4	Calculate the amount of Mg^{++} stock needed per L (0.1 mL for each 1 mg/L needed).	8.7 mg/L • 0.1 mL = 0.87 mL
5	Add 0.87 mL Mg^{++} stock per L.	
6	Check the pH after the cations are added.	The final pH should be 7.2–7.4.

Abbreviation: pH, negative logarithm of hydrogen ion concentration.

NOTE: For testing daptomycin, supplement the broth to 50 mg/L Ca^{++}.

A3.2 Cation-Adjusted Mueller-Hinton Broth With Additives

To prepare CAMHB + 2% NaCl, the steps in section A3.1 are followed, except that for oxacillin testing of staphylococci, 20 g NaCl is added to 1 L of MHB (2% w/v) before autoclaving.

To prepare CAMHB + 2.5% to 5% LHB, the steps in section A3.1 are followed, except that for each liter of broth desired, the media should be prepared with 50 to 100 mL less deionized water to account for the addition of the blood. After autoclaving and cooling, 50 to 100 mL of 50% LHB is added to 1 L broth (see section A1.2 for preparation of LHB). The pH needs to be checked after aseptically adding the blood to the medium. The final pH should be the same as for CAMHB (7.2 to 7.4).

To prepare CAMHB + 0.002% polysorbate 80 (P-80), a 2% solution of P-80 is prepared by diluting full-strength P-80 50-fold (ie, 1 mL P-80 + 49 mL deionized water). 1 mL of the 2% P-80 solution is added per L broth before autoclaving. The final pH should be the same as for CAMHB (7.2 to 7.4).

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Appendix A. (Continued)

A3.3 Iron-Depleted Cation-Adjusted Mueller-Hinton Broth

Iron-depleted cation-adjusted Mueller-Hinton broth (ID-CAMHB) supplemented with calcium (20 to 25 mg/L), magnesium (10 to 12.5 mg/L), and zinc (0.5 to 1.0 mg/L) is used when cefiderocol is tested. See section A3.1 for instructions for preparing calcium and magnesium stock solutions.

The steps for preparing zinc stock solution are listed in the following table.

Step	Action	Comment
1	Dissolve 0.29 g $ZnSO_4 \cdot 7H_2O$ in 100 mL deionized water.	• This solution contains 0.65 mg Zn^{++}/mL (10 mmol Zn^{++}/mL). • Verify that the deionized water has an iron content of ≤ 0.03 mg/L.
2	Sterilize the solution by membrane filtration.	
3	Store the solution at 15 to 25°C in a sterile single-use plastic container.	Previously used glass containers should be avoided to prevent inadvertent iron contamination.

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Appendix A. (Continued)

The steps for preparing ID-CAMHB are listed in the following table.[2]

Step	Action	Comment
1	Prepare the CAMHB.	Follow manufacturer's instructions.
2	Autoclave the media and let cool to room temperature.	
3	Add 100 g chelating resin to 1 L autoclaved CAMHB.[2]	Removes polyvalent metal cations in the medium- to low-level concentrations (range, 0–0.18 mg/L).[2]
4	Stir the solution at room temperature for approximately 6 hours using a magnetic stir bar.	
5	Filter the solution using a 0.2-µm filter.	• Removes the resin. • It is recommended that testing for residual iron levels of the filtrate should be conducted at this step to confirm that the iron content does not exceed 0.03 mg/L. Residual iron content can be measured with a commercial iron detection kit capable of detecting low levels of iron (0.02 mg/L). If iron levels exceed 0.03 mg/L, restart the procedure at the chelation step 3 above.
6	Check the pH to determine whether it is 7.3 ± 0.1.	If the pH is above 7.4, adjust it using 1 or 6 N HCl (use of 6 N HCl will minimize the volume required to adjust the pH). If the pH is below 7.2, use 2.5 N NaOH.
7	Add the cation to achieve final concentrations in the following ranges: • Ca^{++} 20–25 mg/L • Mg^{++} 10–12.5 mg/L • Zn^{++} 0.5–1.0 mg/L	• The final concentration of iron in ID-CAMHB prepared using this method should be ≤ 0.03 mg/L. • Refer to section A3.1 for calculating the amount of Ca^{++}, Mg^{++}, and the table below for calculating the amount of Zn^{++} needed.
8	Check the pH to determine whether it is 7.3 ± 0.1.	If the pH exceeds 7.4, adjust it using 1 or 6 N HCl (use of 6 N HCl will minimize the volume required to adjust the pH). If the pH is below 7.2, use 2.5 N NaOH.
9	Filter the final product using a 0.2-µm filter.	
10	Store the media at 4 to 8°C for up to 2 months.	

Abbreviations: CAMHB, cation-adjusted Mueller-Hinton broth; ID-CAMHB, iron-depleted cation-adjusted Mueller-Hinton broth; pH, negative logarithm of hydrogen ion concentration.

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Appendix A. (Continued)

Example for adding Zn^{++} back to CAMHB that contains below-detectable concentrations (< 0.0001 mg/L) of Zn^{++} after chelation in step 3.[2]

Step	Action	Comment
1	Calculate the amount of Zn^{++} needed using this formula: final amount needed – amount in medium = amount to be added	• For Zn^{++}, the final amount needed is 0.5–1 mg/L. • 1 mg/L – 0 mg/L = 1 mg/L
2	Add 1.54 mL Zn^{++} stock per L (1.54 mL for each 1 mg/L).	• C = concentration, V = volume • $C_1 \cdot V_1$ = desired $C_2 \cdot$ final V_2 • 0.65 mg/mL $Zn^{++} \cdot V_1$ = 1 mg Zn^{++}/1000 mL • 1000 mL • V_1 = 1 mg ÷ 0.65 mg/mL • V_1 = 1.54 mL of Zn^{++} stock
3	Proceed with steps 8 and 9 on the previous page.	

NOTE: Ensure all reagents (eg, deionized water to prepare acid and base and cation solutions) have been verified as having an iron content of ≤ 0.03 mg/L.

A3.4 *Haemophilus* Test Medium Broth

In its broth form, *Haemophilus* test medium (HTM) consists of:

- MHB
- 15 µg/mL NAD
- 15 µg/mL bovine or porcine hematin
- 5 g/L yeast extract
- 0.2 IU/mL thymidine phosphorylase (if sulfonamides or trimethoprim are to be tested)

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Appendix A. (Continued)

The steps for preparing HTM broth are listed in the following table.

Step	Action	Comment
1	Prepare a fresh hematin stock solution by dissolving 50 mg hematin powder in 100 mL 0.01 mol/L NaOH with heat.	Stir the solution until the powder is thoroughly dissolved.
2	Prepare an NAD stock solution by dissolving 50 mg NAD in 10 mL distilled water. Filter to sterilize.	
3	Prepare MHB from a commercially available dehydrated base according to the manufacturer's instructions.	
4	Add 5 g yeast extract and 30 mL hematin stock solution to 1 L MHB.	
5	Autoclave the media.	
6	Cool the media to 45 to 50°C.	
7	Aseptically add cations, if needed, as in CAMHB.	If sulfonamides or trimethoprim are to be tested, aseptically add 0.2 IU thymidine phosphorylase to the medium.
8	Aseptically add 3 mL of the NAD stock solution.	
9	Check the pH of the medium after the NAD is added to the autoclaved and cooled medium.	The pH should be 7.2–7.4.

Abbreviations: ATCC®,[a] American Type Culture Collection; CAMHB, cation-adjusted Mueller-Hinton broth; HTM, *Haemophilus* test medium; MHB, Mueller-Hinton broth; NAD, β-nicotinamide adenine dinucleotide; pH, negative logarithm of hydrogen ion concentration; QC, quality control.
[a] ATCC® is a registered trademark of the American Type Culture Collection.

NOTE: *Haemophilus influenzae* (ATCC® 10211) is recommended as a useful additional QC strain to verify the growth promotion properties of HTM. In particular, HTM manufacturers are encouraged to use *H. influenzae* ATCC® 10211 as a supplemental QC test strain.

A3.5 Mueller-Hinton Fastidious Broth

Mueller-Hinton fastidious (MH-F) broth consists of:

- CAMHB
- 5% LHB
- 20 mg/mL NAD

To prepare MH-F broth for testing *H. influenzae*, the steps for preparing CAMHB in sections A3.1 and A3.2 should be followed, except that for each liter of broth desired, the medium should be prepared with 100 mL less deionized water to account for the addition of the blood (see section A1.2 for preparation of LHB). After the medium is autoclaved and cooled, 1 mL of 20 mg/L β-nicotinamide adenine dinucleotide (NAD) and 100 mL 50% LHB are added to 1 L broth. The pH needs to be checked after the NAD and blood is aseptically added to the medium. The final pH should be the same as for CAMHB (7.2 to 7.4).

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Appendix A. (Continued)

The steps for preparing MH-F broth are listed in the following table.

Step	Action	Comment
1	Prepare an NAD stock solution by dissolving 20 mg NAD per 1 mL distilled water. Filter to sterilize.	1 mL of a 20 mg/mL solution will be needed for each liter of MH-F broth. Weigh > 20 mg and determine the water volume needed by dividing the NAD weight by 20 mg/mL. For example, if 45 mg NAD is weighed, divide 45 by 20 and dissolve NAD in 2.25 mL distilled water. The remaining solution can be stored at ≤ –70°C for 1 year.
2	Prepare MHB from a commercially available dehydrated base according to the manufacturer's instructions. For each liter of broth desired, the medium should be prepared with 100 mL less deionized water.	
3	Autoclave the MHB medium.	
4	Cool the medium to 45 to 50°C.	
5	Aseptically add cations, if needed, as in CAMHB.	
6	Aseptically add 1 mL of the 20 mg/mL NAD stock solution to 1 L CAMHB.	
7	Aseptically add 100 mL 50% LHB to 900 mL CAMHB.	
8	Check the pH.	The pH should be 7.2–7.4.

Abbreviations: CAMHB, cation-adjusted Mueller-Hinton broth; LHB, lysed horse blood; MHB, Mueller-Hinton broth; MH-F, Mueller-Hinton fastidious; NAD, β-nicotinamide adenine dinucleotide; pH, negative logarithm of hydrogen ion concentration.

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Appendix A. (Continued)

A4 Reagents: 0.5 McFarland Turbidity Standard

The steps for preparing a 0.5 McFarland turbidity standard are listed in the following table.

Step	Action	Comment
1	Prepare a 0.048-mol/L $BaCl_2$ (1.175% w/v $BaCl_2 \cdot 2H_2O$) stock solution in laboratory-grade water.	
2	Prepare a 0.18-mol/L (0.36 N) H_2SO_4 (1% v/v) stock solution in laboratory-grade water.	
3	Add 0.5 mL of the $BaCl_2$ solution to 99.5 mL of the H_2SO_4 stock solution and stir constantly.	Stirring helps maintain a suspension.
4	Verify that the density of the turbidity standard is correct by measuring absorbance using a spectrophotometer with a 1-cm light path and matched cuvettes.	The absorbance at 625 nm should be 0.08–0.13 for the 0.5 McFarland standard.
5	Transfer the $BaSO_4$ suspension to tubes in 4- to 6-mL aliquots.	Use screw-cap tubes of the same size as those used for standardizing the bacterial inoculum.
6	Tightly seal the tubes and store in the dark at room temperature.	
7	Vortex the $BaSO_4$ turbidity standard vigorously before each use and inspect for a uniformly turbid appearance.	• Replace the standard if large particles appear. • **NOTE:** McFarland standards made from latex particle suspension are commercially available. When used, they should be mixed by inverting gently (not on a vortex mixer) immediately before use. • $BaSO_4$ standards should be replaced or their densities verified monthly.

References for Appendix A

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Appendix B. Conditions for Dilution Antimicrobial Susceptibility Tests

Conditions needed for performing antimicrobial susceptibility tests on nonfastidious and fastidious organisms are listed in Tables B1 and B2, respectively.

Table B1. Conditions for Dilution Antimicrobial Susceptibility Tests for Nonfastidious Organisms (refer to CLSI M100[1] Tables 2)

Organism/ Organism Group	Medium	0.5 McFarland Inoculum	Incubation	Incubation Time	Routine QC[a]	Comments/ Modifications
Enterobacterales (including *Salmonella* and *Shigella* spp.)	CAMHB; MHA; ID-CAMHB (cefiderocol)	Colony suspension in broth, sterile distilled water, or saline, or broth culture method	35°C ± 2°C; ambient air	16–20 hours	• *Escherichia coli* ATCC®[b] 25922 • *P. aeruginosa* ATCC® 27853 for carbapenems • *Staphylococcus aureus* ATCC® 29213 when testing azithromycin against *Salmonella enterica* ser. Typhi or *Shigella* spp. • For β-lactam combination agents, refer to CLSI M100[1] Tables 5.	
Pseudomonas aeruginosa	CAMHB; MHA; ID-CAMHB (cefiderocol)	Colony suspension in broth, sterile distilled water, or saline, or broth culture method	35°C ± 2°C; ambient air	16–20 hours	• *P. aeruginosa* ATCC® 27853 • For β-lactam combination agents, refer to CLSI M100[1] Tables 5.	
***Acinetobacter* spp.**	CAMHB; MHA; ID-CAMHB (cefiderocol)	Colony suspension in broth, sterile distilled water, or saline, or broth culture method	35°C ± 2°C; ambient air	20–24 hours	• *P. aeruginosa* ATCC® 27853 • *E. coli* ATCC® 25922 for tetracyclines and trimethoprim-sulfamethoxazole • For β-lactam combination agents, refer to CLSI M100[1] Tables 5.	

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Table B1. (Continued)

Organism/ Organism Group	Medium	0.5 McFarland Inoculum	Incubation	Incubation Time	Routine QC[a]	Comments/ Modifications
***Burkholderia cepacia* complex**	CAMHB; MHA	Colony suspension in broth, sterile distilled water, or saline, or broth culture method	35°C ± 2°C; ambient air	20–24 hours	• *P. aeruginosa* ATCC® 27853 • *E. coli* ATCC® 25922 for chloramphenicol, minocycline, and trimethoprim-sulfamethoxazole • For β-lactam combination agents, refer to CLSI M100[1] Tables 5.	
Stenotrophomonas maltophilia	CAMHB; MHA; ID-CAMHB (cefiderocol)	Colony suspension in broth, sterile distilled water, or saline, or broth culture method	35°C ± 2°C; ambient air	20–24 hours	• *P. aeruginosa* ATCC® 27853 • *E. coli* ATCC® 25922 for chloramphenicol, minocycline, and trimethoprim-sulfamethoxazole • For β-lactam combination agents, refer to CLSI M100[1] Tables 5.	
Other non-Enterobacterales	CAMHB; MHA	Colony suspension in broth, sterile distilled water, or saline, or broth culture method	35°C ± 2°C; ambient air	16–20 hours	• *E. coli* ATCC® 25922 for chloramphenicol, tetracyclines, sulfonamides, and trimethoprim-sulfamethoxazole • *P. aeruginosa* ATCC® 27853 • For β-lactam combination agents, refer to CLSI M100[1] Tables 5.	

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Table B1. (Continued)

Organism/ Organism Group	Medium	0.5 McFarland Inoculum	Incubation	Incubation Time	Routine QC[a]	Comments/Modifications
***Staphylococcus* spp.**	CAMHB; CAMHB with 2% NaCl for oxacillin; CAMHB supplemented to 50 µg/mL calcium for daptomycin; MHA; MHA with 2% NaCl for oxacillin salt agar test	Colony suspension in broth, sterile distilled water, or saline	35°C ± 2°C; ambient air	16–20 hours; 24 hours for oxacillin and vancomycin	*S. aureus* ATCC® 29213	• Agar dilution has not been validated for daptomycin. • Testing at temperatures above 35°C may not detect MRS.
***Enterococcus* spp.**	CAMHB; CAMHB supplemented to 50 µg/mL calcium for daptomycin; MHA	Colony suspension in broth, sterile distilled water, or saline, or broth culture method	35°C ± 2°C; ambient air	16–20 hours; 24 hours for vancomycin	*Enterococcus faecalis* ATCC® 29212	Agar dilution has not been validated for daptomycin.
					E. faecalis ATCC® 33186	Alternate to *E. faecalis* ATCC® 29212 to assess suitability of medium for sulfonamide or trimethoprim MIC and disk diffusion tests.

Abbreviations: ATCC®, American Type Culture Collection; CAMHB, cation-adjusted Mueller-Hinton broth; ID-CAMHB, iron-depleted cation-adjusted Mueller-Hinton broth; MHA, Mueller-Hinton agar; MIC, minimal inhibitory concentration; MRS, methicillin (oxacillin)-resistant staphylococci; QC, quality control.

[a] See specific CLSI M100[1] Tables 3 for additional QC recommendations for screening and confirmatory tests.

[b] ATCC® is a registered trademark of the American Type Culture Collection.

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Table B2. Conditions for Dilution Antimicrobial Susceptibility Tests for Fastidious Organisms (refer to CLSI M100[1] Tables 2)

Organism/ Organism Group	Medium	0.5 McFarland Inoculum	Incubation	Incubation Time	Routine QC[a]	Comments/Modifications
Haemophilus influenzae* and *Haemophilus parainfluenzae	HTM broth or MH-F broth for *H. influenzae*	Colony suspension in broth or saline prepared from an overnight (preferably 20- to 24-hour) chocolate agar plate[b]	35°C ± 2°C; ambient air	20–24 hours	• *H. influenzae* ATCC®[c] 49247 and/or *H. influenzae* ATCC® 49766 • For β-lactam combination agents, refer to CLSI M100[1] Tables 5.	Use either *H. influenzae* ATCC® 49247 or *H. influenzae* ATCC® 49766 or both of these strains, based on the antimicrobial agents to be tested. Neither strain has QC ranges for all agents that might be tested against *H. influenzae* or *H. parainfluenzae*.
Neisseria gonorrhoeae	GC agar base with 1% defined growth supplement[d]	Colony suspension in MHB or 0.9% phosphate-buffered saline, pH 7, prepared from an overnight (20- to 24-hour) chocolate agar plate incubated in 5% CO_2	36°C ± 1°C (do not exceed 37°C); 5% CO_2	20–24 hours	*N. gonorrhoeae* ATCC® 49226	The use of a cysteine-free supplement is needed for agar dilution tests with carbapenems and clavulanates. Cysteine-containing defined growth supplements **do not** significantly alter dilution test results with other drugs.
Streptococcus pneumoniae	CAMHB with 2.5% to 5% LHB broth; MHA + 5% sheep blood, or MH-F broth	Colony suspension in broth or saline using colonies from an overnight (18- to 20- hour) sheep blood agar plate	35°C ± 2°C; ambient air (CO_2 if necessary for growth with agar dilution)	20–24 hours	*S. pneumoniae* ATCC® 49619	
***Streptococcus* spp.**	CAMHB with 2.5% to 5% LHB; CAMHB with 2.5% to 5% LHB supplemented to 50 µg/mL calcium for daptomycin; MHA + 5% sheep blood	Colony suspension in broth or saline using colonies from an overnight (18- to 20-hour) sheep blood agar plate	35°C ± 2°C; ambient air (CO_2 if necessary for growth with agar dilution)	20–24 hours	*S. pneumoniae* ATCC® 49619	Agar dilution has not been validated for daptomycin.

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Table B2. (Continued)

Organism/ Organism Group	Medium	0.5 McFarland Inoculum	Incubation	Incubation Time	Routine QC[a]	Comments/ Modifications
Neisseria meningitidis	CAMHB with 2.5% to 5% LHB; MHA with 5% sheep blood	Colony suspension in broth or saline prepared from a 20- to 24-hour chocolate agar plate incubated in 5% CO_2[d]	35°C ± 2°C; 5% CO_2	20–24 hours	• *S. pneumoniae* ATCC® 49619 (ambient air or 5% CO_2; azithromycin QC test must be incubated in ambient air) • *E. coli* ATCC® 25922 (ambient air or 5% CO_2; for ciprofloxacin, nalidixic acid, minocycline, and sulfisoxazole)	**Caution:** Perform all testing in a BSC.
Anaerobes	• Brucella broth with hemin (5 µg/mL), vitamin K_1 (1 µg/mL), and 5% LHB **(for *Bacteroides fragilis* group only)** • Brucella agar with hemin (5 µg/mL), vitamin K_1 (1 µg/mL), and 5% laked sheep blood **(for all anaerobes)** • (See CLSI M11[2] for method of preparation for these media.)	• Colony suspension in prereduced Brucella broth or broth culture method • (See CLSI M11[2] for exact methods for inoculum preparation.)	36°C ± 1°C; anaerobically	Broth, 46–48 hours; agar, 42–48 hours	Test one or more of the following organisms. The choice and number of QC strains tested should be based on obtaining on-scale end points for the antimicrobial agents tested. • *B. fragilis* ATCC® 25285 • *Bacteroides thetaiotaomicron* ATCC® 29741 • *Clostridiodes* (formerly *Clostridium*) *difficile* ATCC® 700057 • *Eggerthella* (*Eubacterium*) *lentum* ATCC® 43055	

Abbreviations: ATCC®, American Type Culture Collection; BSC, biological safety cabinet; CAMHB, cation-adjusted Mueller-Hinton broth; CFU, colony-forming unit(s); HTM, *Haemophilus* test medium; LHB, lysed horse blood; MHA, Mueller-Hinton agar; MHB, Mueller-Hinton broth; MH-F, Mueller-Hinton fastidious; pH, negative logarithm of hydrogen ion concentration; QC, quality control.

[a] See specific CLSI M100[1] Tables 3 for additional QC recommendations for screening and confirmatory tests.

[b] This suspension will contain approximately 1 to 4 × 10^8 CFU/mL. Care should be used in preparing this suspension, because higher inoculum concentrations may lead to false-resistant results with some β-lactam antimicrobial agents, particularly when β-lactamase–producing strains of *H. influenzae* are tested.

[c] ATCC® is a registered trademark of the American Type Culture Collection.

[d] Colonies grown on sheep blood agar may be used for inoculum preparation. However, the 0.5 McFarland suspension obtained from sheep blood agar will contain fewer CFU/mL. This must be considered when the final dilution before panel inoculation is prepared, as guided by colony counts.

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Appendix B. (Continued)

References for Appendix B

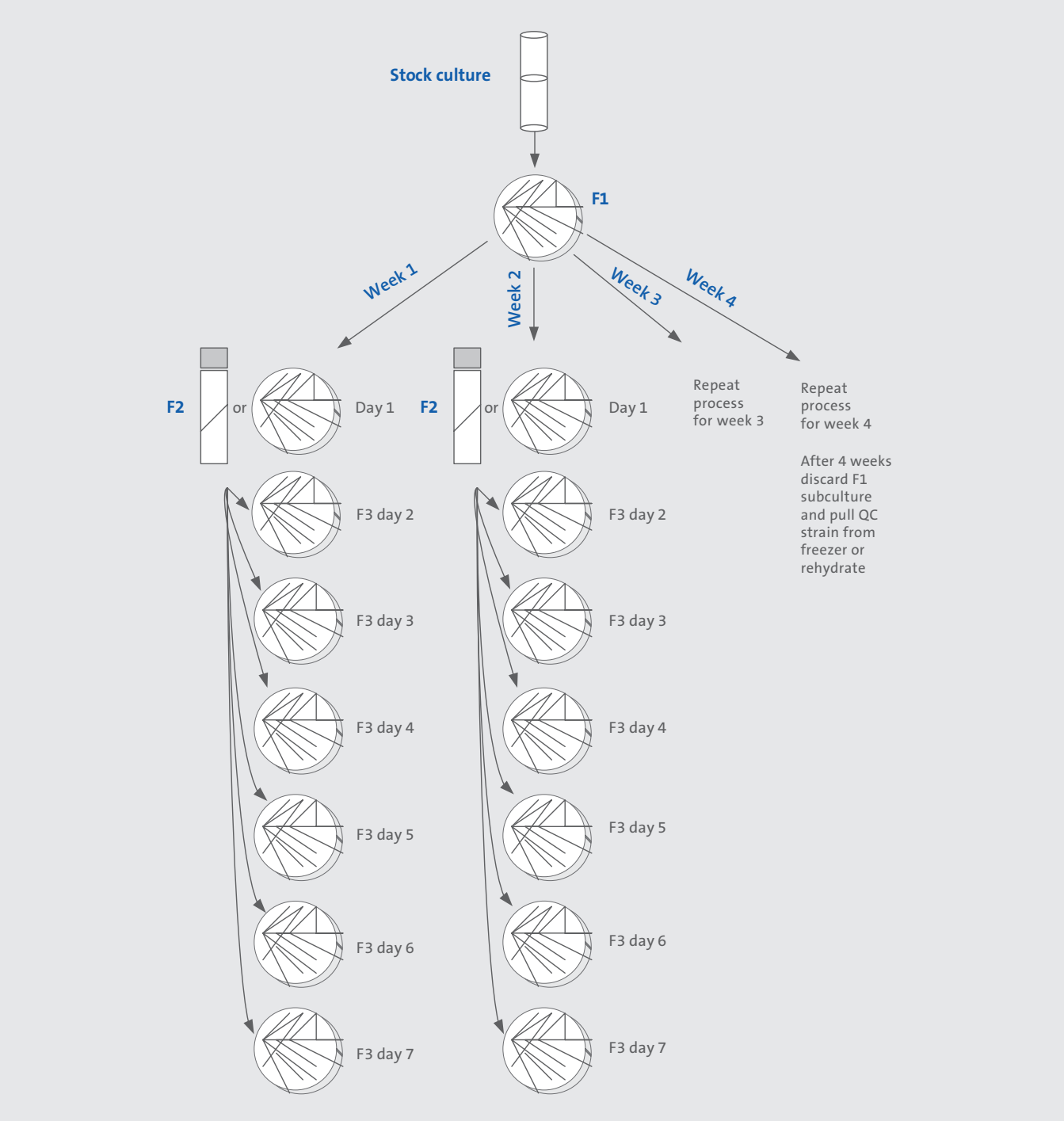
1 CLSI. *Performance Standards for Antimicrobial Susceptibility Testing*. 34th ed. CLSI supplement M100. Clinical and Laboratory Standards Institute; 2024.

2 CLSI. *Methods for Antimicrobial Susceptibility Testing of Anaerobic Bacteria*. 9th ed. CLSI standard M11. Clinical and Laboratory Standards Institute; 2018.

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Appendix C. Quality Control Strain Maintenance

Figure C1 illustrates the workflow for subculturing and using reference strains as described in Subchapter 4.4 of this standard. In Figure C1, "F" indicates the "frozen" or "freeze-dried" state of the stock culture, while "1" indicates the first passage, "2" the second passage, and "3" the third passage from the stock culture.



Abbreviation: QC, quality control.

Figure C1. Workflow for Subculturing and Using Reference Strains

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Appendix C. (Continued)

The steps for maintaining stock cultures are listed in the following table.

Step	Action	Comment
1	Subculture frozen or freeze-dried stock culture to appropriate agar plate. • Tryptic soy or blood agar for nonfastidious QC strains • Blood agar or enriched chocolate for fastidious QC strains	This subculture is designated the F1 subculture ("F" indicates the "frozen" or "freeze-dried" state of the stock culture, while "1" indicates the first passage and "2" the second passage from the stock culture).
2	Incubate under the appropriate conditions for the organism.	Store F1 subculture at 2 to 8°C or as appropriate for the organism type for up to 4 weeks.
3	Subculture from F1 onto an agar slant or agar plate to prepare F2 subculture.	Subculture lyophilized or frozen stock cultures twice before they are used to prepare a QC test inoculum suspension.
4	Incubate the F2 subculture under the appropriate conditions for the organism.	Store the F2 subculture at 2 to 8°C or as appropriate for the organism type for up to 1 week.
5	Subculture from F2 onto an agar plate (not to broth or agar slant) to prepare F3 subcultures.	The number of F3 subcultures needed depends on the laboratory's QC testing schedule.
6	Select isolated colonies from fresh F2 or F3 agar plate (eg, overnight incubation) to prepare the QC test inoculum suspension.	If a QC test appears contaminated or if QC results are questionable, it may be necessary to prepare a new F1 subculture from lyophilized or frozen stock culture.
7	Prepare a new F2 subculture (agar slant or agar plate) each week for weeks 2, 3, and 4.	• Prepare new F1 subcultures at least every 2 weeks for: – *Pseudomonas aeruginosa* ATCC®[a] 27853 – *Enterococcus faecalis* ATCC® 51299 – *Streptococcus pneumoniae* ATCC® 49619 • If held longer than 2 weeks, results for *P. aeruginosa* ATCC® 27853 and *E. faecalis* ATCC® 51299 may fall out of range and *S. pneumoniae* ATCC® 49619 may lose viability.
8	After 4 weeks, prepare a new F1 subculture from frozen or freeze-dried stock culture.	

Abbreviations: ATCC®, American Type Culture Collection; QC, quality control.
[a] ATCC® is a registered trademark of the American Type Culture Collection.

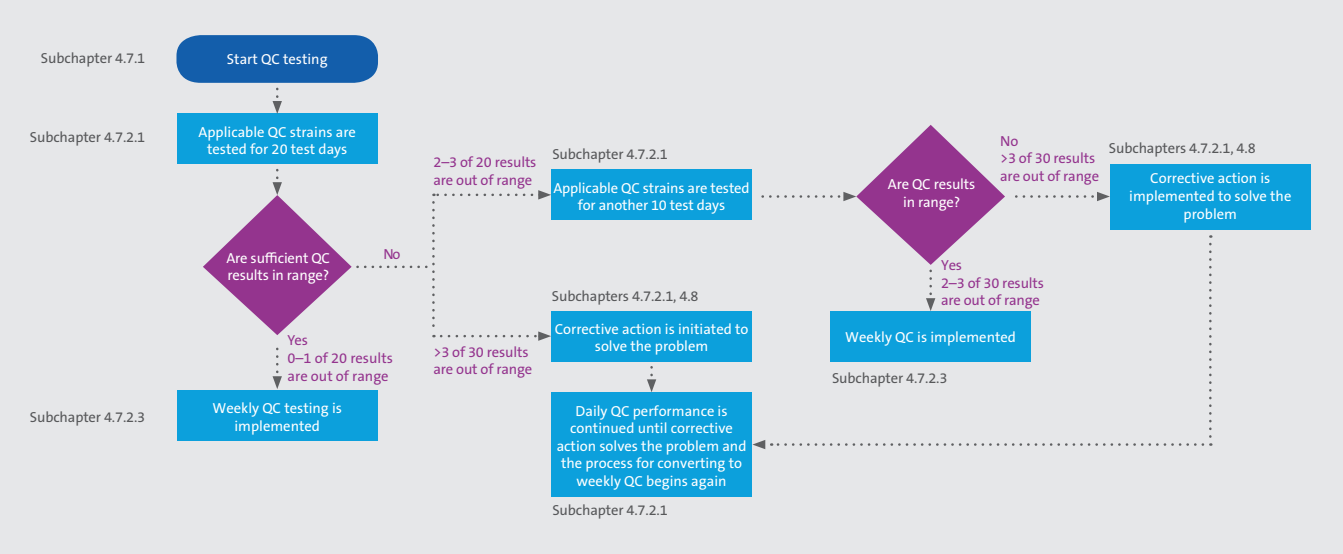
NOTE: The suggestions here apply to daily or weekly QC plans and to routine and supplemental QC strains.

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Appendix D. Quality Control Protocol Flow Charts

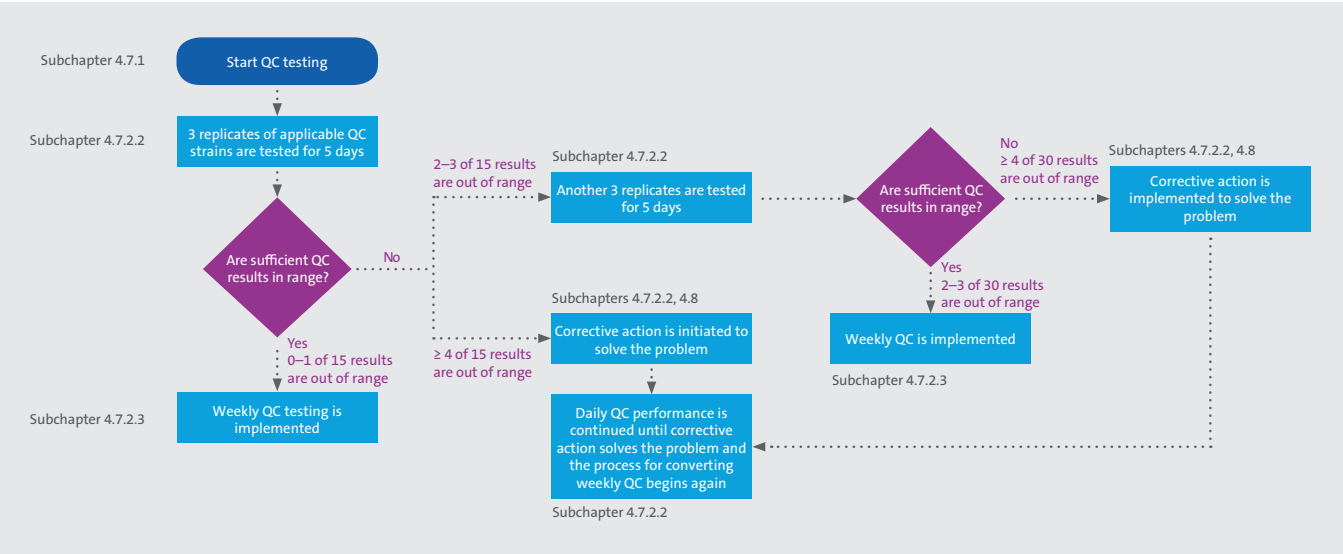
NOTE: All subchapter references are to subchapters in this standard.

D1 Quality Control Protocol: Conversion From Daily to Weekly Testing (20- or 30-Day Plan)



Abbreviation: QC, quality control.

D2 Quality Control Protocol: Conversion From Daily to Weekly Testing (15-Replicate [3- × 5-Day] Plan)

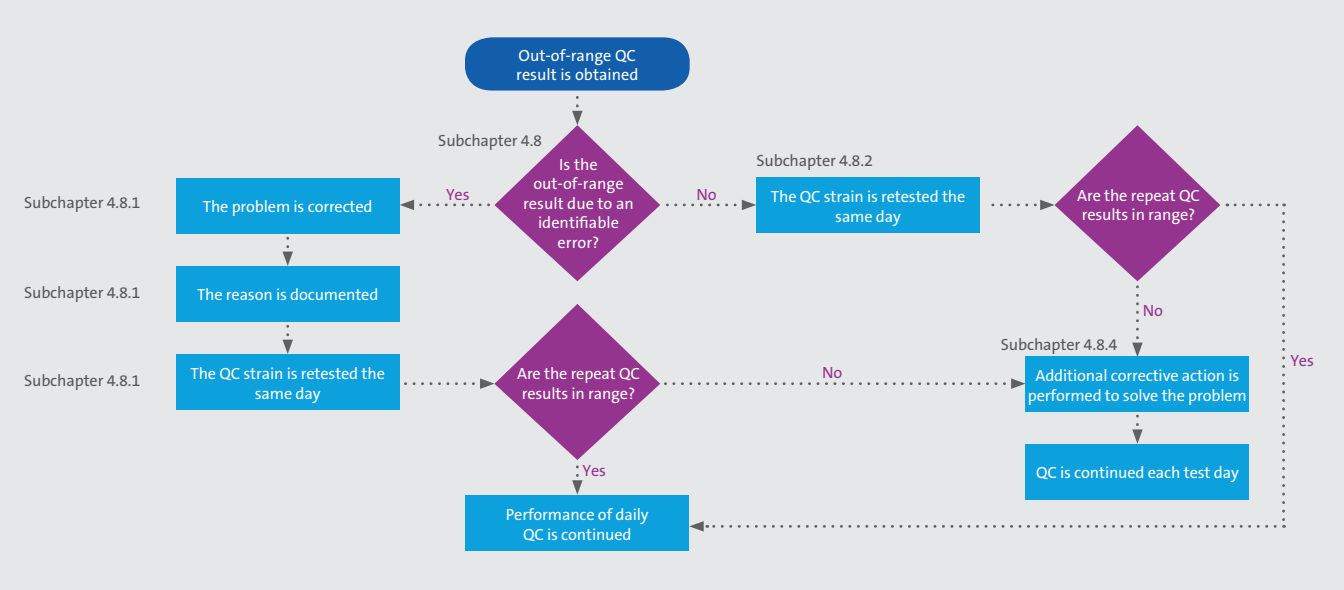


Abbreviation: QC, quality control.

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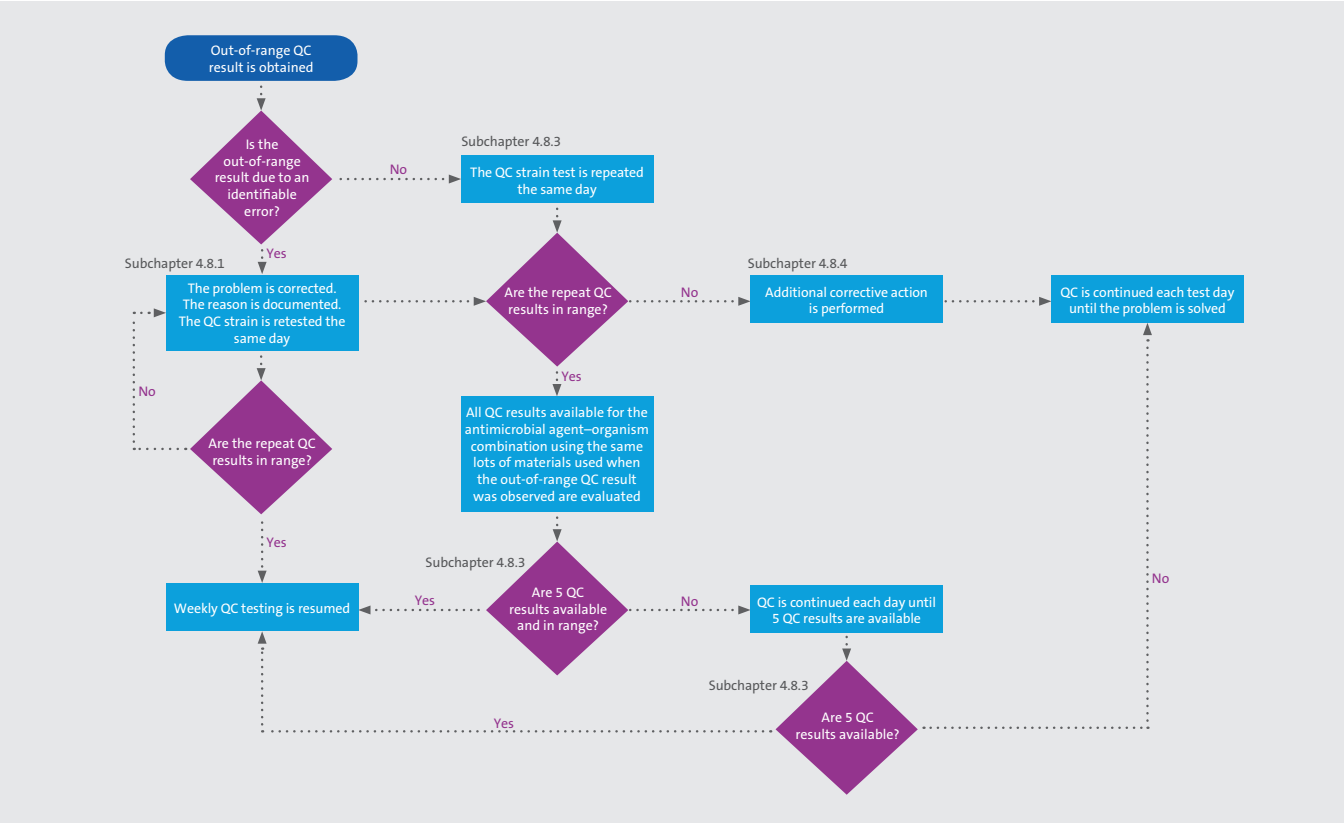
Appendix D. (Continued)

D3 Quality Control Protocol: Daily Quality Control Testing—Corrective Action



Abbreviation: QC, quality control.

D4 Quality Control Protocol: Weekly Quality Control Testing—Corrective Action



Abbreviation: QC, quality control.

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The Quality Management System Approach

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- Organization and Leadership
- Customer Focus
- Facilities and Safety Management
- Personnel Management
- Supplier and Inventory Management
- Equipment Management
- Process Management
- Documents and Records Management
- Information Management
- Nonconforming Event Management
- Assessments
- Continual Improvement

The QSEs covered by CLSI M07 and its related CLSI documents are available on the CLSI website: https://clsi.org/qse

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Minimal Inhibitory Concentration Reading Guide

NOTE: Orange circles throughout this guide indicate where the minimal inhibitory concentration (MIC) end point should be determined.

General Rules for Determining MIC End Points (Figures 1 to 3)

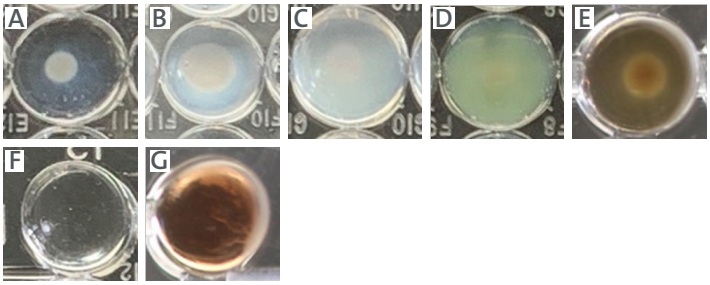


Figure 1. Assessing Growth of Control Wells

For results on a broth microdilution panel to be considered valid, acceptable growth exhibiting a button and/or definite turbidity must occur in the growth control well (see **Figures 1A to 1E**). See CLSI M100 for specific antimicrobial agent interpretations. No growth should be demonstrated in the negative control well (see **Figures 1F and 1G**).

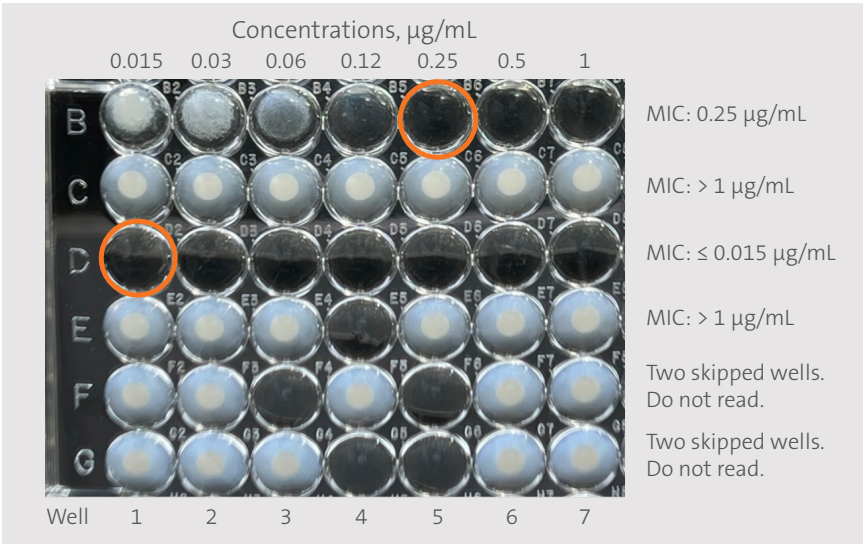


Figure 2. Determining MIC End Points

General Reading Instructions. For many antimicrobial agents, the MIC end point is the lowest concentration of an antimicrobial agent that completely inhibits the growth of the organism in the microdilution well as detected by the unaided eye (see **Figure 2, orange circles**); exceptions are listed in the sections below. Aside from the exceptions, light or faint growth should be considered when determining end points (see **Figure 2, well B4**). Users should read MIC panels against a dark background when determining MIC end points. Users may further examine wells with questionable growth with other laboratorians and/or a supervisor. If a consensus result cannot be determined, repeat the test, and if possible, perform AST by an alternative method (eg, disk diffusion), particularly if critical to patient care (eg, MIC is near the breakpoint).

NOTE: Bubbles can be an artifact of panel preparation and should be ignored when determining MIC end points.

Reading MICs at the Ends of the Dilution Range. If there is growth in all of the concentrations of the antimicrobial agent tested including the highest concentration tested, the MIC is reported as greater than but not equal to (>) the highest concentration tested (see **Figure 2, row C**). If there is no growth in all of the concentrations of antimicrobial agent tested including the lowest concentration tested, the MIC is reported as less than or equal to (≤) the lowest concentration tested (see **Figure 2, row D**).

Skipped Wells. When a single skipped well occurs, ignore the skipped well and read the highest MIC (see **Figure 2, row E**). When multiple skipped wells are observed non-consecutively (see **Figure 2, row F**) or across multiple concentrations consecutively (see **Figure 2, row G**), do not report the results and repeat the test.

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Reference: CLSI M100 | *Performance Standards for Antimicrobial Susceptibility Testing, 34th Edition*

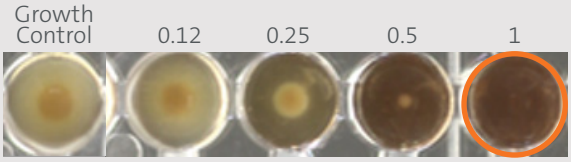
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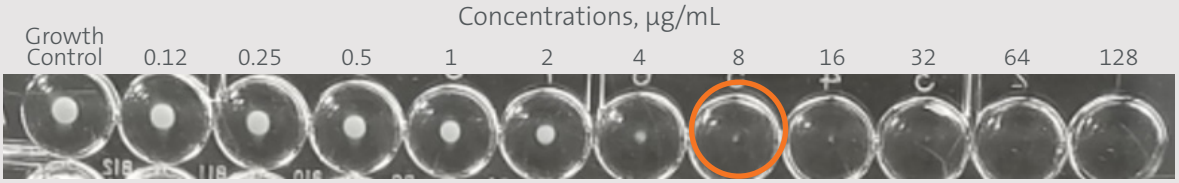
Proteus mirabilis and tobramycin. MIC: 2 µg/mL.



Streptococcus pneumoniae (ATCC® 49619) and levofloxacin. MIC: 1 µg/mL.

Figure 3. Other Examples of Reading Complete Inhibition

Determining MIC End Points in Special Situations (Figures 4 to 8)



Shigella sonnei and azithromycin. MIC: 8 µg/mL



Staphylococcus aureus (ATCC® 29213) and azithromycin. MIC: 2 µg/mL.

Figure 4. Trailing End Points

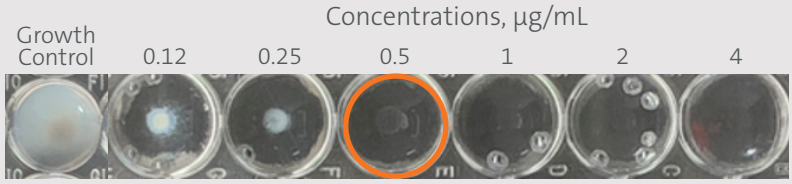
For reading certain classes of antimicrobials or antimicrobial agents (eg, lincosamides, tetracyclines, macrolides, phenicols, oxazolidinones, streptogramins, fusidic acid), trailing growth can occur. Read the MIC as the first well where trailing in growth begins.

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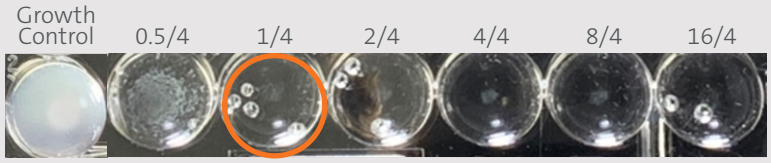
NOTE: Orange circles throughout this guide indicate where the minimal inhibitory concentration (MIC) end point should be determined.



Proteus mirabilis and meropenem (trailing growth starts at 0.5 µg/mL). MIC: 0.5 µg/mL.



Proteus mirabilis and piperacillin-tazobactam (extremely light growth in all wells). MIC: ≤ 4/4 µg/mL.



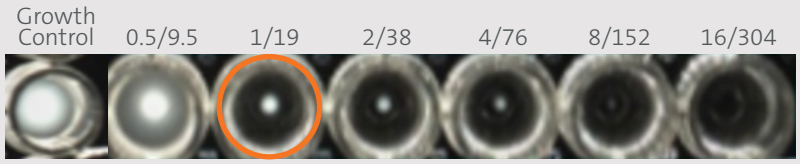
Proteus mirabilis and ceftazidime-avibactam (extremely light growth in 1/4 – 16/4 µg/mL). MIC: 1/4 µg/mL.

Figure 5. Ghosting Growth Pattern of *Proteus mirabilis*

Certain antimicrobials among a panel of agents exhibit a ghosting pattern. Swarming *Proteus* spp. may present as trailing or "ghosting" on an MIC panel. Obvious trailing or consistent light growth (ie, ghosting) should be ignored. Sometimes, *Proteus* spp. trailing or ghosting make determining MIC end points difficult; if so, repeat the test and/or use another method (eg, disk diffusion).



Acinetobacter baumannii and trimethoprim-sulfamethoxazole. MIC: 8/152 µg/mL.



Escherichia coli and trimethoprim-sulfamethoxazole. MIC: 1/19 µg/mL.

Figure 6. Trimethoprim-sulfamethoxazole or sulfisoxazole

Trimethoprim-sulfamethoxazole or sulfisoxazole MIC end points should be read at 80% inhibition.

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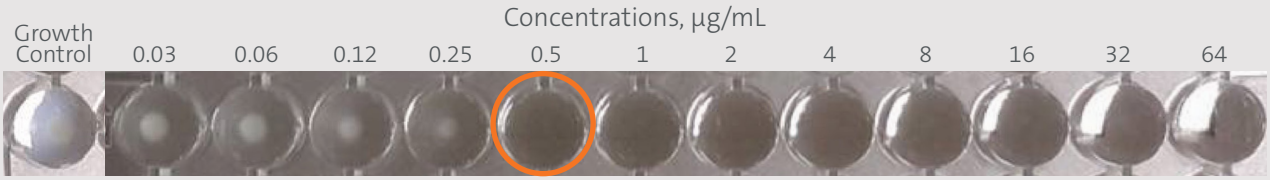
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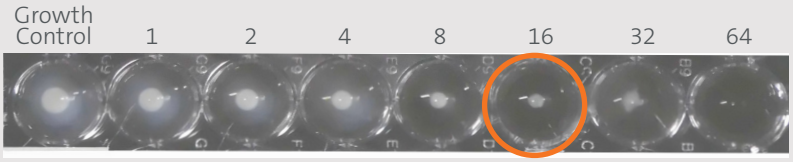
Cefiderocol test with a clear end point. MIC: 0.5 µg/mL.



Cefiderocol growth button with a 1 mm diameter in proportion to the 7 mm diameter of a well in a 96-well MIC panel.



Cefiderocol test with trailing end point. MIC: 0.25 µg/mL.



Cefiderocol test with trailing end point. MIC: 16 µg/mL. The button of the 16 µg/mL well shows a reduction in growth compared to the growth control well that corresponds to a growth button of approximately 1 mm diameter.



Cefiderocol test with light haze or faint turbidity. MIC: 32 µg/mL.



Cefiderocol test with light haze or faint turbidity. MIC: 4 µg/mL.

Figure 7. Cefiderocol

Compare the amount of growth in the wells containing cefiderocol with the amount of growth in the growth control well containing no antimicrobial agent. Read the MIC as the lowest concentration of cefiderocol (first clear well) where no trailing (button ≤ 1 mm) or light haziness is observed. If reduced growth is observed, read the MIC as the lowest concentration of cefiderocol in which the reduction of growth compared to the growth control well corresponds to:

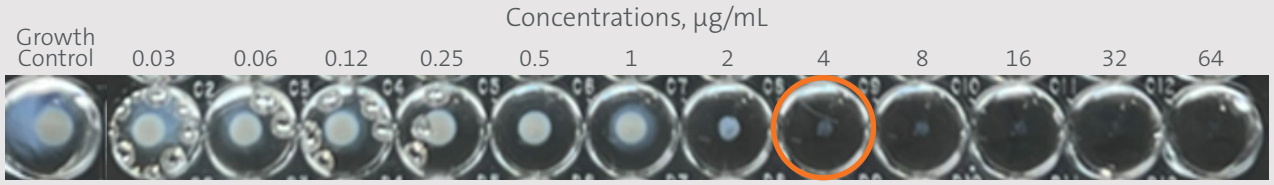
- A button of approximately ≤ 1 mm, or
- a light haze or faint turbidity with a significant (eg, 80%) reduction of growth compared to the growth control well.

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Reference: CLSI M100 | *Performance Standards for Antimicrobial Susceptibility Testing, 34th Edition*

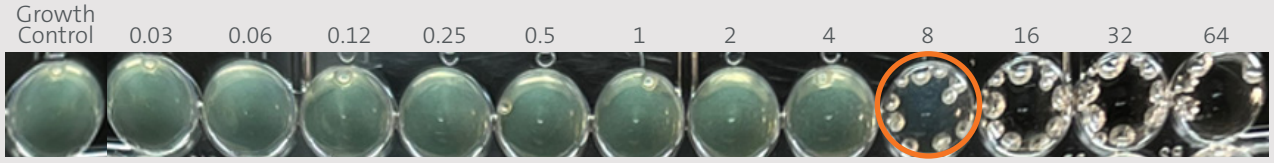
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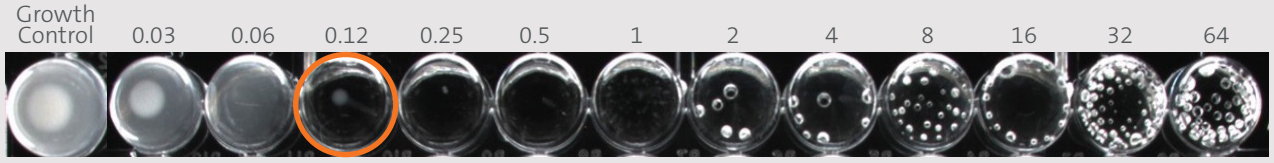
Klebsiella pneumoniae (trailing is observed in wells with 4 - 16 µg/mL, with the growth in 4 µg/mL corresponding to a button of ≤ 1 mm). MIC: 4 µg/mL.



Pseudomonas aeruginosa (light haze in 8 µg/mL demonstrates significant (eg, 80%) reduction in growth compared to the growth control well). MIC: 8 µg/mL.



Acinetobacter baumannii (trailing is observed in wells with 0.5 - 2 µg/mL, with the growth in 2 µg/mL corresponding to a button of ≤ 1 mm). MIC: 2 µg/mL.



Acinetobacter baumannii (growth in 0.12 µg/mL demonstrates a significant (eg, 80%) reduction in growth compared to the growth control well). MIC: 0.12 µg/mL.

Figure 8. Other Examples of Reading Cefiderocol

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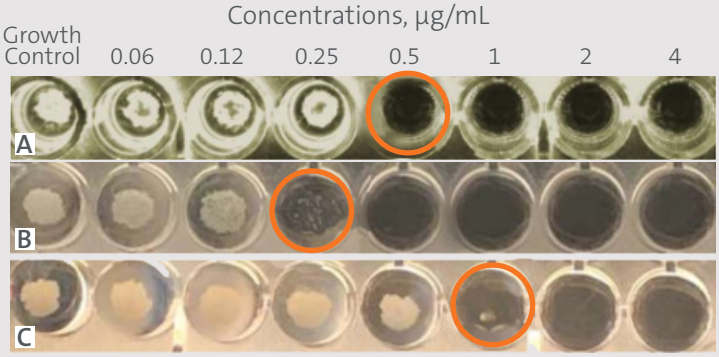
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M07-Ed12-QG

Minimal Inhibitory Concentration Reading Guide

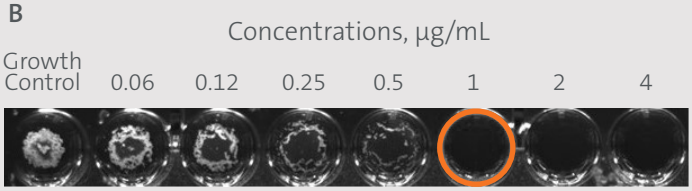
NOTE: Orange circles throughout this guide indicate where the minimal inhibitory concentration (MIC) end point should be determined.

A



Staphylococcus aureus incubated in ambient conditions for 16-20 hours. MICs are shown in orange circles (A = 0.5, B = 0.25, and C = 1 µg/mL).

B



Staphylococcus spp. other than *S. aureus* incubated in 5% CO_2 for 20-24 hours. MIC: 1 µg/mL.

Figure 9. Exebacase

For *Staphylococcus aureus* with exebacase, most MIC end points will be read as complete inhibition of growth compared with the growth control well (see **Figure 9A, row A**). In some cases, a marked reduction in growth compared with the growth control is observed (see **Figure 9A, rows B and C**). When there is a marked reduction in growth, read the MIC as the first well in which growth is significantly reduced. For *Staphylococcus* spp. other than *S. aureus,* a marked reduction in growth compared with the growth control will be frequently observed but the MIC should be read at complete inhibition (see **Figure 9B**). For β-hemolytic streptococci, the MIC should be read at complete inhibition.

Source Document: CLSI M07 | *Methods for Dilution Antimicrobial Susceptibility Tests for Bacteria That Grow Aerobically, 12th Edition*
Reference: CLSI M100 | *Performance Standards for Antimicrobial Susceptibility Testing, 34th Edition*

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